Chapter 1

Vector Analysis

Problem 1.1

Show that the dot product and cross product is distributive

Solution:

Let the three vectors be

$$\boldsymbol{v}_1 = [a, b, c]$$
$$\boldsymbol{v}_2 = [d, e, f]$$
$$\boldsymbol{v}_3 = [g, h, i]$$

dot product:

Show

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = \boldsymbol{v_1} \cdot \boldsymbol{v}_2 + \boldsymbol{v_1} \cdot \boldsymbol{v}_3$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d + g, e + h, f + i]$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = ad + be + cf$$
$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_3 = ag + bh + ci$$

LHS:

$$\boldsymbol{v}_1 \cdot (\boldsymbol{v}_2 + \boldsymbol{v}_3) = [a, b, c] \cdot [d + g, e + h, f + i] =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

RHS:

$$\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 + \boldsymbol{v}_1 + \boldsymbol{v}_3 = ad + be + cf + ag + bh + ci =$$
$$= a(d + g) + b(e + h) + c(f + i)$$

cross product: Show

$$\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) = (\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3)$$

Using

$$\boldsymbol{v}_2 + \boldsymbol{v}_3 = [d+g, e+h, f+i]$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_2 = bf - ce + cd - af + ae - bd$$
$$\boldsymbol{v}_1 \times \boldsymbol{v}_3 = bi - ch + cg - ai + ah - bg$$

which gives

$$(\boldsymbol{v}_1 \times \boldsymbol{v}_2) + (\boldsymbol{v}_1 \times \boldsymbol{v}_3) = a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h)$$

and LHS:

$$\begin{aligned}\boldsymbol{v}_1 \times (\boldsymbol{v}_2 + \boldsymbol{v}_3) &= b(f+i) - c(e+h) + c(d+g) - a(f+i) + a(e+h) - b(d+g) = \\ &= bf + bi - ce - ch + cd + cg - af - ai + ae + ah - bd - bg = \\ &= a(e+h-f-i) + b(f+i-d-g) + c(d+g-e-h)\end{aligned}$$

□

Problem 1.2

Is the cross product associative?

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} \stackrel{?}{=} \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})$$

Solution:

No, for example set

$$\boldsymbol{A} = \hat{\boldsymbol{x}}$$
$$\boldsymbol{B} = \hat{\boldsymbol{x}}$$
$$\boldsymbol{C} = \hat{\boldsymbol{z}}$$

Then, the left hand side above becomes

$$(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = (\hat{\boldsymbol{x}} \times \hat{\boldsymbol{x}}) \times \hat{\boldsymbol{z}} = 0$$

while the right hand side is

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \hat{\boldsymbol{x}} \times (\hat{\boldsymbol{x}} \times \hat{\boldsymbol{z}}) = \hat{\boldsymbol{x}} \times -\hat{\boldsymbol{y}} = -\hat{\boldsymbol{z}}$$

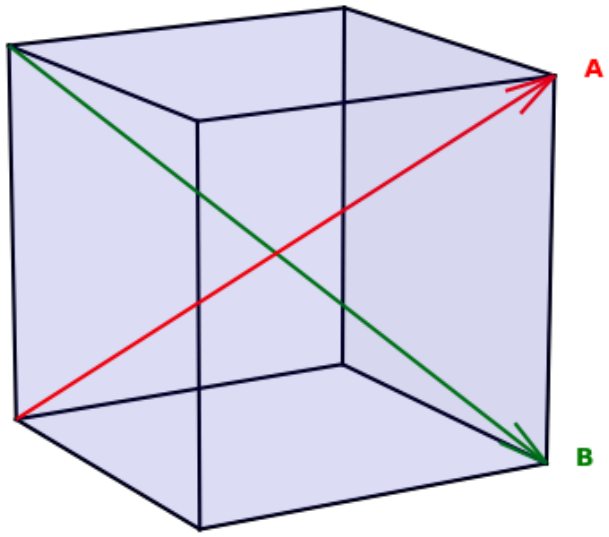
□

Problem 1.3

Find the angle between the body diagonals of a cube.

Solution:

The body diagonals $\boldsymbol{A} = [1, 1, 1]$ and $\boldsymbol{B} = [1, 1, -1]$



$|\boldsymbol{A}| = |\boldsymbol{B}| = \sqrt{3}$

$\boldsymbol{A} \cdot \boldsymbol{B} = 1 + 1 - 1 = 1$

$\boldsymbol{A} \cdot \boldsymbol{B} = |\boldsymbol{A}|\ |\boldsymbol{B}| \cos\theta = 3\cos\theta = 1 \rightarrow \theta = \arccos\left(\frac{1}{3}\right) \approx 70.53°$ □

Problem 1.4

Use the cross product to find the components of the unit vector $\hat{\boldsymbol{n}}$ perpendicular to the shaded plane in Fig.

Solution:

Two of the plane vectors are $\boldsymbol{v}_1 = [-1, 2, 0]$ and $\boldsymbol{v}_2 = [-1, 0, 3]$. $\boldsymbol{v}_1 \times \boldsymbol{v}_2 = n = [6, 3, 2]$ and $|\boldsymbol{n}| = 7$ gives $\hat{\boldsymbol{n}} = \frac{1}{7}[6, 3, 2]$ □

Problem 1.5

Prove the **BAC-CAB** rule,

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})$$

by writing out both sides in component form.

Solution:

Writing out the vectors in component form

$$\boldsymbol{A} = A_x\hat{\boldsymbol{x}} + A_y\hat{\boldsymbol{y}} + A_z\hat{\boldsymbol{z}}$$
$$\boldsymbol{B} = B_x\hat{\boldsymbol{x}} + B_y\hat{\boldsymbol{y}} + B_z\hat{\boldsymbol{z}}$$
$$\boldsymbol{C} = C_x\hat{\boldsymbol{x}} + C_y\hat{\boldsymbol{y}} + C_z\hat{\boldsymbol{z}}$$

starting with lhs

$$\boldsymbol{B} \times \boldsymbol{C} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \big(B_yC_z - B_zC_y\big)\hat{\boldsymbol{x}} + (B_zC_x - B_xC_z)\hat{\boldsymbol{y}} + \big(B_xC_y - B_yC_x\big)\hat{\boldsymbol{z}}$$

rewriting with a new matrix $\boldsymbol{D}$ where

$$D_x = B_yC_z - B_zC_y$$
$$D_y = B_zC_x - B_xC_z$$
$$D_z = B_xC_y - B_yC_x$$

gives

$$\boldsymbol{D} = \boldsymbol{B} \times \boldsymbol{C} = D_x\hat{\boldsymbol{x}} + D_y\hat{\boldsymbol{y}} + D_z\hat{\boldsymbol{z}}$$

Now

$$\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{A} \times \boldsymbol{D} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{vmatrix} =$$

$$= \big(A_yD_z - A_zD_y\big)\hat{\boldsymbol{x}} + (A_zD_x - A_xD_z)\hat{\boldsymbol{y}} + \big(A_xD_y - A_yD_x\big)\hat{\boldsymbol{z}}$$

RHS:

$$\boldsymbol{A} \cdot \boldsymbol{C} = A_xC_x + A_yC_y + A_zC_z = P$$
$$\boldsymbol{A} \cdot \boldsymbol{B} = A_xB_x + A_yB_y + A_zB_z = Q$$
$$\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) = B_xP\hat{\boldsymbol{x}} + B_yP\hat{\boldsymbol{y}} + B_zP\hat{\boldsymbol{z}}$$
$$\boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) = C_xQ\hat{\boldsymbol{x}} + C_yQ\hat{\boldsymbol{y}} + C_zQ\hat{\boldsymbol{z}}$$
$$\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) = (B_xP - C_xQ)\hat{\boldsymbol{x}} + \big(B_yP - C_yQ\big)\hat{\boldsymbol{y}} + (B_zP - C_zQ)\hat{\boldsymbol{z}}$$

Now final step is checking if

$$\textbf{1}\text{: } A_yD_z - A_zD_y = B_xP - C_xQ$$
$$\textbf{2}\text{: } A_zD_x - A_xD_z = B_yP - C_yQ$$
$$\textbf{3}\text{: } A_xD_y - A_yD_x = B_zP - C_zQ$$

For **1**:

$$\begin{aligned}\text{LHS: } & A_y(B_xC_y - B_yC_x) - A_z(B_zC_x - B_xC_z)\\ &= A_yB_xC_y - A_yB_yC_x - A_zB_zC_x + A_zB_xC_z\\ &= B_x(A_yC_y + A_zC_z) - C_x(A_yB_y + A_zB_z)\end{aligned}$$

$$\begin{aligned}\text{RHS: } & B_x(A_xC_x + A_yC_y + A_zC_z) - C_x(A_xB_x + A_yB_y + A_zB_z)\\ &= B_xA_xC_x + B_x(A_yC_y + A_zC_z) - C_xA_xB_x - C_x(A_yB_y + A_zB_z)\\ &= B_x(A_yC_y + A_zC_z) - C_x(A_yB_y + A_zB_z) = \text{LHS } \checkmark\end{aligned}$$

For **2**:

$$\begin{aligned}\text{LHS: } & A_z(B_yC_z - B_zC_y) - A_x(B_xC_y - B_yC_x)\\ &= A_zB_yC_z - A_zB_zC_y - A_xB_xC_y + A_xB_yC_x\\ &= B_y(A_zC_z + A_xC_x) - C_y(A_zB_z + A_xB_x)\end{aligned}$$

$$\begin{aligned}\text{RHS: } & B_y(A_xC_x + A_yC_y + A_zC_z) - C_y(A_xB_x + A_yB_y + A_zB_z)\\ &= B_yA_yC_y + B_y(A_xC_x + A_zC_z) - C_yA_yB_y - C_y(A_xB_x + A_zB_z)\\ &= B_y(A_xC_x + A_zC_z) - C_y(A_xB_x + A_zB_z) = \text{LHS } \checkmark\end{aligned}$$

For **3**:

$$\begin{aligned}\text{LHS: } & A_x(B_zC_x - B_xC_z) - A_y(B_yC_z - B_zC_y)\\ &= A_xB_zC_x - A_xB_xC_z - A_yB_yC_z + A_yB_zC_y\\ &= B_z(A_xC_x + A_yC_y) - C_z(A_xB_x + A_yB_y)\end{aligned}$$

$$\begin{aligned}\text{RHS: } & B_z(A_xC_x + A_yC_y + A_zC_z) - C_z(A_xB_x + A_yB_y + A_zB_z)\\ &= B_zA_zC_z + B_z(A_xC_x + A_yC_y) - C_zA_zB_z - C_z(A_xB_x + A_yB_y)\\ &= B_z(A_xC_x + A_yC_y) - C_z(A_xB_x + A_yB_y) = \text{LHS } \checkmark\end{aligned}$$

Since all three components are equal, the BAC-CAB rule is proven. □

Problem 1.6

Prove that

$$[\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})] + [\boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A})] + [\boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B})] = 0$$

Under what conditions does $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$?

Solution:

Using the BAC-CAB rule, rewrite the triple cross product as

$$(1)\ \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = \boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B})$$
$$(2)\ \boldsymbol{B} \times (\boldsymbol{C} \times \boldsymbol{A}) = \boldsymbol{C}(\boldsymbol{B} \cdot \boldsymbol{A}) - \boldsymbol{A}(\boldsymbol{B} \cdot \boldsymbol{C})$$
$$(3)\ \boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})$$

Using the commutative property of the dot product, $\boldsymbol{v}_1 \cdot \boldsymbol{v}_2 = \boldsymbol{v}_2 \cdot \boldsymbol{v}_1$, it is seen that:

- The $\boldsymbol{B}$-part in (1) cancels out the $\boldsymbol{B}$-part in (3)
- The $\boldsymbol{C}$-part in (1) cancels out the $\boldsymbol{C}$-part in (2)
- The $\boldsymbol{A}$-part in (2) cancels out the $\boldsymbol{A}$-part in (3)

The conditions when $\boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C}) = (\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C}$ is translated with the BAC-CAB rule into $\boldsymbol{B}(\boldsymbol{A} \cdot \boldsymbol{C}) - \boldsymbol{C}(\boldsymbol{A} \cdot \boldsymbol{B}) = -\boldsymbol{A}(\boldsymbol{C} \cdot \boldsymbol{B}) + \boldsymbol{B}(\boldsymbol{C} \cdot \boldsymbol{A})$ which gives that either $\boldsymbol{A} = \boldsymbol{C}$ or $\boldsymbol{B} \perp (\boldsymbol{A}$ and $\boldsymbol{C})$ □

Problem 1.7

Find the separation vector $\boldsymbol{\mathscr{r}}rs$ from the source point (2, 8, 7) to the field point (4, 6, 8). Determine its magnitude ($\mathscr{r}$), and construct the unit vector $\hat{\boldsymbol{\mathscr{r}}}$.

Solution:

The separation vector $\boldsymbol{\mathscr{r}}$ is the difference between the field point and the source point.

$$\boldsymbol{\mathscr{r}} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

The magnitude is

$$|\boldsymbol{\mathscr{r}}| = \mathscr{r} = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

and the unit vector is

$$\frac{\boldsymbol{\mathscr{r}}}{\mathscr{r}} = \hat{\boldsymbol{\mathscr{r}}} = \frac{1}{3}(2, -2, 1)$$

□

Problem 1.8

$$\begin{pmatrix} \overline{A}_y \\ \overline{A}_z \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix} \tag{1.}$$

$$\begin{pmatrix} \overline{A}_x \\ \overline{A}_y \\ \overline{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{xx} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \tag{2.}$$

(a) Prove that the two-dimensional rotation matrix (Equation 1) dot products. (That is, show that $\overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z = A_yB_y + A_zB_z$).

(b) What constraints must the elements (R_{ij}) of the three-dimensional rotation matrix (Equation 2) satisfy in order to preserve the length of $\mathbf{A}$ (for all vectors $\mathbf{A}$)?

Solution:

(a)

$$\begin{pmatrix} \overline{A}_y \\ \overline{A}_z \end{pmatrix} \cdot \begin{pmatrix} \overline{B}_y \\ \overline{B}_z \end{pmatrix} = \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix} \cdot \begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} B_y \\ B_z \end{pmatrix}$$

$$\downarrow$$

$$\overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z = \begin{pmatrix} \cos\phi A_y + \sin\phi A_z \\ -\sin\phi A_y + \cos\phi A_z \end{pmatrix} \cdot \begin{pmatrix} \cos\phi B_y + \sin\phi B_z \\ -\sin\phi B_y + \cos\phi B_z \end{pmatrix} =$$

$$= (\cos\phi A_y + \sin\phi A_z)(\cos\phi B_y + \sin\phi B_z) + (-\sin\phi A_y + \cos\phi A_z)(-\sin\phi B_y + \cos\phi B_z) =$$

$$= \cos^2\phi A_yB_y + \cos\phi\sin\phi A_yB_z + \sin\phi\cos\phi A_zB_y + \sin^2 A_zB_z +$$
$$+ \sin^2\phi A_yB_y - \sin\phi\cos\phi A_yB_z - \cos\phi\sin\phi A_zB_y + \cos^2\phi A_zB_z =$$

$$= (\cos^2\phi + \sin^2\phi)A_yB_y + (\cos^2\phi + \sin^2\phi)A_zB_z +$$
$$+\underbrace{(\cos\phi\sin\phi - \sin\phi\cos\phi)}_{=0}A_yB_z + \underbrace{(\sin\phi\cos\phi - \cos\phi\sin\phi)}_{=0}A_zB_y =$$

$$= A_yB_y + A_zB_z = \overline{A}_y\overline{B}_y + \overline{A}_z\overline{B}_z$$

(b) We want to preserve the length which means

$$\|\mathbf{R}\cdot\mathbf{A}\| = \|\mathbf{A}\|$$

Squaring both sides gives

$$(\mathbf{RA})^\mathsf{T}(\mathbf{RA}) = \mathbf{A}^\mathsf{T}\mathbf{A}$$
$$\downarrow$$
$$\mathbf{A}^\mathsf{T}\mathbf{R}^\mathsf{T}\mathbf{RA} = \mathbf{A}^\mathsf{T}\mathbf{A}$$

which gives

$$\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

□

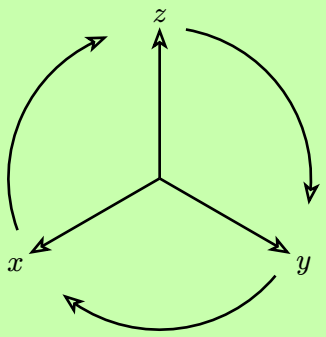
Problem 1.9

Find the transformation matrix R that describes a rotation by 120° about an axis from the origin through the point $(1, 1, 1)$. The rotation is clockwise as you look down the axis toward the origin.

Solution:

Looking down the axis from $(1, 1, 1)$ to the origin, the rotation (clockwise) takes the z-axis to the y-axis, y-axis to the x-axis, and the x-axis to the z-axis. The transformation matrix is then

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$



□

Problem 1.10

(a) How do the components of a vector transform under a translation of coordinates ($x = x, y = y - a, z = z$, Fig. 1.16a)?

(b) How do the components of a vector transform under an inversion of coordinates ($x = -x, y = -y, z = -z$, Fig. 1.16b)?

(c) How do the components of a cross product (Eq. 1.13) transform under inversion? [The cross-product of two vectors is properly called a pseudovector because of this "anomalous" behavior.] Is the cross product of two pseudovectors a vector, or a pseudovector? Name two pseudovector quantities in classical mechanics.

(d) How does the scalar triple product of three vectors transform under inversions? (Such an object is called a pseudoscalar.)

Solution:

(a) It will still point in the same direction, with the same components. Therefore no change.

(b) Simply inverse all the coordinates gives

$$\begin{gathered} v_x \to -\overline{v}_x \\ v_y \to -\overline{v}_y \\ v_z \to -\overline{v}_z \\ \downarrow \\ \mathbf{v} = -\overline{\mathbf{v}} \end{gathered}$$

(c)

$$\mathbf{A} \times \mathbf{B} \to (-\mathbf{A}) \times (-\mathbf{B}) = \mathbf{A} \times \mathbf{B}$$

So nothing happens when inversion on cross product.

(d)

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \to (-\mathbf{A}) \cdot ((-\mathbf{B}) \times (-\mathbf{C})) = (-\mathbf{A}) \cdot (\mathbf{B} \times \mathbf{C})$$

so a minus sign is introduced. □

Problem 1.11

Find the gradients of the following functions:

$$\begin{aligned} &\text{(a) } f(x, y, z) = x^2 + y^3 + z^4 \\ &\text{(b) } f(x, y, z) = x^2 y^3 z^4 \\ &\text{(c) } f(x, y, z) = e^x \sin(y) \ln(z) \end{aligned}$$

Solution:

The gradient of a function T is defined as

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial T}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial T}{\partial z}\hat{\boldsymbol{z}}$$

This results in

(a) $2x\hat{\boldsymbol{x}} + 3y^2\hat{\boldsymbol{y}} + 4z^3\hat{\boldsymbol{z}}$

(b) $2xy^3z^4\hat{\boldsymbol{x}} + 3x^2y^2z^4\hat{\boldsymbol{y}} + 4x^2y^3z^3\hat{\boldsymbol{z}}$

(c) $e^x\left(\sin(y)\ln(z)\hat{\boldsymbol{x}} + \cos(y)\ln(z)\hat{\boldsymbol{y}} + \dfrac{\sin(y)}{z}\hat{\boldsymbol{z}}\right)$

□

Problem 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in miles) north, x the distance east, of South Hadley.

(a) Where is the top of the hill located?

(b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and 1 mile east of South Hadley? In what direction is the slope steepest, at that point?

Solution:

The gradient of $h(x, y)$ is

$$\boldsymbol{\nabla} h = (20y - 60x - 180)\hat{\boldsymbol{x}} + (20x - 80y + 280)\hat{\boldsymbol{y}}$$

(a) The location of the top is calculated by setting the gradient to 0. This means

$$20y - 60x - 180 = 0$$
$$20x - 80y + 280 = 0$$

which has the solution $(x, y) = (-2, 3)$ miles.

(b) Putting the values given by (a) into the height function $h(x, y)$, the height of the hill is $h(-2, 3) = 720$ft.

(c) At the point $(x, y) = (1, 1)$, the steepness and direction is obtained by putting this point into the gradient. The slope is then $\sqrt{2} \cdot 220$ ft/mile in the direction $-\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}}$, which corresponds to northwest. □

Problem 1.13

Let $\boldsymbol{\imath}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z), and let $\imath$ be its length. Show that

$$\text{(a) } \boldsymbol{\nabla}(\imath^2) = 2\boldsymbol{\imath}$$

(b) $\boldsymbol{\nabla}\left(\frac{1}{\imath}\right) = -\hat{\boldsymbol{\imath}}/\imath^2$

(c) What is the general formula for $\boldsymbol{\nabla}(\imath^n)$?

Solution:

$$\boldsymbol{\imath} = (x - x', y - y', z - z')$$
$$\imath = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

(a)

$$\boldsymbol{\nabla}(\imath^2) = \boldsymbol{\nabla}\left((x - x')^2 + (y - y')^2 + (z - z')^2\right) =$$
$$= 2(x - x')\hat{\boldsymbol{x}} + 2(y - y')\hat{\boldsymbol{y}} + 2(z - z')\hat{\boldsymbol{z}} = 2\boldsymbol{\imath}$$

(b)

$$\boldsymbol{\nabla}\left(\frac{1}{\imath}\right) = \boldsymbol{\nabla}\left(\frac{1}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}\right) =$$
$$= -\frac{1}{2}\frac{2(x - x') + 2(y - y') + 2(z - z')}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}^3} = -\frac{\boldsymbol{\imath}}{\imath^3} = -\frac{\hat{\boldsymbol{\imath}}}{\imath^2}$$

(c)

$$\boldsymbol{\nabla}(\imath^n) = \boldsymbol{\nabla}\left(\left((x - x')^2 + (y - y')^2 + (z - z')^2\right)^{\frac{n}{2}}\right) =$$
$$= \frac{n}{2}(2(x - x')\hat{\boldsymbol{x}} + 2(y - y')\hat{\boldsymbol{y}} + 2(z - z')\hat{\boldsymbol{z}})\left[(x - x')^2 + (y - y')^2 + (z - z')^2\right]^{\frac{n-2}{2}} =$$
$$= n\imath^{n-2}\boldsymbol{\imath}$$

□

Problem 1.14

Problem 1.15

Calculate the divergence of the following vector functions:

(a) $\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\boldsymbol{x}} + (2xy + z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

Solution:

The divergence of a vector $\boldsymbol{v}$ is defined as

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

(a) $\boldsymbol{\nabla} \cdot \boldsymbol{v}_a = 2x - 2x = 0$

(b) $\boldsymbol{\nabla} \cdot \boldsymbol{v}_b = y + 2z + 3x$

(c) $\boldsymbol{\nabla} \cdot \boldsymbol{v}_c = 2x + 2y$

□

Problem 1.16

Sketch the vector function

$$\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2},$$

and compute its divergence.

Solution:

The divergence of the r-component in spherical coordinates is

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r)$$

Using this, the divergence is $\boldsymbol{\nabla} \cdot \boldsymbol{v} = 0$. From the sketch, clearly the divergence is not 0 at the origin. The problem arises at the point $\boldsymbol{r} = 0$ where $\boldsymbol{v} \to \infty$. At all other points $\boldsymbol{r} \neq 0$, the divergence is 0.

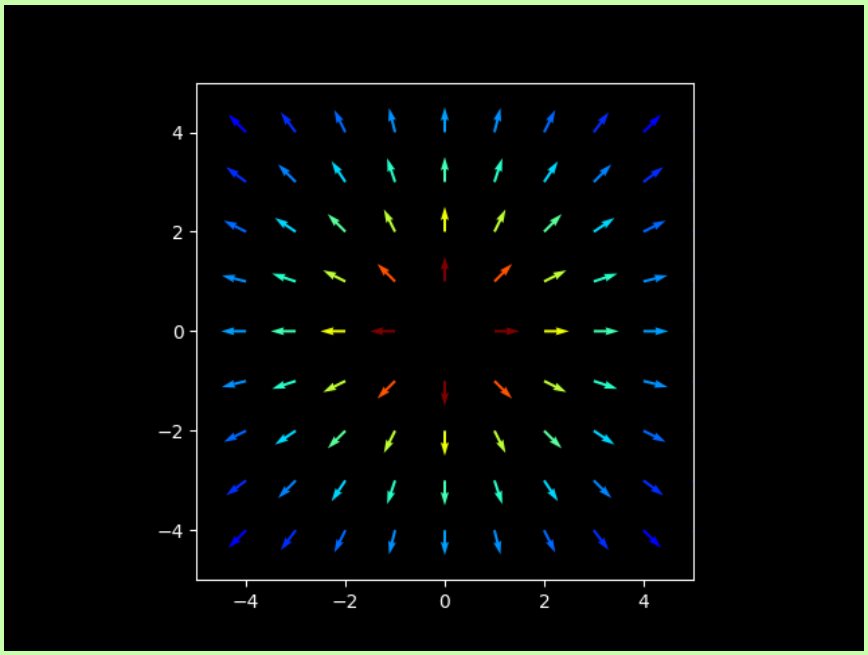


Figure 3: Sketch of the function $\boldsymbol{v} = \frac{\hat{\boldsymbol{r}}}{r^2}$

□

Problem 1.17

Problem 1.18

Calculate the curls of the vector functions

(a) $\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

(b) $\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$

(c) $\boldsymbol{v}_c = y^2\hat{\boldsymbol{x}} + (2xy + z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

Solution:

The definition of the curl of a vector function $\boldsymbol{v}$ is

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

This gives

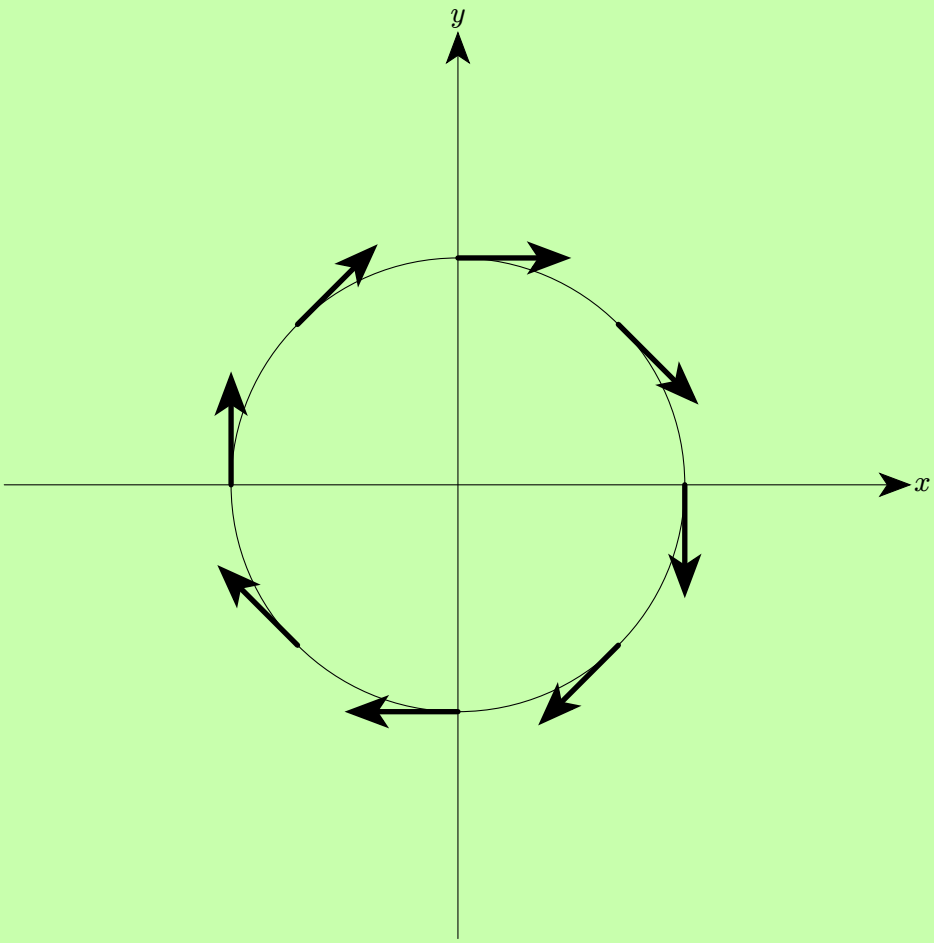
(a) $\boldsymbol{\nabla} \times \boldsymbol{v}_a = -6xz\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{y}} + 3z^2\hat{\boldsymbol{z}}$
(b) $\boldsymbol{\nabla} \times \boldsymbol{v}_b = -2y\hat{\boldsymbol{x}} + -3z\hat{\boldsymbol{y}} + -x\hat{\boldsymbol{z}}$
(c) $\boldsymbol{\nabla} \times \boldsymbol{v}_c = 0$

□

Problem 1.19

Draw a circle in the x-y plane. At a few representative points draw the unit vector $\hat{\boldsymbol{v}}$ tangent to the circle, pointing in the clockwise direction. By comparing adjacent vectors, determine the signs of $\frac{\partial v_x}{\partial y}$ and $\frac{\partial v_y}{\partial x}$. According to The definition of the curl, what is the direction of $\boldsymbol{\nabla} \times \hat{\boldsymbol{v}}$?

Solution:



Going from one arrow to the next, it is seen that

$$\frac{\partial v_x}{\partial y} = \text{positive}$$
$$\frac{\partial v_y}{\partial x} = \text{negative}$$

For example, consider the arrow at 12 o'clock with $x = 0, y = 1, v_x = 1, v_y = 0$, and the arrow after it at around 2 o'clock with $x = 0.7, y = 0.7, v_x = 0.7, v_y = -0.7$. This confirms the values of $\frac{\partial v_x}{\partial y}$ and $\frac{\partial v_y}{\partial x}$ above.

□

Problem 1.20

Construct a vector function that has a zero divergence and zero curl everywhere.

Solution:

The vector function $\boldsymbol{R}$ should satisfy

$$\boldsymbol{\nabla} \cdot \boldsymbol{R} = 0$$
$$\boldsymbol{\nabla} \times \boldsymbol{R} = 0$$

Using the definitions of the divergence and curl, the conditions are

$$\boldsymbol{\nabla} \cdot \boldsymbol{R} = \frac{\partial R_x}{\partial x} + \frac{\partial R_y}{\partial y} + \frac{\partial R_z}{\partial z} = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{R} = \left(\frac{\partial R_z}{\partial y} - \frac{\partial R_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial R_x}{\partial z} - \frac{\partial R_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial R_y}{\partial x} - \frac{\partial R_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

Want to fulfill:

$$\frac{\partial R_z}{\partial y} = \frac{\partial R_y}{\partial z}$$
$$\frac{\partial R_x}{\partial z} = \frac{\partial R_z}{\partial x}$$
$$\frac{\partial R_y}{\partial x} = \frac{\partial R_x}{\partial y}$$

An example is $\boldsymbol{R} = yz\hat{\boldsymbol{x}} + xz\hat{\boldsymbol{y}} + xy\hat{\boldsymbol{z}}$

□

Problem 1.21

Prove the product rules

$$\text{(i)}\quad \boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla}g + g\boldsymbol{\nabla}f$$
$$\text{(iv)}\quad \boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})$$
$$\text{(v)}\quad \boldsymbol{\nabla}\times(f\boldsymbol{A}) = f(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\times(\boldsymbol{\nabla}f)$$

Solution:

(i)

$$\boldsymbol{\nabla}(fg) = \frac{\partial}{\partial x}(fg)\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}(fg)\hat{\boldsymbol{y}} + \frac{\partial}{\partial z}(fg)\hat{\boldsymbol{z}}$$
$$\downarrow$$
$$\frac{\partial}{\partial x}(fg)\hat{\boldsymbol{x}} = \left[g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}\right]\hat{\boldsymbol{x}}$$
$$\frac{\partial}{\partial y}(fg)\hat{\boldsymbol{y}} = \left[g\frac{\partial f}{\partial y} + f\frac{\partial g}{\partial y}\right]\hat{\boldsymbol{y}}$$
$$\frac{\partial}{\partial z}(fg)\hat{\boldsymbol{z}} = \left[g\frac{\partial f}{\partial z} + f\frac{\partial g}{\partial z}\right]\hat{\boldsymbol{z}}$$
$$\downarrow$$
$$\boxed{\boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla}g + g\boldsymbol{\nabla}f}$$

(iv) Expand the cross product $\mathbf{A}\times\mathbf{B}$ and the curls $\boldsymbol{\nabla}\times\mathbf{A}, \boldsymbol{\nabla}\times\mathbf{B}$

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_yB_z - A_zB_y)\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{z}}$$

$$\boldsymbol{\nabla}\times\mathbf{A} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

$$\boldsymbol{\nabla}\times\mathbf{B} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix} = \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

Now the divergence of $\mathbf{A}\times\mathbf{B}$ is

$$\boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B})=\frac{\partial}{\partial x}(A_yB_z-A_zB_y)+\frac{\partial}{\partial y}(A_zB_x-A_xB_z)+\frac{\partial}{\partial z}(A_xB_y-A_yB_x)=$$

$$\begin{aligned}&=B_z\frac{\partial A_y}{\partial x}+A_y\frac{\partial B_z}{\partial x}-B_y\frac{\partial A_z}{\partial x}-A_z\frac{\partial B_y}{\partial x}+\\&+B_x\frac{\partial A_z}{\partial y}+A_z\frac{\partial B_x}{\partial y}-B_z\frac{\partial A_x}{\partial y}-A_x\frac{\partial B_z}{\partial y}+\\&+B_y\frac{\partial A_x}{\partial y}+A_x\frac{\partial B_y}{\partial y}-B_x\frac{\partial A_y}{\partial y}-A_y\frac{\partial B_x}{\partial y}\end{aligned}$$

while

$$\mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})=B_x\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+B_y\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+B_z\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)$$

$$\mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})=A_x\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)+A_y\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)+A_z\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)$$

Taking rule (iv) and matching the terms containing $\frac{\partial}{\partial x},\frac{\partial}{\partial y},\frac{\partial}{\partial z}$:

$$\boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B})=\mathbf{B}\cdot(\boldsymbol{\nabla}\times\mathbf{A})-\mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})$$

$$\downarrow$$

$$B_z\frac{\partial A_y}{\partial x}+A_y\frac{\partial B_z}{\partial x}-B_y\frac{\partial A_z}{\partial x}-A_z\frac{\partial B_y}{\partial x}=-B_y\frac{\partial A_z}{\partial x}+B_z\frac{\partial A_y}{\partial x}-\left(-A_y\frac{\partial B_z}{\partial x}+A_z\frac{\partial B_y}{\partial x}\right)\checkmark$$

$$B_x\frac{\partial A_z}{\partial y}+A_z\frac{\partial B_x}{\partial y}-B_z\frac{\partial A_x}{\partial y}-A_x\frac{\partial B_z}{\partial y}=B_x\frac{\partial A_z}{\partial y}-B_z\frac{\partial A_x}{\partial y}-\left(A_x\frac{\partial B_z}{\partial y}-A_z\frac{\partial B_x}{\partial y}\right)\checkmark$$

$$B_y\frac{\partial A_x}{\partial y}+A_x\frac{\partial B_y}{\partial y}-B_x\frac{\partial A_y}{\partial y}-A_y\frac{\partial B_x}{\partial y}=-B_x\frac{\partial A_y}{\partial z}+B_y\frac{\partial A_x}{\partial z}-\left(-A_x\frac{\partial B_y}{\partial z}+A_y\frac{\partial B_x}{\partial z}\right)\checkmark$$

(v) Expanding the first term $\boldsymbol{\nabla}\times(f\mathbf{A})$:

$$\boldsymbol{\nabla}\times(f\mathbf{A})=\left(\frac{\partial fA_z}{\partial y}-\frac{\partial fA_y}{\partial z}\right)\hat{\boldsymbol{x}}+\left(\frac{\partial fA_x}{\partial z}-\frac{\partial fA_z}{\partial x}\right)\hat{\boldsymbol{y}}+\left(\frac{\partial fA_y}{\partial x}-\frac{\partial fA_x}{\partial y}\right)\hat{\boldsymbol{z}}=$$

$$\begin{aligned}&=\left(f\frac{\partial A_z}{\partial y}+A_z\frac{\partial f}{\partial y}-f\frac{\partial A_y}{\partial z}-A_y\frac{\partial f}{\partial z}\right)\hat{\boldsymbol{x}}+\\&+\left(f\frac{\partial A_x}{\partial z}+A_x\frac{\partial f}{\partial z}-f\frac{\partial A_z}{\partial x}-A_z\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{y}}+\\&+\left(f\frac{\partial A_y}{\partial x}+A_y\frac{\partial f}{\partial x}-f\frac{\partial A_x}{\partial y}-A_x\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{z}}\end{aligned}$$

Expanding the second term $f(\boldsymbol{\nabla} \times \mathbf{A})$:

$$f(\boldsymbol{\nabla} \times \mathbf{A}) = f\left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}}\right]$$

Expanding the third term $\mathbf{A} \times (\boldsymbol{\nabla} f)$

$$\mathbf{A} \times (\boldsymbol{\nabla} f) = \mathbf{A} \times \left(\frac{\partial f}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial f}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{z}}\right) = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} =$$

$$= \left(A_y\frac{\partial f}{\partial z} - A_z\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{x}} + \left(A_z\frac{\partial f}{\partial x} - A_x\frac{\partial f}{\partial z}\right)\hat{\boldsymbol{y}} + \left(A_x\frac{\partial f}{\partial y} - A_y\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{z}}$$

Now match each component of both sides $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$:

$$\boldsymbol{\nabla} \times (f\boldsymbol{A}) = f(\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \times (\boldsymbol{\nabla} f)$$

$$\downarrow$$

$$\left(f\frac{\partial A_z}{\partial y} + A_z\frac{\partial f}{\partial y} - f\frac{\partial A_y}{\partial z} - A_y\frac{\partial f}{\partial z}\right)\hat{\boldsymbol{x}} = f\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{x}} - \left(A_y\frac{\partial f}{\partial z} - A_z\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{x}} \checkmark$$

$$\left(f\frac{\partial A_x}{\partial z} + A_x\frac{\partial f}{\partial z} - f\frac{\partial A_z}{\partial x} - A_y\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{y}} = f\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\boldsymbol{y}} - \left(A_z\frac{\partial f}{\partial x} - A_x\frac{\partial f}{\partial z}\right)\hat{\boldsymbol{x}} \checkmark$$

$$\left(f\frac{\partial A_y}{\partial x} + A_y\frac{\partial f}{\partial x} - f\frac{\partial A_x}{\partial y} - A_x\frac{\partial f}{\partial y}\right)\hat{\boldsymbol{z}} = f\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\boldsymbol{z}} - \left(A_x\frac{\partial f}{\partial y} - A_y\frac{\partial f}{\partial x}\right)\hat{\boldsymbol{z}} \checkmark$$

□

Problem 1.22

$$\hat{\boldsymbol{r}} = \frac{\mathbf{r}}{r} = \frac{x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2 + y^2 + z^2}} \tag{3.}$$

(a) If $\mathbf{A}$ and $\mathbf{B}$ are two vector functions, what does the expression $(\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B}$ mean? (That is, what are its $x, y,$ and z components, in terms of the Cartesian components of $\mathbf{A}, \mathbf{B},$ and $\boldsymbol{\nabla}$?)

(b) Compute $(\hat{\boldsymbol{r}} \cdot \boldsymbol{\nabla})\hat{\boldsymbol{r}}$, where $\hat{\boldsymbol{r}}$ is the unit vector defined in Equation 3

(c) For the functions in Prob. 1.15,

$$\boldsymbol{v}_a = x^2\hat{\boldsymbol{x}} + 3xz^2\hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$$
$$\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$$

evaluate $(\mathbf{v}_a \cdot \boldsymbol{\nabla})\mathbf{v}_b$

Solution:

The definition of the gradient, $\boldsymbol{\nabla}$, is

$$\boldsymbol{\nabla} = \frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial}{\partial z}\hat{\boldsymbol{z}}$$

This gives the dot product between $\mathbf{A}$ and $\boldsymbol{\nabla}$

$$\mathbf{A}\cdot\boldsymbol{\nabla} = \left(A_x\hat{\boldsymbol{x}} + A_y\hat{\boldsymbol{y}} + A_z\hat{\boldsymbol{z}}\right)\cdot\left(\frac{\partial}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial}{\partial z}\hat{\boldsymbol{z}}\right) =$$
$$= A_x\frac{\partial}{\partial x} + A_y\frac{\partial}{\partial y} + A_z\frac{\partial}{\partial z}$$

Now multiply by $\mathbf{B}$ on the right side

$$(\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B} = \left(A_x\frac{\partial}{\partial x} + A_y\frac{\partial}{\partial y} + A_z\frac{\partial}{\partial z}\right)\mathbf{B} =$$
$$= A_x\frac{\partial\mathbf{B}}{\partial x} + A_y\frac{\partial\mathbf{B}}{\partial y} + A_z\frac{\partial\mathbf{B}}{\partial z}$$

What this means then, is the derivative of $\mathbf{B}$ in the direction of $\mathbf{A}$. Physically, if you think about yourself standing in a vector field $\mathbf{B}$, and taking a small step in the direction of $\mathbf{A}$, the change you experience in $\mathbf{B}$ is proportional to $(\mathbf{A}\cdot\boldsymbol{\nabla})\mathbf{B}$.

(b) Using the result from (a), we get

$$(\hat{\boldsymbol{r}}\cdot\boldsymbol{\nabla})\hat{\boldsymbol{r}} = r_x\frac{\partial\hat{\boldsymbol{r}}}{\partial x} + r_y\frac{\partial\hat{\boldsymbol{r}}}{\partial y} + r_z\frac{\partial\hat{\boldsymbol{r}}}{\partial z}$$

Starting with the x-component:

$$r_x\frac{\partial\hat{\boldsymbol{r}}}{\partial x} = \frac{x}{\sqrt{x^2+y^2+z^2}}\frac{\partial}{\partial x}\left(\frac{x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2+y^2+z^2}}\right) =$$
$$= \frac{x}{\sqrt{x^2+y^2+z^2}}\frac{\partial}{\partial x}\left(\frac{x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2+y^2+z^2}}\right)$$

Using the quotient derivative rule on the x-component and regular derivative on the y- and z-components

$$\frac{\partial}{\partial x}\left(\frac{x}{\sqrt{x^2+y^2+z^2}}\right)=\left(\frac{\sqrt{x^2+y^2+z^2}-x\frac{2x}{2\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}\right)=$$

$$=\frac{1}{\sqrt{x^2+y^2+z^2}}-\frac{x^2}{\sqrt{x^2+y^2+z^2}^3}=\frac{y^2+z^2}{\sqrt{x^2+y^2+z^2}^3}$$

$$\frac{\partial}{\partial x}\left(\frac{y}{\sqrt{x^2+y^2+z^2}}\right)=-\frac{xy}{\sqrt{x^2+y^2+z^2}^3}$$

$$\frac{\partial}{\partial x}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)=-\frac{xz}{\sqrt{x^2+y^2+z^2}^3}$$

The final form of the x-component is then

$$r_x\frac{\partial}{\partial x}\hat{\boldsymbol{r}}=\frac{x}{\sqrt{x^2+y^2+z^2}}\frac{y^2+z^2-xy-xz}{\sqrt{x^2+y^2+z^2}^3}$$

Doing the same calculations on the y-component gives

$$r_y\frac{\partial}{\partial y}\hat{\boldsymbol{r}}=\frac{y}{\sqrt{x^2+y^2+z^2}}\frac{x^2+z^2-yx-yz}{\sqrt{x^2+y^2+z^2}^3}$$

and on the z-component

$$r_z\frac{\partial}{\partial z}\hat{\boldsymbol{r}}=\frac{z}{\sqrt{x^2+y^2+z^2}}\frac{x^2+y^2-zx-zy}{\sqrt{x^2+y^2+z^2}^3}$$

Adding them all up gives

$$\frac{(xy^2+xz^2-x^2y-x^2z)+(yx^2+yz^2-y^2x-y^2z)+(zx^2+zy^2-z^2x-z^2y)}{(x^2+y^2+z^2)^2}=$$

$$=\frac{(xy^2-y^2x)+(xz^2-z^2x)+(yx^2-x^2y)+(yz^2-z^2y)+(zx^2-x^2z)+(zy^2-y^2z)}{(x^2+y^2+z^2)^2}=0$$

$$\downarrow$$

$$(\hat{\boldsymbol{r}}\cdot\boldsymbol{\nabla})\hat{\boldsymbol{r}}=0$$

(c)

$$\boldsymbol{v}_a=x^2\hat{\boldsymbol{x}}+3xz^2\hat{\boldsymbol{y}}-2xz\hat{\boldsymbol{z}}$$
$$\boldsymbol{v}_b=xy\hat{\boldsymbol{x}}+2yz\hat{\boldsymbol{y}}+3xz\hat{\boldsymbol{z}}$$

$$(\mathbf{v}_a \cdot \boldsymbol{\nabla})\mathbf{v}_b = v_{ax}\frac{\partial \mathbf{v}_b}{\partial x} + v_{ay}\frac{\partial \mathbf{v}_b}{\partial y} + v_{az}\frac{\partial \mathbf{v}_b}{\partial z} =$$
$$= x^2(y\hat{\boldsymbol{x}} + 3z\hat{\boldsymbol{z}}) + 3xz^2(x\hat{\boldsymbol{x}} + 2z\hat{\boldsymbol{y}}) - 2xz(2y\hat{\boldsymbol{y}} + 3x\hat{\boldsymbol{z}}) =$$
$$= x^2(y + 3z^2)\hat{\boldsymbol{x}} + 2xz(3z^2 - 2y)\hat{\boldsymbol{y}} - (3x^2 z)\hat{\boldsymbol{z}}$$

□

Problem 1.23

Problem 1.24

Problem 1.25

(ii) $\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B}) = \boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B}) + \boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A}) + (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}$

(iv) $\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A}) - \boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})$

(vi) $\boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B}) = (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A} - (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B} + \boldsymbol{A}(\boldsymbol{\nabla}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{\nabla}\cdot\boldsymbol{A})$

(a) Check product rule (iv) for the functions

$$\boldsymbol{A} = x\hat{\boldsymbol{x}} + 2y\hat{\boldsymbol{y}} + 3z\hat{\boldsymbol{z}}\ ,\quad \boldsymbol{B} = 3y\hat{\boldsymbol{x}} - 2x\hat{\boldsymbol{y}}$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

Solution:

(a) Start with LHS:

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}\times\boldsymbol{B}) = \boldsymbol{\nabla}\cdot\big((A_yB_z - A_zB_y)\hat{\boldsymbol{x}} + (A_zB_x - A_xB_z)\hat{\boldsymbol{y}} + (A_xB_y - A_yB_x)\hat{\boldsymbol{z}}\big) =$$
$$= \frac{\partial}{\partial x}(A_yB_z - A_zB_y) + \frac{\partial}{\partial y}(A_zB_x - A_xB_z) + \frac{\partial}{\partial z}(A_xB_y - A_yB_x) =$$
$$= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z = \boxed{15z}$$

RHS:

$$\underbrace{\boldsymbol{B}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})}_{(1)}-\underbrace{\boldsymbol{A}\cdot(\boldsymbol{\nabla}\times\boldsymbol{B})}_{(2)}=$$

$$(1)\ \boldsymbol{B}\cdot[\boldsymbol{\nabla}\times(A)]=$$

$$=B_x\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)+B_y\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)+B_z\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)=0$$

$$(2)\ \boldsymbol{A}\cdot[\boldsymbol{\nabla}\times(B)]=$$

$$=A_x\left(\frac{\partial B_z}{\partial y}-\frac{\partial B_y}{\partial z}\right)+A_y\left(\frac{\partial B_x}{\partial z}-\frac{\partial B_z}{\partial x}\right)+A_z\left(\frac{\partial B_y}{\partial x}-\frac{\partial B_x}{\partial y}\right)=$$

$$=3z(-5)=-15z$$

$$(1)-(2)=0-(-15z)=\boxed{15z}$$

(b) LHS:

$$\boldsymbol{\nabla}(\boldsymbol{A}\cdot\boldsymbol{B})=\boldsymbol{\nabla}\big(A_xB_x+A_yB_y+A_zB_z\big)=\boldsymbol{\nabla}(3xy-4xy)=$$

$$=\frac{\partial(-xy)}{\partial x}\hat{\boldsymbol{x}}+\frac{\partial(-xy)}{\partial y}\hat{\boldsymbol{y}}=\boxed{-y\hat{\boldsymbol{x}}-x\hat{\boldsymbol{y}}}$$

RHS:

$$\underbrace{\boldsymbol{A}\times(\boldsymbol{\nabla}\times\boldsymbol{B})}_{(1)}+\underbrace{\boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{A})}_{(2)}+\underbrace{(\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}}_{(3)}+\underbrace{(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}}_{(4)}$$

$$(1)\ \boldsymbol{A}\times[\boldsymbol{\nabla}\times(B)]=$$

$$=(x\hat{\boldsymbol{x}}+2y\hat{\boldsymbol{y}}+3z\hat{\boldsymbol{z}})\times(-5\hat{\boldsymbol{z}})=-10y\hat{\boldsymbol{x}}+5x\hat{\boldsymbol{y}}$$

$$(2)\ \boldsymbol{B}\times[\boldsymbol{\nabla}\times(A)]=0$$

$$(3)\ (\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}=$$

$$=\left(A_x\frac{\partial B_x}{\partial x}+A_y\frac{\partial B_x}{\partial y}+A_z\frac{\partial B_x}{\partial z}\right)\hat{\boldsymbol{x}}+\left(A_x\frac{\partial B_y}{\partial x}+A_y\frac{\partial B_y}{\partial y}+A_z\frac{\partial B_y}{\partial z}\right)\hat{\boldsymbol{y}}+\left(A_x\frac{\partial B_z}{\partial x}+A_y\frac{\partial B_z}{\partial y}+A_z\frac{\partial B_z}{\partial z}\right)\hat{\boldsymbol{z}}=$$

$$=6y\hat{\boldsymbol{x}}-2x\hat{\boldsymbol{y}}$$

$$(4)\ (\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}=$$

$$=\left(B_x\frac{\partial A_x}{\partial x}+B_y\frac{\partial A_x}{\partial y}+B_z\frac{\partial A_x}{\partial z}\right)\hat{\boldsymbol{x}}+\left(B_x\frac{\partial A_y}{\partial x}+B_y\frac{\partial A_y}{\partial y}+B_z\frac{\partial A_y}{\partial z}\right)\hat{\boldsymbol{y}}+\left(B_x\frac{\partial A_z}{\partial x}+B_y\frac{\partial A_z}{\partial y}+B_z\frac{\partial A_z}{\partial z}\right)\hat{\boldsymbol{z}}=$$

$$=3y\hat{\boldsymbol{x}}-4x\hat{\boldsymbol{y}}$$

$$(1)+(2)+(3)+(4)=(-10y+6y+3y)\hat{\boldsymbol{x}}+(5x-2x-4x)\hat{\boldsymbol{y}}=\boxed{-y\hat{\boldsymbol{x}}-x\hat{\boldsymbol{y}}}$$

(c) LHS:

$$\begin{aligned}\boldsymbol{\nabla}\times(\boldsymbol{A}\times\boldsymbol{B}) &= \boldsymbol{\nabla}\times\left[(A_yB_z-A_zB_y)\hat{\boldsymbol{x}}+(A_zB_x-A_xB_z)\hat{\boldsymbol{y}}+(A_xB_y-A_yB_x)\hat{\boldsymbol{z}}\right]= \\ &= \boldsymbol{\nabla}\times\left[6xz\hat{\boldsymbol{x}}+9yz\hat{\boldsymbol{y}}-(2x^2+6y^2)\hat{\boldsymbol{z}}\right]=\boldsymbol{\nabla}\times\boldsymbol{C}= \\ &= \boldsymbol{\nabla}\times(C)= \\ &= (-12y-9y)\hat{\boldsymbol{x}}+(6x+4x)\hat{\boldsymbol{y}}= \boxed{-21y\hat{\boldsymbol{x}}+10x\hat{\boldsymbol{y}}}\end{aligned}$$

RHS: From (b) it was calculated that $(\boldsymbol{A}\cdot\boldsymbol{\nabla})\boldsymbol{B}=6y\hat{\boldsymbol{x}}-2x\hat{\boldsymbol{y}}$ and $(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{A}=3y\hat{\boldsymbol{x}}-4x\hat{\boldsymbol{y}}$. The remaining parts of (vi) RHS is

$$\boldsymbol{A}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{B})}_{=0}-\boldsymbol{B}\underbrace{(\boldsymbol{\nabla}\cdot\boldsymbol{A})}_{=6}=-6\boldsymbol{B}=-18y\hat{\boldsymbol{x}}+12x\hat{\boldsymbol{y}}$$

Summing it all up according to (vi), the result is

$$(3y-6y-18y)\hat{\boldsymbol{x}}+(-4x+2x+12x)\hat{\boldsymbol{y}}=\boxed{-21y\hat{\boldsymbol{x}}+10x\hat{\boldsymbol{y}}}$$

□

Problem 1.26

Calculate the Laplacian of the following functions:

(a) $T_a=x^2+2xy+3z+4$

(b) $T_b=\sin x\sin y\sin z$

(c) $T_c=e^{-5x}\sin 4y\cos 3z$

(d) $\boldsymbol{v}=x^2\hat{\boldsymbol{x}}+3xz^2\hat{\boldsymbol{y}}-2xz\hat{\boldsymbol{z}}$

Solution:

The Laplacian of a function T is defined as

$$\boldsymbol{\nabla}^2T=\frac{\partial T}{\partial x}+\frac{\partial T}{\partial y}+\frac{\partial T}{\partial z}$$

and the Laplacian of a vector $\boldsymbol{v}$ is defined as

$$\boldsymbol{\nabla}^2\boldsymbol{v}=(\boldsymbol{\nabla}^2v_x)\hat{\boldsymbol{x}}+(\boldsymbol{\nabla}^2v_y)\hat{\boldsymbol{y}}+(\boldsymbol{\nabla}^2v_z)\hat{\boldsymbol{z}}$$

Using this, the result is

(a) $\boldsymbol{\nabla}^2T_a=2$

(b) $\boldsymbol{\nabla}^2 T_b = -3 \sin x \sin y \sin z$

(c) $\boldsymbol{\nabla}^2 T_c = e^{-5x} \sin 4y \cos 3z (25 - 16 - 9) = 0$

(d) $\boldsymbol{\nabla}^2 \boldsymbol{v} = 2\hat{\boldsymbol{x}} + 6x\hat{\boldsymbol{y}}$

□

Problem 1.27

Prove that the divergence of a curl is always zero.

Solution:

For a vector function $\boldsymbol{v}$, show

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{v}) = 0$$

Using the definition of the divergence and curl:

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\boldsymbol{z}}$$

the result is

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{v}) = \frac{\partial}{\partial x} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) =$$

$$= \frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_y}{\partial x \partial z} + \frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_z}{\partial y \partial x} + \frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_x}{\partial z \partial y} =$$

$$= \underbrace{\frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_z}{\partial y \partial x}}_{=0} + \underbrace{\frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_x}{\partial z \partial y}}_{=0} + \underbrace{\frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_y}{\partial x \partial z}}_{=0}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.28

Prove that the curl of a gradient is always zero.

Solution:

For a function T, show

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} T) = 0$$

Using the definitions of the gradient and curl

$$\boldsymbol{\nabla} T = \frac{\partial T}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial T}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial T}{\partial z}\hat{\boldsymbol{z}}$$

$$\boldsymbol{\nabla} \times \boldsymbol{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{\boldsymbol{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\hat{\boldsymbol{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{\boldsymbol{z}}$$

results in

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} T) = \underbrace{\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y}\right)}_{=0}\hat{\boldsymbol{x}} + \underbrace{\left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z}\right)}_{=0}\hat{\boldsymbol{y}} + \underbrace{\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x}\right)}_{=0}\hat{\boldsymbol{z}}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.29

Calculate the line integral of the function $\boldsymbol{v} = x^2\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + y^2\hat{\boldsymbol{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

(a) $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$

(b) $(0, 0, 0) \to (0, 0, 1) \to (0, 1, 1) \to (1, 1, 1)$

(c) The direct straight line

(d) What is the line integral around the closed loop that goes out along path (a) and back along path (b)?

Solution:

The line integral has the from

$$\int_a^b \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l}$$

where $d\boldsymbol{l} = dx\hat{\boldsymbol{x}} + dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}}$

(a)

$(0,0,0) \to (1,0,0)$ has $d\boldsymbol{l} = dx\hat{\boldsymbol{x}}$ and x: $0 \to 1, y = 0, z = 0$.

$(1,0,0) \to (1,1,0)$ has $d\boldsymbol{l} = dy\hat{\boldsymbol{y}}$ and y: $0 \to 1, x = 1, z = 0$.

$(1,1,0) \to (1,1,1)$ has $d\boldsymbol{l} = dz\hat{\boldsymbol{z}}$ and z: $0 \to 1, x = 1, y = 1$.

$$\int_0^1 x^2 \,\mathrm{d}x + \int_0^1 2yz \,\mathrm{d}y + \int_0^1 y^2 \,\mathrm{d}z = \int_0^1 x^2 \,\mathrm{d}x + \int_0^1 1 \,\mathrm{d}z = \frac{1}{3} + 1 = \boxed{\frac{4}{3}}$$

(b)

$(0,0,0) \to (0,0,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\boldsymbol{z}}$ and z: $0 \to 1, x = 0, y = 0$.

$(0,0,1) \to (0,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\boldsymbol{y}}$ and y: $0 \to 1, x = 0, z = 1$.

$(0,1,1) \to (1,1,1)$ has $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\boldsymbol{x}}$ and x: $0 \to 1, y = 1, z = 1$.

$$\int_0^1 y^2 \,\mathrm{d}z + \int_0^1 2yz \,\mathrm{d}y + \int_0^1 x^2 \,\mathrm{d}x = \int_0^1 2y \,\mathrm{d}y + \int_0^1 x^2 \,\mathrm{d}x = 1 + \frac{1}{3} = \boxed{\frac{4}{3}}$$

(c) The direct straight line is obtained by setting $x = y = z = t$ and letting t: $0 \to 1$. Also, $\mathrm{d}\boldsymbol{l} = \mathrm{d}t\hat{\boldsymbol{t}}$, giving $\boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l} = 4t^2 \,\mathrm{d}t$.

$$\int_0^1 4t^2 \,\mathrm{d}t = \boxed{\frac{4}{3}}$$

(d) Since the result of the line integral of $\boldsymbol{v}$ is independent of the path, going from the origin to the point $(1,1,1)$ via the path in (a), and then from the point $(1,1,1)$ to the origin via the path in (b) gives

$$\underbrace{\int_0^1 x^2 \,\mathrm{d}x + \int_0^1 1 \,\mathrm{d}z}_{(a)} - \left[\underbrace{\int_1^0 2y \,\mathrm{d}y + \int_1^0 x^2 \,\mathrm{d}x}_{(b) \text{ in reverse}}\right] = \boxed{\frac{4}{3} - \frac{4}{3} = 0}$$

□

Problem 1.30

Calculate the surface integral of the function

$$\boldsymbol{v} = 2xz\hat{\boldsymbol{x}} + (x+2)\hat{\boldsymbol{y}} + y(z^2-3)\hat{\boldsymbol{z}}$$

over the bottom of the box. For consistency, let "upward" be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the closed surface of the box (including the bottom)?

Solution:

For the bottom of the box, x, y: $0 \to 2$, $z = 0$ and $\mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}y\hat{\boldsymbol{z}}$

$$\int_0^2 \int_0^2 -3y\,\mathrm{d}x\,\mathrm{d}y = \int_0^2 \left(\int_0^2 -3y\,\mathrm{d}y \right) \mathrm{d}x = \int_0^2 -6\,\mathrm{d}x = \boxed{-12}$$

The total outward flux from all faces exluding the bottom was 20 (result from Ex. 1.7). The resulting total outward flux is then $20 - (-12) = 32$ (since the result from the bottom surface was calculated in the "wrong" direction). The surface integral does not depend only on the boundary line, since the bottom surface shares the boundary line with the combined 5 other surfaces, and the calculations gave different values (20 for the 5 surfaces, -12 for the bottom surface).

□

Problem 1.31

Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

Solution:

The boundaries are

$x : 0 \to 1 - y - z$

$y : 0 \to 1 - z$

$z : 0 \to 1$

$$\int_0^1 z^2 \left(\int_0^{1-z} \left(\int_0^{1-y-z} \mathrm{d}x \right) \mathrm{d}y \right) \mathrm{d}z =$$
$$= \int_0^1 z^2 \left(\int_0^{1-z} (1-y-z)\, \mathrm{d}y \right) \mathrm{d}z =$$
$$= \int_0^1 \frac{1}{2}(z^2 - 2z^3 + z^4)\, \mathrm{d}z$$
$$= \frac{1}{2}\left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right) = \frac{1}{2}\left(\frac{10}{30} - \frac{15}{30} + \frac{6}{30}\right) = \frac{1}{60}$$

□

Problem 1.32

Check the fundamental theorem for gradients, using $T = x^2 + 4xy + 2yz^3$, the points $\boldsymbol{a} = (0,0,0)$, $\boldsymbol{b} = (1,1,1)$, and the three paths:

(a) $(0,0,0) \to (1,0,0) \to (1,1,0) \to (1,1,1)$

(b) $(0,0,0) \to (0,0,1) \to (0,1,1) \to (1,1,1)$

(c) the parabolic path $z = x^2$, $y = x$

Solution:

The fundamental theorem for gradients says

$$\int_a^b (\boldsymbol{\nabla} T) \cdot \mathrm{d}\boldsymbol{l} = T(\boldsymbol{b}) - T(\boldsymbol{a})$$

$$\boldsymbol{\nabla} T = (2x + 4y)\hat{\boldsymbol{x}} + (4x + 2z^3)\hat{\boldsymbol{y}} + (6yz^2)\hat{\boldsymbol{z}}$$

RHS for all paths is

$$T(\boldsymbol{b}) - T(\boldsymbol{a}) = 7 - 0 = \boxed{7}$$

which is what we want to check for the LHS of the different paths

(a)

$(0,0,0) \to (1,0,0)$ has $x : 0 \to 1$, $y = z = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\boldsymbol{x}}$

$$\text{(i)} \int_0^1 2x\, \mathrm{d}x = 1$$

$(1,0,0) \to (1,1,0)$ has $y : 0 \to 1$, $x = 1$, $z = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\boldsymbol{y}}$

$$\text{(ii)} \int_0^1 4\,\mathrm{d}y = 4$$

$(1,1,0) \to (1,1,1)$ has $z : 0 \to 1$, $x = 1$, $y = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\boldsymbol{z}}$

$$\text{(iii)} \int_0^1 6z^2\,\mathrm{d}z = 2$$

$$\text{(i)} + \text{(ii)} + \text{(iii)} = 1 + 4 + 2 = \boxed{7}$$

(b)

$(0,0,0) \to (0,0,1)$ has $z : 0 \to 1$, $x = y = 0$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\boldsymbol{z}}$

$$\text{(i)} \int_0^1 0\,\mathrm{d}y = 0$$

$(0,0,1) \to (0,1,1)$ has $y : 0 \to 1$, $x = 0$, $z = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}y\hat{\boldsymbol{y}}$

$$\text{(ii)} \int_0^1 2\,\mathrm{d}y = 2$$

$(0,1,1) \to (1,1,1)$ has $x : 0 \to 1$, $y = 1$, $z = 1$ and $\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\boldsymbol{x}}$

$$\text{(iii)} \int_0^1 2x + 4\,\mathrm{d}x = 5$$

$$\text{(i)} + \text{(ii)} + \text{(iii)} = 0 + 2 + 5 = \boxed{7}$$

(c) Parametrization $x = y = t$, $z = t^2$ gives $\boldsymbol{r}(t) = (t, t, t^2)$, $t : 0 \to 1$. This means $\mathrm{d}\boldsymbol{l} = \boldsymbol{r}'(t)\,\mathrm{d}t = (1,1,2t)\,\mathrm{d}t$

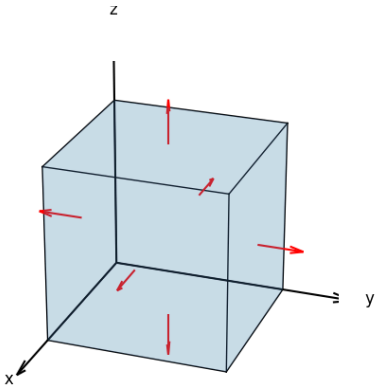
$$\int_0^1 (6t, 4t + 2t^6, 6t^5) \cdot (1,1,2t)\,\mathrm{d}t = \int_0^1 (6t + 4t + 2t^6 + 12t^6)\,\mathrm{d}t =$$

$$= \int_0^1 10t + 2t^6 + 12t^6\,\mathrm{d}t = 5 + \frac{2}{7} + \frac{12}{7} = \boxed{7}$$

□

Problem 1.33

Test the divergence theorem for the function $\boldsymbol{v} = (xy)\hat{\boldsymbol{x}} + (2yz)\hat{\boldsymbol{y}} + (3xz)\hat{\boldsymbol{z}}$. Take as your volume the cube with corners $(0,0,0)$,$(0,0,2)$,$(0,2,0)$,$(2,0,0)$



Solution:

The divergence theorem says

$$\int_V (\boldsymbol{\nabla} \cdot \boldsymbol{v})\, \mathrm{d}\tau = \oint_S \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{a}$$

The divergence of $\boldsymbol{v}$ is

$$\boldsymbol{\nabla} \cdot \boldsymbol{v} = y + 2z + 3x$$

and the resulting LHS is

$$\iiint_0^2 (y + 2z + 3x)\, \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z = \iint_0^2 \left[yx + 2zx + \frac{3x^2}{2} \right]_0^2 \mathrm{d}y\, \mathrm{d}z =$$

$$= \iint_0^2 2y + 4z + 6\, \mathrm{d}y\, \mathrm{d}z = \int_0^2 \left[y^2 + 4yz + 6y \right]_0^2 \mathrm{d}z =$$

$$= \int_0^2 16 + 8z\, \mathrm{d}z = \left[16z + 4z^2 \right]_0^2 = \boxed{48}$$

Top face:

$$x,y : 0 \to 2 \ , \ z = 2 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\, \mathrm{d}y \hat{\boldsymbol{z}}$$

$$\int_S 6x\, \mathrm{d}x\, \mathrm{d}y = 24$$

Bottom face:

$$x,y : 0 \to 2 \ , \ z = 0 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\, \mathrm{d}y (-\hat{\boldsymbol{z}})$$

$$\int_S 0\, \mathrm{d}x\, \mathrm{d}y = 0$$

Left face:

$$x,z : 0 \to 2 \ , \ y = 0 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}z(-\hat{\boldsymbol{y}})$$

$$\int_S 0\,\mathrm{d}x\,\mathrm{d}z = 0$$

Right face:

$$x,z : 0 \to 2 \ , \ y = 2 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}x\,\mathrm{d}z\hat{\boldsymbol{y}}$$

$$\int_S 4z\,\mathrm{d}x\,\mathrm{d}z = 16$$

Front face:

$$y,z : 0 \to 2 \ , \ x = 2 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{x}}$$

$$\int_S 2y\,\mathrm{d}y\,\mathrm{d}z = 8$$

Back face:

$$y,z : 0 \to 2 \ , \ x = 0 \ , \mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z(-\hat{\boldsymbol{x}})$$

$$\int_S 0\,\mathrm{d}y\,\mathrm{d}z = 0$$

Total is 48

□

Problem 1.34

Test Stokes' theorem for the function $\boldsymbol{v} = (xy)\hat{\boldsymbol{x}} + (2yz)\hat{\boldsymbol{y}} + (3xz)\hat{\boldsymbol{z}}$, using the triangular area with points $(0,0,0)$, $(0,1,0)$,$(0,0,1)$.

Solution:

Stokes's theorem

$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{v}) \cdot \mathrm{d}\boldsymbol{a} = \oint_P \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{l}$$

$$\nabla \times \boldsymbol{v} = (-2y)\hat{\boldsymbol{x}} + (-3z)\hat{\boldsymbol{y}} + (-x)\hat{\boldsymbol{z}}$$
$$\mathrm{d}\boldsymbol{a} = \mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{x}}$$
$$y : 0 \to 2 - z$$
$$z : 0 \to 2$$
$$\int_0^2 \left(\int_0^{2-z} -2y\,\mathrm{d}y \right) \mathrm{d}z =$$
$$= \int_0^2 \left[-y^2\right]_0^{2-z} \mathrm{d}z = \int_0^2 (-4 + 4z - z^2) = -8 + 8 - \frac{8}{3} = \boxed{-\frac{8}{3}}$$

Line 1:

$$(0,0,0) \to (0,2,0)$$
$$\mathrm{d}\boldsymbol{l} = \mathrm{d}x\hat{\boldsymbol{x}}$$
$$x = z = 0$$
$$\int_0^2 0\,\mathrm{d}x = 0$$

Line 2:

$$(0,2,0) \to (0,0,2)$$
$$\boldsymbol{r}(t) = (0, 2-t, t),\ t : 0 \to 2$$
$$\mathrm{d}\boldsymbol{l} = \boldsymbol{r}'(t)\,\mathrm{d}t$$
$$\int_0^2 (0, 2(2-t)(t), 0) \cdot (0,-1,1)\,\mathrm{d}t = \int_0^2 -4t + 2t^2\,\mathrm{d}t = -2t^2 + \frac{2}{3}t^3 = -8 + \frac{16}{3} = -\frac{8}{3}$$

Line 3:

$$(0,0,2) \to (0,0,0)$$
$$\mathrm{d}\boldsymbol{l} = \mathrm{d}z\hat{\boldsymbol{z}}$$
$$y = x = 0$$
$$\int_2^0 0\,\mathrm{d}z = 0$$

All line integrals combined gives $\boxed{-\frac{8}{3}}$

□

Problem 1.35

(a) Show that

$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \oint_P f\boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}$$

(b) Show that

$$\int_V \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A})\, \mathrm{d}\tau = \int_V \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})\, \mathrm{d}\tau + \oint_S (\boldsymbol{A} \times \boldsymbol{B}) \cdot \mathrm{d}\boldsymbol{a}$$

Solution:

(a) Using product rule (7) (from the cover of the book)

$$(7)\ \boldsymbol{\nabla} \times (f\boldsymbol{A}) = f(\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \times (\boldsymbol{\nabla} f)$$
$$\downarrow$$
$$f(\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{A} \times (\boldsymbol{\nabla} f) + \boldsymbol{\nabla} \times (f\boldsymbol{A})$$

Integrating this expression over a surface gives

$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \underbrace{\int_S \boldsymbol{\nabla} \times (f\boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a}}_{\text{use Stokes' Theorem}}$$
$$\downarrow$$
$$\int_S f(\boldsymbol{\nabla} \times \boldsymbol{A}) \cdot \mathrm{d}\boldsymbol{a} = \int_S [\boldsymbol{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\boldsymbol{a} + \oint_P f\boldsymbol{A} \cdot \mathrm{d}\boldsymbol{l}$$

(b) Using product rule (6)

$$\boldsymbol{\nabla} \cdot (\boldsymbol{A} \times \boldsymbol{B}) = \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) - \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})$$
$$\downarrow$$
$$\boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B}) + \boldsymbol{\nabla} \cdot (\boldsymbol{A} \times \boldsymbol{B})$$

Integrating this expression over a volume gives

$$\int_V \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A})\, \mathrm{d}\tau = \int_V \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})\, \mathrm{d}\tau + \underbrace{\int_V \boldsymbol{\nabla} \cdot (\boldsymbol{A} \times \boldsymbol{B})\, \mathrm{d}\tau}_{\text{use Gauss' Theorem}}$$
$$\downarrow$$
$$\int_V \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A})\, \mathrm{d}\tau = \int_V \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B})\, \mathrm{d}\tau + \oint_S (\boldsymbol{A} \times \boldsymbol{B}) \cdot \mathrm{d}\boldsymbol{a}$$

□

Problem 1.36

(a) Show that

$$\int_{\mathcal{S}} f(\boldsymbol{\nabla} \times \mathbf{A}) \cdot \mathrm{d}\mathbf{a} = \int_{\mathcal{S}} [\mathbf{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{d}\mathbf{a} + \oint_{\mathcal{P}} f\mathbf{A} \cdot \mathrm{dl}$$

(b) Show that

$$\int_{\mathcal{V}} \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A})\, \mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B})\, \mathrm{d}\tau + \oint_{\mathcal{S}} (\mathbf{A} \times \mathbf{B}) \cdot \mathrm{da}$$

Solution:

(a) Using product rule (v):

$$\text{(v) } \boldsymbol{\nabla} \times (f\mathbf{A}) = f(\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \times (\boldsymbol{\nabla} f)$$
$$\downarrow$$
$$f(\boldsymbol{\nabla} \times \mathbf{A}) = \mathbf{A} \times (\boldsymbol{\nabla} f) + \boldsymbol{\nabla} \times (f\mathbf{A})$$

Integrating this expression over a surface $\mathcal{S}$ yields

$$\int_{\mathcal{S}} f(\boldsymbol{\nabla} \times \mathbf{A}) \cdot \mathrm{da} = \int_{\mathcal{S}} \mathbf{A} \times (\boldsymbol{\nabla} f) \cdot \mathrm{da} + \int_{\mathcal{S}} \boldsymbol{\nabla} \times (f\mathbf{A}) \cdot \mathrm{da}\ .$$

Using Stokes' Theoreom on the last part gives the answer

$$\boxed{\int_{\mathcal{S}} f(\boldsymbol{\nabla} \times \mathbf{A}) \cdot \mathrm{da} = \int_{\mathcal{S}} [\mathbf{A} \times (\boldsymbol{\nabla} f)] \cdot \mathrm{da} + \oint_{\mathcal{P}} f\mathbf{A} \cdot \mathrm{dl}}$$

(b) Using product rule (iv)

$$\text{(iv) } \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B})$$
$$\downarrow$$
$$\mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) + \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B})$$

Integrating this expression over a volume $\mathcal{V}$ yields

$$\int_{\mathcal{V}} \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A})\, \mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B})\, \mathrm{d}\tau + \int_{\mathcal{V}} \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B})\, \mathrm{d}\tau\ .$$

Using Gauss' Theorem / Divergence Theorem on the last part gives the answer

$$\boxed{\int_{\mathcal{V}} \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A})\, \mathrm{d}\tau = \int_{\mathcal{V}} \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B})\, \mathrm{d}\tau + \oint_{\mathcal{S}} (\mathbf{A} \times \mathbf{B}) \cdot \mathrm{da}}$$

□

Problem 1.37

Find formulas for r, θ, ϕ in terms of x, y, z.

Solution:

In spherical coordinates we have

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

r-component:

From Pythagoras we know that $r = \sqrt{x^2+y^2+z^2} \longleftrightarrow r^2 = x^2+y^2+z^2$. This leads us to start with the r-component, by squaring each relation and adding them up:

$$x^2 = r^2\sin^2\theta\cos^2\phi$$
$$y^2 = r^2\sin^2\theta\sin^2\phi$$
$$z^2 = r^2\cos^2\theta$$
$$\downarrow$$
$$x^2+y^2+z^2 = r^2(\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta)$$

All the trigonometric functions above should then be equal to one, which is confirmed by rearranging and simplifying

$$= r^2\left(\sin^2\theta\underbrace{(\cos^2\phi+\sin^2\phi)}_{=1} + \cos^2\theta\right) = r^2\underbrace{(\sin^2\theta+\cos^2\theta)}_{=1} = r^2$$
$$\downarrow$$
$$\boxed{r = \sqrt{x^2+y^2+z^2}}$$

θ-component:

Since θ is alone in the relation for z, we can use z to solve for θ. Rewriting the term for z leads to

$$z = r\cos\theta \rightarrow \cos\theta = \frac{z}{r} = \frac{z}{\sqrt{x^2+y^2+z^2}}$$
$$\downarrow$$
$$\boxed{\theta = \arccos\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)}$$

ϕ-component: The x and y components both share a factor of $r\sin\theta$, which makes the ϕ component appear by itself if we divide y by x (or x by y)

$$\frac{y}{x} = \frac{r\sin\theta\sin\phi}{r\sin\theta\cos\phi} = \frac{\sin\phi}{\cos\phi} = \tan\phi$$

$$\downarrow$$

$$\boxed{\phi = \arctan\left(\frac{y}{x}\right)}$$

The relations for r, θ, ϕ in terms of x, y, z are then

$$\boxed{\begin{aligned} r &= \sqrt{x^2+y^2+z^2} \\ \theta &= \arccos\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right) \\ \phi &= \arctan\left(\frac{y}{x}\right) \end{aligned}}$$

□

Problem 1.38

Express the unit vectors $\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ in terms of $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$. Check your answers several ways ($\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}} \stackrel{?}{=} 1, \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} \stackrel{?}{=} 0, \hat{\boldsymbol{r}} \times \hat{\boldsymbol{\theta}} \stackrel{?}{=} \hat{\boldsymbol{\phi}}, ...$). Also work out the inverse formulas, giving $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$ in terms of $\hat{\boldsymbol{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ (and ϕ, θ).

Problem 1.39

(a) Check the divergence theorem for the function $\mathbf{v}_1 = r^2\hat{\boldsymbol{r}}$, using as your volume the sphere of radius R, centered at the origin.

(b) Do the same for $\mathbf{v}_2 = (1/r^2)\hat{\boldsymbol{r}}$.

Solution:

The divergence theorem is

$$\int_{\mathcal{V}} \boldsymbol{\nabla} \cdot r^2\hat{\boldsymbol{r}}\, \mathrm{d}\tau = \oint_{\mathcal{S}} r^2\hat{\boldsymbol{r}} \cdot \mathrm{d}\mathbf{a}$$

Taking the divergence of $\mathbf{v}_1$ in spherical coordinates gives

$$\boldsymbol{\nabla} \cdot r^2\hat{\boldsymbol{r}} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_{1r}) = \frac{1}{r^2}\frac{\partial r^4}{\partial r} = \frac{4r^3}{r^2} = 4r$$

In spherical coordinates,

$$d\tau = r^2 \sin\theta \, dr \, d\theta \, d\varphi$$
$$\mathbf{da} = r^2 \sin\theta \, d\theta \, d\varphi \hat{\mathbf{r}}$$

LHS of the divergence theorem becomes

$$\int_{\mathcal{V}} 4r r^2 \sin\theta \, dr \, d\theta \, d\varphi = 4 \int_0^R r^3 \int_0^\pi \sin\theta \int_0^{2\pi} dr \, d\theta \, d\varphi =$$
$$= 8\pi \int_0^R r^3 \int_0^\pi \sin\theta \, dr \, d\theta = 16\pi \int_0^R r^3 \, dr = \boxed{4\pi R^4}$$

and RHS

$$\oint_{\mathcal{S}} r^2 r^2 \sin\theta \, d\theta \, d\varphi = r^4 \int_0^\pi \sin\theta \int_0^{2\pi} d\theta \, d\varphi = 2\pi r^4 \int_0^\pi \sin\theta \, d\theta = 4\pi r^4 \Big|_{r=R} = \boxed{4\pi R^4}$$

which confirms the divergence theorem for $\mathbf{v}_1$.

(b)

The divergence of $\mathbf{v}_2$ is

$$\boldsymbol{\nabla} \cdot \mathbf{v}_2 = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_{2r}) = \frac{1}{r^2} \frac{\partial 1}{\partial r} = \boxed{0}$$

so LHS of divergence theorem is zero (which is not completely true, it's zero everywhere except at $r = 0$, so the calculation here is not fully correct). The RHS is

$$\oint_{\mathcal{S}} = \frac{1}{r^2} r^2 \sin\theta \, d\theta \, d\varphi = \int_0^\pi \sin\theta \int_0^{2\pi} d\theta \, d\varphi = 2\pi \int_0^\pi \sin\theta \, d\theta = \boxed{4\pi}$$

□

Problem 1.40

Compute the divergence of

$$\mathbf{v} = (r\cos\theta)\hat{\mathbf{r}} + (r\sin\theta)\hat{\boldsymbol{\theta}} + (r\sin\theta\cos\varphi)\hat{\boldsymbol{\varphi}}$$

Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R, resting on the xy plane and centered at the origin.

Solution:

The divergence of $\mathbf{v}$ is

$$\boldsymbol{\nabla}\cdot\mathbf{v}=\frac{1}{r^2}\frac{\partial}{\partial r}(r^2v_r)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta v_\theta)+\frac{1}{r\sin\theta}\frac{\partial v_\varphi}{\partial\varphi}=$$
$$=3\cos\theta+2\cos\theta-\sin\varphi=5\cos\theta-\sin\varphi$$

The divergence theorem is

$$\int_{\mathcal{V}}\boldsymbol{\nabla}\cdot\mathbf{v}\,\mathrm{d}\tau=\oint_{\mathcal{S}}\mathbf{v}\cdot\mathrm{d}\mathbf{a}$$

The volume integral is

$$\int_0^R\int_0^{\frac{\pi}{2}}\int_0^{2\pi}(5\cos\theta-\sin\varphi)r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi=$$
$$=\int_0^R r^2\left(\int_0^{\frac{\pi}{2}}\sin\theta\left(\int_0^{2\pi}(5\cos\theta-\sin\varphi)\,\mathrm{d}\varphi\right)\mathrm{d}\theta\right)\mathrm{d}r=$$
$$=\int_0^R r^2\left(\int_0^{\frac{\pi}{2}}\sin\theta([5\cos\theta\varphi+\cos\varphi]_0^{2\pi})\,\mathrm{d}\theta\right)\mathrm{d}r=$$
$$=\int_0^R r^2\left(\int_0^{\frac{\pi}{2}}\sin\theta(10\pi\cos\theta+1-1)\,\mathrm{d}\theta\right)\mathrm{d}r=$$
$$=\int_0^R r^2\left(\int_0^{\frac{\pi}{2}}10\pi\sin\theta\cos\theta\,\mathrm{d}\theta\right)\mathrm{d}r=$$
$$=\int_0^R r^2\left(\int_0^{\frac{\pi}{2}}5\pi\sin2\theta\,\mathrm{d}\theta\right)\mathrm{d}r=$$
$$=\int_0^R r^2\left[\frac{-5\pi\cos2\theta}{2}\right]_0^{\frac{\pi}{2}}\mathrm{d}r=\int_0^R r^2\left[\frac{5\pi}{2}-\frac{-5\pi}{2}\right]\mathrm{d}r=\int_0^R 5\pi r^2\,\mathrm{d}r=$$
$$=\boxed{\frac{5\pi R^3}{3}}$$

The closed surface integral is split into the circular disc ($\mathcal{S}_1$) and the half surface of the sphere ($\mathcal{S}_2$)

$$\oint_{\mathcal{S}}=\int_{\mathcal{S}_1}+\int_{\mathcal{S}_2}$$

For $\mathcal{S}_1$:

$$\mathrm{d}\mathbf{a}=\hat{\boldsymbol{\theta}}r\,\mathrm{d}r\,\mathrm{d}\varphi,\quad\theta=\frac{\pi}{2},\quad r:0\to R,\quad\varphi:0\to2\pi$$

so the integral becomes

$$\int_0^R \int_0^{2\pi} r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\varphi = \int_0^R r^2 \left(\int_0^{2\pi} \mathrm{d}\varphi \right) \mathrm{d}r = 2\pi \int_0^R r^2 \, \mathrm{d}r = \frac{2\pi R^3}{3}$$

For $\mathcal{S}_2$:

$$\mathrm{d}\mathbf{a} = \hat{\boldsymbol{r}} r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\varphi, \quad r = R, \quad \theta : 0 \to \frac{\pi}{2}, \quad \varphi : 0 \to 2\pi$$

so the integral becomes

$$\int_0^{\frac{\pi}{2}} \int_0^{2\pi} r \cos\theta r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\varphi = R^3 \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta \left(\int_0^{2\pi} \mathrm{d}\varphi \right) \mathrm{d}\theta =$$

$$= 2\pi R^3 \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta \, \mathrm{d}\theta = 2\pi R^3 \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin 2\theta \, \mathrm{d}\theta =$$

$$= 2\pi R^3 \left[-\frac{\cos 2\theta}{4} \right]_0^{\frac{\pi}{2}} = 2\pi R^3 \left(\frac{1}{4} - \frac{-1}{4} \right) = \pi R^3 = \frac{3\pi R^3}{3}$$

Adding these results we get

$$\frac{2\pi R^3}{3} + \frac{3\pi R^3}{3} = \boxed{\frac{5\pi R^3}{3}}$$

which is the same result we got from the volume integral.

□

Problem 1.41

Problem 1.42

Problem 1.43

Problem 1.44

Evaluate the following integrals:

(a) $\int_2^6 (3x^2 - 2x - 1)\delta(x - 3) \, \mathrm{d}x$

(b) $\int_0^5 \cos(x)\delta(x - \pi) \, \mathrm{d}x$

(c) $\int_0^3 x^3 \delta(x + 1) \, \mathrm{d}x$

(d) $\int_{-\infty}^{\infty} \ln(x+3)\delta(x+2)\,\mathrm{d}x$

Solution:

Integral of delta-dirac

$$\int_a^b \delta(x-p)\,\mathrm{d}x = \begin{cases} 1, \text{ if } p \in (a,b) \\ 0, \text{ otherwise} \end{cases}$$

(a) Zero everywhere except at $x = 3$ which gives

$$20 \underbrace{\int_2^6 \delta(x-3)\,\mathrm{d}x}_{=1 \text{ at } x=3} = 20$$

(b) Zero everywhere except at $x = \pi$ which gives

$$-1 \int_0^5 \delta(x-\pi)\,\mathrm{d}x = -1$$

(c) Zero everywhere except at $x = -1$, which is not included in the integral bounds, therefore it is zero.

(d) Zero everywhere except at $x = -2$ which gives

$$\ln(1) \int_{-\infty}^{\infty} \delta(x+2)\,\mathrm{d}x = 0$$

□

Problem 1.45

Evaluate the following integrals:

(a) $\int_{-2}^{2} (2x+3)\delta(3x)\,\mathrm{d}x$

(b) $\int_0^2 (x^3+3x+2)\delta(1-x)\,\mathrm{d}x$

(c) $\int_{-1}^{1} 9x^2\delta(3x+1)\,\mathrm{d}x$

(d) $\int_{-\infty}^{a} \delta(x-b)\,\mathrm{d}x$

Solution:

For the following problems we use

$$\delta(kx) = \frac{1}{|k|}\delta(x)$$

(a)

$$\int_{-2}^{2}(2x+3)\delta(3x)\,\mathrm{d}x = 3\int_{-2}^{2}\frac{1}{3}\delta(x)\,\mathrm{d}x = 1$$

(b)

$$\int_{0}^{2}(x^3+3x+2)\delta(1-x)\,\mathrm{d}x = \int_{0}^{2}(x^3+3x+2)\delta(x-1)\,\mathrm{d}x = 6\int_{0}^{2}\delta(x-1)\,\mathrm{d}x = 6$$

(c)

$$\delta(3x+1) = \frac{1}{3}\delta\left(x+\frac{1}{3}\right)$$

$$\downarrow$$

$$\int_{-1}^{1}9x^2\delta(3x+1)\,\mathrm{d}x = \int_{-1}^{1}\frac{1}{3}\delta\left(x+\frac{1}{3}\right)\mathrm{d}x = \frac{1}{3}$$

(d)

$$\int_{-\infty}^{a}\delta(x-b)\,\mathrm{d}x = \begin{cases}1, \text{ if } a > b\\ 0, \text{ if } a < b\end{cases}$$

□

Problem 1.46

(a) Show that

$$x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x)) = -\delta(x)$$

(b) Let $\theta(x)$ be the **step function**:

$$\theta(x) = \begin{cases}1, & \text{if} > 0\\ 0, & \text{if } x \leq 0\end{cases}$$

Show that $\mathrm{d}\theta/\,\mathrm{d}x = \delta(x)$

Solution:

(a)

$$\int_{-\infty}^{\infty} x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x))\,\mathrm{d}x = \underbrace{[x\delta(x)]_{-\infty}^{\infty}}_{=0} - \int_{-\infty}^{\infty}\delta(x)\,\mathrm{d}x$$

$$\downarrow$$

$$\int_{-\infty}^{\infty} x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x))\,\mathrm{d}x = -\int_{-\infty}^{\infty}\delta(x)\,\mathrm{d}x \rightarrow \boxed{x\frac{\mathrm{d}}{\mathrm{d}x}(\delta(x)) = -\delta(x)}$$

(b) Using test function $f(x)$ and integrating:

$$f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} \rightarrow \int_{-\infty}^{\infty} f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x}\,\mathrm{d}x = \underbrace{[f(x)\theta(x)]_{-\infty}^{\infty}}_{=f(\infty)} - \int_{-\infty}^{\infty} f'(x)\theta(x)\,\mathrm{d}x =$$

$$= f(\infty) - \int_{0}^{\infty} f'(x)\,\mathrm{d}x = f(\infty) - f(\infty) + f(0) = f(0) = f(x)\delta(x)$$

$$\downarrow$$

$$f(x)\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} = f(x)\delta(x) \rightarrow \boxed{\frac{\mathrm{d}\theta(x)}{\mathrm{d}x} = \delta(x)}$$

□

Problem 1.47

Problem 1.48

Problem 1.49

Problem 1.50

(a) Let $\boldsymbol{F}_1 = x^2\hat{\boldsymbol{z}}$ and $\boldsymbol{F}_2 = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$. Calculate the divergence and curl of $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$. Which one can be written as the gradient of a scalar? Find a scalar potential that does the job. Which one can be written as the curl of a vector? Find a suitable vector potential.

(b) Show that $\boldsymbol{F}_3 = yz\hat{\boldsymbol{x}} + zx\hat{\boldsymbol{y}} + xy\hat{\boldsymbol{z}}$ can be written both as the gradient of a scalar and as the curl of a vector. Find scalar and vector potentials for this function.

Solution:

(a)

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{F}_1 &= 0\\ \boldsymbol{\nabla}\cdot\boldsymbol{F}_2 &= 3\\ \boldsymbol{\nabla}\times\boldsymbol{F}_1 &= -2x\hat{\boldsymbol{y}}\\ \boldsymbol{\nabla}\times\boldsymbol{F}_2 &= 0\end{aligned}$$

This means that

$$\begin{aligned}\boldsymbol{F}_1 &= \boldsymbol{\nabla}\times\boldsymbol{A}\\ \boldsymbol{F}_2 &= -\boldsymbol{\nabla}V\end{aligned}$$

To find $\boldsymbol{A}$, the requirements are

$$\boldsymbol{\nabla}\times(A) = x^2\hat{\boldsymbol{z}}$$
$$\downarrow$$
$$\left(\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}\right)=\left(\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}\right)=0,\quad\left(\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y}\right)=x^2$$

One of the vectors that satisfy this is

$$\boldsymbol{A}=\frac{-x^2y}{2}\hat{\boldsymbol{x}}+\frac{x^3}{6}\hat{\boldsymbol{y}}$$

To find V, the requirements are

$$-(\boldsymbol{\nabla}(V)) = x\hat{\boldsymbol{x}}+y\hat{\boldsymbol{y}}+z\hat{\boldsymbol{z}}$$
$$\downarrow$$
$$\begin{cases}-\frac{\partial V}{\partial x}=x\\ -\frac{\partial V}{\partial y}=y\\ -\frac{\partial V}{\partial z}=z\end{cases}$$

One of the scalars that satisfy this is

$$V=-\frac{1}{2}(x^2+y^2+z^2)$$

(b) The divergence and curl of $\boldsymbol{F}_3$ is

$$\begin{aligned}\boldsymbol{\nabla}\cdot\boldsymbol{F}_3 &= 0\\ \boldsymbol{\nabla}\times\boldsymbol{F}_3 &= 0\end{aligned}$$

This means that $\boldsymbol{F}_3$ can be written both as the gradient of a scalar, and as the curl of a vector.

$$\begin{aligned}\boldsymbol{F}_3 &= -\boldsymbol{\nabla}V\\ \boldsymbol{F}_3 &= \boldsymbol{\nabla}\times\boldsymbol{A}\end{aligned}$$

The conditions to be met are then

$$-(\boldsymbol{\nabla}(V)) = yz\hat{\boldsymbol{x}} + xz\hat{\boldsymbol{y}} + xy\hat{\boldsymbol{z}}$$

which is satisfied by, for example

$$V = -xyz$$

The conditions for the vector are

$$\boldsymbol{\nabla} \times (A) = yz\hat{\boldsymbol{x}} + xz\hat{\boldsymbol{y}} + xy\hat{\boldsymbol{z}}$$

which is satisfied by, for example

$$\boldsymbol{A} = \frac{xz^2}{2}\hat{\boldsymbol{x}} + \frac{yx^2}{2}\hat{\boldsymbol{y}} + \frac{zy^2}{2}\hat{\boldsymbol{z}}$$

□

Problem 1.51

Curl-less fields:

(a) $\boldsymbol{\nabla} \times \boldsymbol{F} = 0$ everywhere.

(b) $\int_a^b \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l}$ is independent of path, for any given end points.

(c) $\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0$ for any closed loop.

(d) $\boldsymbol{F}$ is the gradient of some scalar function: $\boldsymbol{F} = -\boldsymbol{\nabla} V$

Show (d) ⇒ (a), (a) ⇒ (c), (c) ⇒ (b), (b) ⇒ (c), and (c) ⇒ (a).

Solution:

(d) ⇒ (a):

$$\boldsymbol{\nabla} \times (-\boldsymbol{\nabla} V) = \boldsymbol{\nabla} \times \left(\frac{\partial V}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial V}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial V}{\partial z}\hat{\boldsymbol{z}} \right) =$$

$$= \left(\frac{\partial \frac{\partial V}{\partial z}}{\partial y} - \frac{\partial \frac{\partial V}{\partial y}}{\partial z} \right)\hat{\boldsymbol{x}} + \left(\frac{\partial \frac{\partial V}{\partial x}}{\partial z} - \frac{\partial \frac{\partial V}{\partial z}}{\partial x} \right)\hat{\boldsymbol{y}} + \left(\frac{\partial \frac{\partial V}{\partial y}}{\partial x} - \frac{\partial \frac{\partial V}{\partial x}}{\partial y} \right)\hat{\boldsymbol{z}} =$$

$$= \left(\frac{\partial^2 V}{\partial y \partial z} - \frac{\partial^2 V}{\partial z \partial y} \right)\hat{\boldsymbol{x}} + \left(\frac{\partial^2 V}{\partial z \partial x} - \frac{\partial^2 V}{\partial x \partial z} \right)\hat{\boldsymbol{y}} + \left(\frac{\partial^2 V}{\partial x \partial y} - \frac{\partial^2 V}{\partial y \partial x} \right)\hat{\boldsymbol{z}} = 0 \quad \square$$

(a) ⇒ (c)

Stokes' Theorem

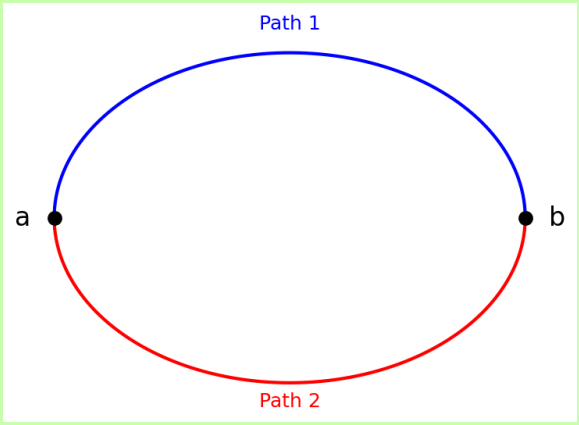
$$\int_S (\boldsymbol{\nabla} \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{a} = \oint_P \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0 \quad \square$$

(c) $\Rightarrow$ (b)

$$\int_{a_{\text{Path 1}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} + \int_{b_{\text{Path 2}}}^{a} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \oint_P \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = 0$$

$$\downarrow$$

$$\int_a^b \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} + \int_b^a \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \to \int_a^b \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} - \int_a^b \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \to \int_{a_{\text{Path 1}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} = \int_{a_{\text{Path 2}}}^{b} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{l} \quad \square$$



(b) $\Rightarrow$ (c), same as (c) $\Rightarrow$ (b), but reversed $\square$

(c) $\Rightarrow$ (a), same as (a) $\Rightarrow$ (c) (Stokes' Theorem) $\square$

$\square$

Problem 1.52

Divergence-less fields

(a) $\boldsymbol{\nabla} \cdot \boldsymbol{F} = 0$ everywhere.

(b) $\int \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a}$ is independent of surface, for any given boundary line.

(c) $\oint \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a} = 0$ for any closed surface.

(d) $\boldsymbol{F}$ is the curl of some vector function: $\boldsymbol{F} = \boldsymbol{\nabla} \times \boldsymbol{A}$

Show (d) $\Rightarrow$ (a), (a) $\Rightarrow$ (c), (c) $\Rightarrow$ (b), (b) $\Rightarrow$ (c), and (c) $\Rightarrow$ (a).

Solution:

(d) ⇒ (a)

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times (A)) =$$

$$= \frac{\partial^2 A_z}{\partial y \partial x} - \frac{\partial^2 A_y}{\partial z \partial x} + \frac{\partial^2 A_x}{\partial z \partial y} - \frac{\partial^2 A_z}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial y \partial z} =$$

$$= \left(\frac{\partial^2 A_z}{\partial y \partial x} - \frac{\partial^2 A_z}{\partial x \partial y} \right) + \left(\frac{\partial^2 A_x}{\partial z \partial y} - \frac{\partial^2 A_x}{\partial y \partial z} \right) + \left(\frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_y}{\partial z \partial x} \right) = 0 \quad \square$$

(a) ⇒ (c)

Divergence Theorem

$$\int_V \boldsymbol{\nabla} \cdot \boldsymbol{F} \, \mathrm{d}\tau = \oint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{a} = 0 \quad \square$$

(c) ⇒ (b) For a boundary line that splits a closed surface into two...

(b) ⇒ (c), same as (c) ⇒ (b), but reversed □

(c) ⇒ (a), same as (a) ⇒ (c) (Stokes' Theorem) □

□

Problem 1.53

(a) Which of the vectors in Problem 1.15 can be expressed as the gradient of a scalar? Find a scalar function that does the job.

(b) Which can be expressed as the curl of a vector? Find such a vector.

$\boldsymbol{v}_a = x^2 \hat{\boldsymbol{x}} + 3xz^2 \hat{\boldsymbol{y}} - 2xz\hat{\boldsymbol{z}}$

$\boldsymbol{v}_b = xy\hat{\boldsymbol{x}} + 2yz\hat{\boldsymbol{y}} + 3xz\hat{\boldsymbol{z}}$

$\boldsymbol{v}_c = y^2 \hat{\boldsymbol{x}} + (2xy + z^2)\hat{\boldsymbol{y}} + 2yz\hat{\boldsymbol{z}}$

Solution:

(a)

$$\boldsymbol{\nabla} \times \mathbf{v}_a \neq 0$$
$$\boldsymbol{\nabla} \times \mathbf{v}_b \neq 0$$
$$\boldsymbol{\nabla} \times \mathbf{v}_c = 0$$

This means $\mathbf{v}_c$ can be expressed as the gradient of a scalar. We want

$$\mathbf{v}_c = \boldsymbol{\nabla} f = \frac{\partial f}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial f}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{z}}$$

$$\frac{\partial f}{\partial x} = y^2 \rightarrow f = y^2 x + \rho(y,z)$$

$$\frac{\partial f}{\partial y} = (2xy + z^2) \rightarrow f = (xy^2 + yz^2) + \rho(x,z)$$

$$\frac{\partial f}{\partial z} = 2yz \rightarrow f = yz^2 + \rho(x,y)$$

$$\downarrow$$

$$\rho(x,y) = xy^2, \quad \rho(x,z) = 0, \quad \rho(y,z) = yz^2,$$

$$\downarrow$$

$$\boxed{f = xy^2 + yz^2}$$

(b)

$$\boldsymbol{\nabla} \cdot \mathbf{v}_a = 0$$
$$\boldsymbol{\nabla} \cdot \mathbf{v}_b \neq 0$$
$$\boldsymbol{\nabla} \cdot \mathbf{v}_c \neq 0$$

This means $\mathbf{v}_a$ can be expressed as the curl of a vector. We want

$$\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right) = x^2$$
$$\left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right) = 3xz^2$$
$$\left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right) = -2xz$$

Setting $V_x = 0$ gives

$$\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right) = x^2$$

$$\frac{\partial V_z}{\partial x} = -3xz^2 \rightarrow V_z = -\frac{3}{2}x^2 z^2$$

$$\frac{\partial V_y}{\partial x} = -2xz \rightarrow V_y = -x^2 z$$

$$\downarrow$$

$$\boxed{\mathbf{V} = (-x^2 z)\hat{\boldsymbol{y}} + \left(-\frac{3}{2}x^2 z^2\right)\hat{\boldsymbol{z}}}$$

□

Problem 1.54

Problem 1.55

Problem 1.56

Problem 1.57

Problem 1.58

Problem 1.59

Problem 1.60

Problem 1.61

Problem 1.62

Problem 1.63

Problem 1.64

Problem 1.65

Chapter 2

Electrostatics

Problem 2.1

(a) Twelve equal charges, q, are situated at the corners of a regular 12-sided polygon (for instance, one on each numeral of a clock face). What is the net force on a test charge Q at the center?

(b) Suppose on of your 12 qs is removed (the one at "6 o'clock"). What is the force on Q? Explain your reasoning carefully.

(c) Now 13 equal charges q, are placed at the corners of a regular 13-sided polygon. What is the force on a test charge Q at the center?

(d) If one of the 13 qs is removed, what is the force on Q? Explain your reasoning.

Solution:

(a) Due to the symmetry, the net force on a test charge at the center is zero. This is because each opposite charge q on the 12-sided polygon cancel each others force at the center. The force from the charge at 1 o'clock cancels the force from the charge at 7 o'clock, etc.

(b) If the charge at 6 o'clock is removed, the net force on a test charge at the center is

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

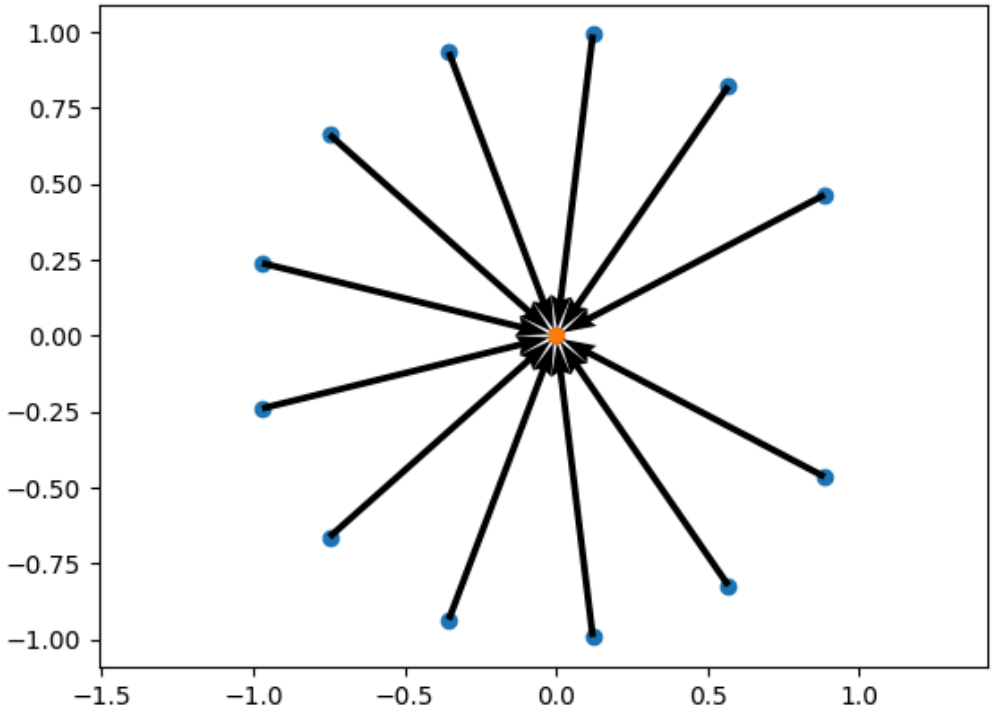
directed straight down (straight up) if the charge q and the test charge Q are both positive or both negative (if they are positive/negative or negative/positive respectively).

(c) Again, due to symmetry, the net force at the center is zero.

(d) The force on the center would again be

$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}$$

directed towards the removed charge (or opposite direction, if the charges q and the test charge Q are of opposite signs). This can either be explained by simply adding a charge of opposite sign at the removed charge, or by the same idea as in (b), from the following image



All vertical components of the forces cancel, giving the total force as the sum of the individual charges:

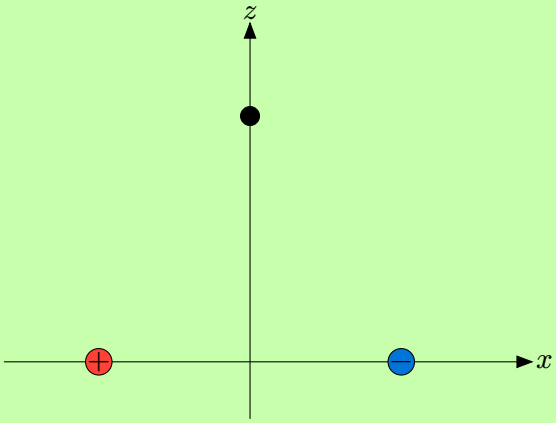
$$\frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\sum_{k=1}^{13}\cos\left(\frac{2\pi k}{13}\right) = \frac{1}{4\pi\epsilon_0}\frac{qQ}{r^2}\hat{\boldsymbol{r}} \text{ with } \hat{\boldsymbol{r}} \text{ pointing towards the right}$$

□

Problem 2.2

Find the electric field (magnitude and direction) a distance z above the mid-point between equal and opposite charges ($\pm q$), a distance d apart.

Solution:



The electric field from a point charge is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$

where r is the distance between the charge and the field point and $\hat{\boldsymbol{r}}$ is the unit separation vector from the charge to the field point. The field at our measurement point will have no vertical (z-directed) component since the positive charge will create a field in the positive z-direction with the same magnitude that the negative charge will create a field in the negative z-direction. Therefore the total field will be directed towards $+x$ with a magnitude twice that of what the positive charge creates. The separation vector magnitude becomes

$$r = \sqrt{z^2 + \left(\frac{d}{2}\right)^2}$$

and

$$\boldsymbol{r} = \frac{d}{2}\hat{\boldsymbol{x}}$$
$$\downarrow$$
$$\hat{\boldsymbol{r}} = \frac{\boldsymbol{r}}{r} = \frac{d}{2r}\hat{\boldsymbol{x}}$$

resulting in the total electric field from both charges

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{qd}{\sqrt{z^2 + \left(\frac{d}{2}\right)^2}^3}\hat{\boldsymbol{x}}$$

□

Problem 2.3

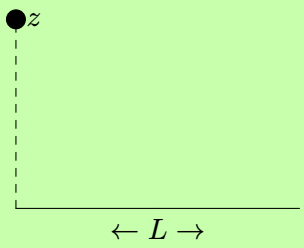
Find the electric field a distance z above one end of a straight line segment of length L that carries a uniform line charge λ. Check that your formula in consistent with what you would expect for the case $z \gg L$.

Solution:

The electric field from a continuous line charge distribution is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda(\mathbf{r}')}{\boldsymbol{\imath}^2}\,\mathrm{d}l'$$

The setup is shown in the figure below



The separation vector $\boldsymbol{\imath}$ becomes

$$-x\hat{\boldsymbol{x}} + z\hat{\boldsymbol{z}}$$

which gives

$$\hat{\boldsymbol{\imath}} = \frac{-x\hat{\boldsymbol{x}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2+z^2}}$$

This means our integral becomes, since λ is constant,

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{-x\hat{\boldsymbol{x}} + z\hat{\boldsymbol{z}}}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x$$

The following standard integrals will be used in the calculations:

$$\int \frac{1}{\sqrt{x^2+a^2}^3}\,\mathrm{d}x = \frac{1}{a^2}\frac{x}{\sqrt{x^2+a^2}}$$

$$\int \frac{x}{\sqrt{x^2+a^2}}\,\mathrm{d}x = -\frac{1}{\sqrt{x^2+a^2}}$$

Starting with the x-component gives

$$E_x = -\frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{x}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x = -\frac{\lambda}{4\pi\epsilon_0}\left[-\frac{1}{\sqrt{x^2+z^2}}\right]_0^L =$$

$$= \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z} - \frac{1}{\sqrt{L^2+z^2}}\right)$$

And the z-component

$$E_z = \frac{\lambda}{4\pi\epsilon_0}\int_0^L \frac{z}{\sqrt{x^2+z^2}^3}\,\mathrm{d}x = \frac{\lambda z}{4\pi\epsilon_0}\left[\frac{1}{z^2}\frac{x}{\sqrt{x^2+z^2}}\right]_0^L = \\ = \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)$$

The electric field is then

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z}-\frac{1}{\sqrt{L^2+z^2}}\right)\hat{\boldsymbol{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{\sqrt{L^2+z^2}}\right)\hat{\boldsymbol{z}}$$

In the case that $z \gg L$, we get

$$\mathbf{E} = \frac{\lambda}{4\pi\epsilon_0}\left(\frac{1}{z}-\frac{1}{z}\right)\hat{\boldsymbol{x}} + \frac{\lambda}{4\pi\epsilon_0 z}\left(\frac{L}{z}\right)\hat{\boldsymbol{z}} = \frac{\lambda L}{4\pi\epsilon_0 z^2}\hat{\boldsymbol{z}}$$

which is of the same form as the field at a point z from a point charge of magnitude λL. □

Problem 2.4

Find the electric field a distance z above the center of a square loop (side length d) carrying a uniform line charge λ.

Solution:

Due to the symmetry, it is enough to calculate the contribution from one of the sides, and multiplying this result by 4. Only the z-component will be calculated since the x- and y-components will cancel.

Starting with determining the separation vector for one of the sides ($x : -\frac{a}{2} \to \frac{a}{2}, y = \frac{a}{2}$)

$$\boldsymbol{\mathscr{r}} = x\hat{\boldsymbol{x}} + \frac{a}{2}\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}} = z\hat{\boldsymbol{z}}$$

and the magnitude

$$\mathscr{r} = \sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}$$

gives

$$\hat{\boldsymbol{\mathscr{r}}} = \frac{z\hat{\boldsymbol{z}}}{\sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}}$$

Using these, the electric field is given by

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{\lambda z \hat{\boldsymbol{z}}}{\sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}^3}\,\mathrm{d}x$$

$$E_x = \frac{\lambda z}{4\pi\epsilon_0}\int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{\sqrt{x^2 + \sqrt{\left(\frac{a}{2}\right)^2 + z^2}^2}^3}\,\mathrm{d}x = \frac{\lambda z}{4\pi\epsilon_0}\left[\frac{x}{\left(\left(\frac{a}{2}\right)^2 + z^2\right)\sqrt{x^2 + \left(\frac{a}{2}\right)^2 + z^2}}\right]_{-\frac{a}{2}}^{\frac{a}{2}} =$$

$$= \frac{\lambda z}{4\pi\epsilon_0}\frac{a}{\left(\left[\frac{a^2}{4} + z^2\right]\sqrt{\frac{a^2}{2} + z^2}\right)}$$

Multiplying this by four gives the final result

$$\boxed{\mathbf{E} = \frac{\lambda z a}{\pi\epsilon_0\left(\frac{a^2}{4} + z^2\right)\sqrt{\frac{a^2}{2} + z^2}}\hat{\boldsymbol{z}}}$$

□

Problem 2.5

Find the electric field a distance z above the center of a circular loop of radius r that carries a uniform line charge λ.

Solution:

Using cylindrical coordinates, the separation vector $\boldsymbol{\imath}$ is

$$\boldsymbol{\imath} = r\hat{\boldsymbol{r}} + z\hat{\boldsymbol{z}} = [\text{symmetry}] = z\hat{\boldsymbol{z}}$$

and the magnitude

$$\imath = \sqrt{r^2 + z^2}$$

gives

$$\hat{\boldsymbol{\imath}} = \frac{z}{\sqrt{r^2 + z^2}}\hat{\boldsymbol{z}}$$

The circular loop at radius r has the integral form $\int_0^{2\pi} r\,\mathrm{d}\theta$. The integral that gives the resulting electric field is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int_0^{2\pi} \frac{\lambda z \hat{\boldsymbol{z}}}{\sqrt{r^2 + z^2}^3} r\,\mathrm{d}\theta = \boxed{\frac{\lambda z r}{2\epsilon_0\sqrt{r^2 + z^2}^3}\hat{\boldsymbol{z}}}$$

□

Problem 2.6

Find the electric field a distance z above the center of a flat circular disk of radius R that carries a uniform surface charge σ. What does your formula give in the limit $R \to \infty$? Also check the case $z \gg R$.

Solution:

The electric field for a surface charge is given by

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int_{\mathcal{S}} \frac{\sigma(\mathbf{r}')}{r\boldsymbol{\imath}^2}\hat{\boldsymbol{\imath}}\,\mathrm{d}a'$$

Our disk is given by the cylindrical coordinates s and φ

$$s : 0 \to R, \quad \varphi : 0 \to 2\pi$$

The integration is

$$\mathrm{d}a' = s\,\mathrm{d}s\,\mathrm{d}\varphi$$

The separation vector is

$$\begin{gathered}\boldsymbol{\imath} = z\hat{\boldsymbol{z}} - s\hat{\boldsymbol{s}} \\ \downarrow \\ \imath = \sqrt{z^2+s^2} \\ \downarrow \\ \hat{\boldsymbol{\imath}} = \frac{z\hat{\boldsymbol{z}} - s\hat{\boldsymbol{s}}}{\sqrt{z^2+s^2}}\end{gathered}$$

Due to the symmetry of the problem, the s-dependence will add up to zero, so we will ignore the s-component in the unit separation vector. The electric field at the z-axis is then

$$\begin{aligned}\mathbf{E}(z) &= \frac{1}{4\pi\epsilon_0}\int_0^R\int_0^{2\pi}\frac{\sigma z\hat{\boldsymbol{z}}}{\sqrt{z^2+s^2}^3}s\,\mathrm{d}s\,\mathrm{d}\varphi = \\ &= \frac{2\pi z\sigma\hat{\boldsymbol{z}}}{4\pi\epsilon_0}\int_0^R\frac{s}{\sqrt{z^2+s^2}^3}\,\mathrm{d}s = \frac{2\pi z\sigma\hat{\boldsymbol{z}}}{4\pi\epsilon_0}\left[-\frac{1}{\sqrt{z^2+s^2}}\right]_0^R = \\ &= \frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}}\right]\hat{\boldsymbol{z}}\end{aligned}$$

In the limit that $R \to \infty$, the electric field is given by

$$\frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z}-\frac{1}{R}\right]\hat{\boldsymbol{z}}=\frac{\sigma}{2\epsilon_0}\hat{\boldsymbol{z}}$$

When $z \gg R$, naïvely setting the square root term to $\frac{1}{\sqrt{z^2+R^2}} = \frac{1}{\sqrt{z^2}} = \frac{1}{z}$ would give the (somewhat) incorrect result zero. The square root expression needs to be Taylor expanded with the rule (assuming $|x| \ll 1$)

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + ...$$

Rewriting the square root term

$$\frac{1}{\sqrt{z^2+R^2}} = \frac{1}{z}\frac{1}{\sqrt{1+\frac{R^2}{z^2}}} = \frac{1}{z}\left(1+\frac{R^2}{z^2}\right)^{-\frac{1}{2}}$$

$$\downarrow$$

$$\frac{1}{\sqrt{z^2+R^2}} \approx \frac{1}{z}\left(1-\frac{1}{2}\frac{R^2}{z^2}\right)$$

which gives the electric field

$$\frac{z\sigma}{2\epsilon_0}\left[\frac{1}{z}-\frac{1}{z}\left(1-\frac{1}{2}\frac{R^2}{z^2}\right)\right]\hat{\boldsymbol{z}} = \frac{z\sigma}{2\epsilon_0}\frac{1}{2}\frac{R^2}{z^3}\hat{\boldsymbol{z}}$$

Expressing the charge density and the radius in terms of the total charg Q would give $Q = \sigma\pi R^2$ which results in the electric field

$$\mathbf{E}(z) = \frac{Q}{4\pi\epsilon_0 z^2}\hat{\boldsymbol{z}}$$

□

Problem 2.7

Problem 2.8

Problem 2.9

Suppose the electric field in some region is found to be $\mathbf{E} = kr^3\mathbf{r}$, in spherical coordinates (k is some constant).

(a) Find the charge density ρ.

(b) Find the total charge contained in a sphere of radius R, centered at the origin. (Do it two different ways.)

Solution:

(a) Gauss's law in differential form is

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho \rightarrow \rho = \epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E}$$

The divergence of $\mathbf{E}$ is

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \boldsymbol{\nabla} \cdot kr^3\hat{\boldsymbol{r}} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 kr^3) = 5kr^2$$

so the charge density ρ is

$$\rho = \epsilon_0 5kr^2$$

(b) The total charge Q can be obtained in two ways; integrating the charge density over the volume, or integrating the electric field over the bounding surface.

Integrating the charge density over the volume of a sphere with radius R gives

$$Q = \int_{\mathcal{V}} \epsilon_0 5kr^2 r^2 \sin\theta \,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

$$\downarrow$$

$$5k\epsilon_0 \int_0^R r^4 \left(\int_0^\pi \sin\theta \left(\int_0^{2\pi} \mathrm{d}\varphi \right) \mathrm{d}\theta \right) \mathrm{d}r =$$

$$= 20k\pi\epsilon_0 \int_0^R r^4 \,\mathrm{d}r = \boxed{Q = 4\pi\epsilon_0 kR^5}$$

Integrating the electric field over the bounding surface of the sphere with radius R gives

$$\frac{Q}{\epsilon_0} = \int_{\mathcal{S}} kr^3 r^2 \sin\theta \,\mathrm{d}\theta\,\mathrm{d}\varphi$$

$$\downarrow$$

$$Q = \epsilon_0 kR^5 \int_0^\pi \sin\theta \left(\int_0^{2\pi} \mathrm{d}\varphi \right) \mathrm{d}\theta \rightarrow \boxed{Q = 4\pi\epsilon_0 kR^5}$$

□

Problem 2.10

A charge q sits at the back corner of a cube, as shown in the figure. What is the flux of $\mathbf{E}$ through the shaded side?

Solution:

Adding 7 cubes to totally enclose the charge gives a total of 24 faces that the flux is passing through. This means that the flux of this one surface is 1/24 of the total flux given by the charge q.

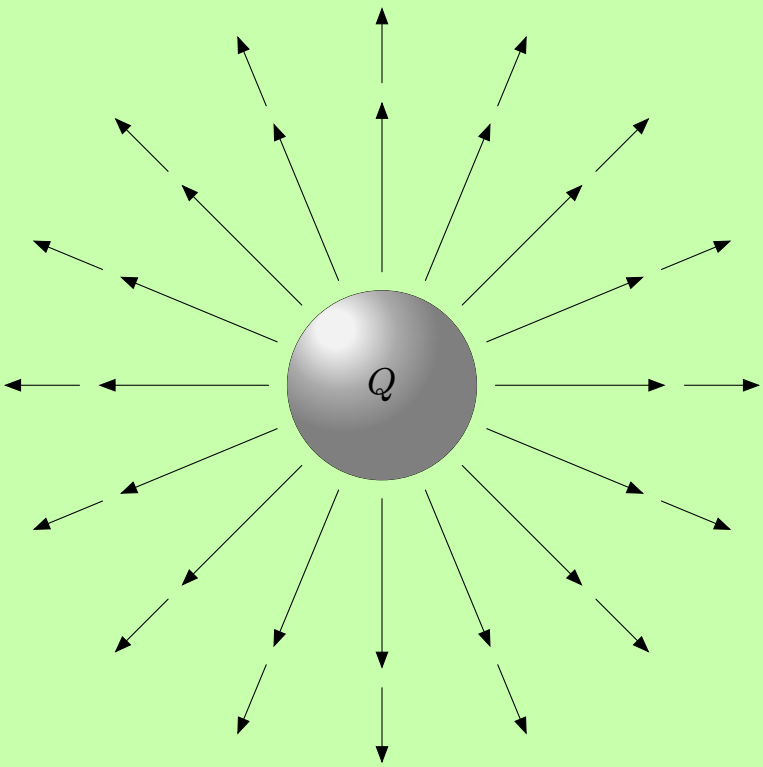
$$[\text{Flux through the shaded surface}] = \frac{q}{24\epsilon_0}$$

□

Problem 2.11

Draw a field-line diagram for a uniformly charged solid sphere.

Solution:



□

Problem 2.12

Use Gauss's law to find the electric field inside and outside a spherical shell of radius R that carries a uniform surface charge density σ.

Solution:

Gauss's law states that the closed surface integral of the electric field is equal to the charge enclosed in that surface.

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

Using the Gaussian surface of a sphere, the charge enclosed inside the spherical shell is zero, and outside it is

$$Q_{\text{enc}} = 4\pi R^2 \sigma$$

The electric field is always directed towards $\hat{\boldsymbol{r}}$ and $\mathrm{d}\mathbf{a} = r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\varphi$ gives

$$Er^2 4\pi = \frac{4\pi R^2 \sigma}{\epsilon_0}$$

$$\downarrow$$

$$\boxed{\mathbf{E} = \frac{R^2 \sigma}{\epsilon_0 r^2} \hat{\boldsymbol{r}}, \quad r > R}$$

where R is the radius of the spherical shell and r is the distance from the center □

Problem 2.13

Use Gauss's law to find the electric field inside a uniformly charged solid sphere (charge density ρ).

Solution:

For our Gaussian surface, we will use the surface of a sphere. Gauss's law is

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

While our Gaussian surface is inside the uniformly charged sphere, the charged enclosed is the volume integral of the charge density, where the volume is a sphere. Let a be the radius of our Gaussian surface.

$$Ea^2 \int_0^{\pi} \int_0^{2\pi} \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi = \frac{1}{\epsilon_0} \int_0^{a} \int_0^{\pi} \int_0^{2\pi} \rho r^2 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\varphi$$

$$\downarrow$$

$$Ea^2 4\pi = \frac{\rho a^3 4\pi}{3\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho a}{3\epsilon_0} \hat{\boldsymbol{r}}$$

□

Problem 2.14

Find the electric field a distance s from an infinitely long straight wire that carries a uniform line charge λ.

Solution:

The Gaussian surface for this problem is a cylinder. The enclosed charge of this cylinder is

$$Q_{\text{enc}} = \lambda z$$

where z is the height of the cylinder. The top and bottom disks of the cylinder are not accounted for since the field there is zero due to symmetry. Gauss's law becomes

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$\int_0^{z} \int_0^{2\pi} Es \, \mathrm{d}\varphi \, \mathrm{d}\theta = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$Esz2\pi = \frac{\lambda z}{\epsilon_0}$$

$$\downarrow$$

$$E = \frac{\lambda}{2\pi s \epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\lambda}{2\pi s \epsilon_0} \hat{\boldsymbol{s}}$$

□

Problem 2.15

Find the electric field inside a sphere that carries a charge density proportional to the distance from the center, $\rho = kr$, for some constant k.

Solution:

The Gaussian surface for this problem is the surface of a sphere. The enclosed charge is obtained by integrating the charge density. Let r be the radius of our Gaussian surface, that is, $r : 0 \to R$, where R is the radius of the charged sphere. Gauss's law becomes

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$\downarrow$$

$$\int_0^{\pi} \int_0^{2\pi} Er^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi = \frac{1}{\epsilon_0} \int_0^{r} \int_0^{\pi} \int_0^{2\pi} kr^3 \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}\varphi$$

$$\downarrow$$

$$Er^2 4\pi = \frac{kr^4 4\pi}{4\epsilon_0}$$

$$\downarrow$$

$$E = \frac{kr^2}{4\epsilon_0}$$

$$\downarrow$$

$$\boxed{\mathbf{E} = \frac{kr^2}{4\epsilon_0} \hat{\boldsymbol{r}}}$$

□

Problem 2.16

A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2}, \quad (a \leq r \leq b)$$

Find the electric field in the following three regions:

(i) $r < a$, (ii) $a < r < b$, (iii) $r > b$

Plot $|\mathbf{E}|$ as a function of r, for the case $b = 2a$.

Solution:

Inside region (i), there is no enclosed charge, therefore the electric field is zero in this region. In region (ii), we construct a spherical Gaussian surface, and integrate the charge density from a to r ($a < r < b$)

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$\downarrow$$

$$\int_0^{\pi} \int_0^{2\pi} E r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\varphi = \frac{1}{\epsilon_0} \int_a^r \int_0^{\pi} \int_0^{2\pi} \frac{k}{r^2} r^2 \sin\theta \, \mathrm{d}r \mathrm{d}\theta \mathrm{d}\varphi$$

$$\downarrow$$

$$E r^2 4\pi = \frac{k 4\pi}{\epsilon_0} (r - a)$$

$$\downarrow$$

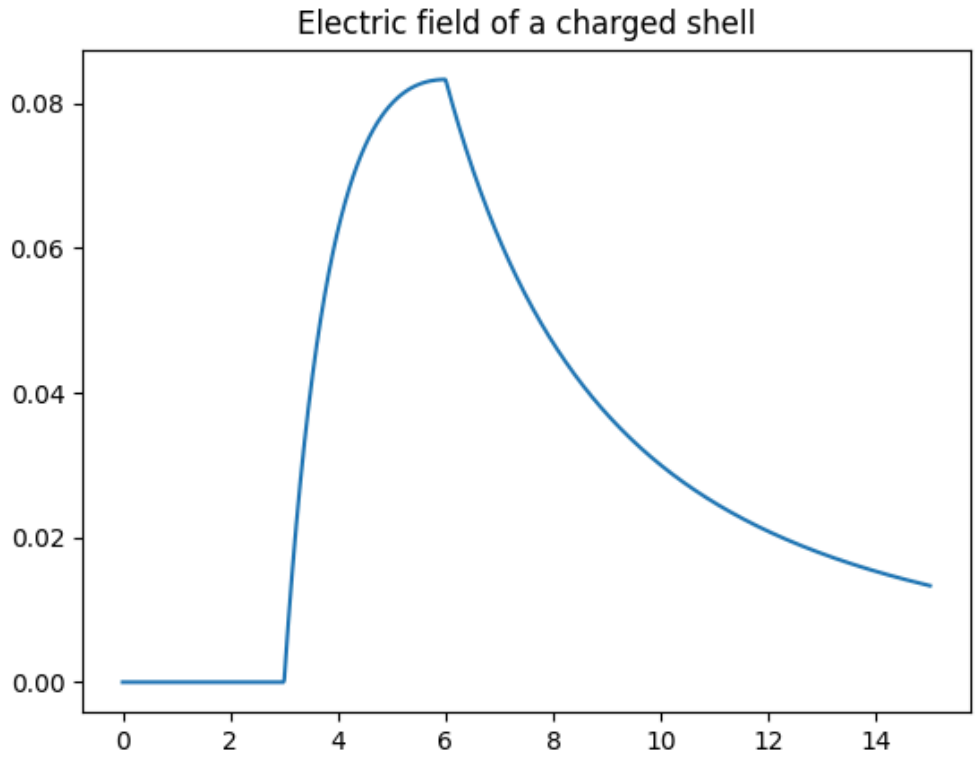
$$E = \frac{k(r-a)}{r^2 \epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{k(r-a)}{r^2 \epsilon_0} \hat{\boldsymbol{r}}$$

In region (iii), the total charge enclosed is obtained by integrating the charge density. We get the same result as in (ii) but now $(r - a) \to (b - a)$ since the integral is taken from a to b.

$$\mathbf{E} = \frac{k(b-a)}{r^2 \epsilon_0}, \quad r > b$$



□

Problem 2.17

A long coaxial cable carries a uniform volume charge density ρ on the inner cylinder (radius a), and a uniform surface charge density on the outer cylindrical shell (radius b). This surface charge is negative and is of just the right magnitude that the cable as a whole is electrically neutral. Find the electric field in each of the three regions:

(i) inside the inner cylinder ($s < a$),

(ii) between the cylinders ($a < s < b$),

(iii) outside the cable ($s > b$).

Plot $|\mathbf{E}|$ as a function of s.

Solution:

Using a cylindrical Gaussian surface gives for region (i)

$$\oint \mathbf{E} \cdot \mathrm{d}\mathbf{a} = Q_{\text{enc}}$$

$$\downarrow$$

$$E 2\pi s z = \frac{1}{\epsilon_0} \int_0^a \rho s \, \mathrm{d}s \mathrm{d}\varphi \mathrm{d}z$$

$$\downarrow$$

$$E 2\pi s z = \frac{\rho s^2 2\pi z}{2\epsilon_0}$$

$$\downarrow$$

$$E = \frac{\rho s}{2\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho s}{2\epsilon_0} \hat{\boldsymbol{s}}, \quad s < a$$

In region (ii), the total enclosed charge is the charge density multiplied by the volume, which is

$$Q_{\text{enc}} = \rho \pi a^2 z$$

which gives the electric field

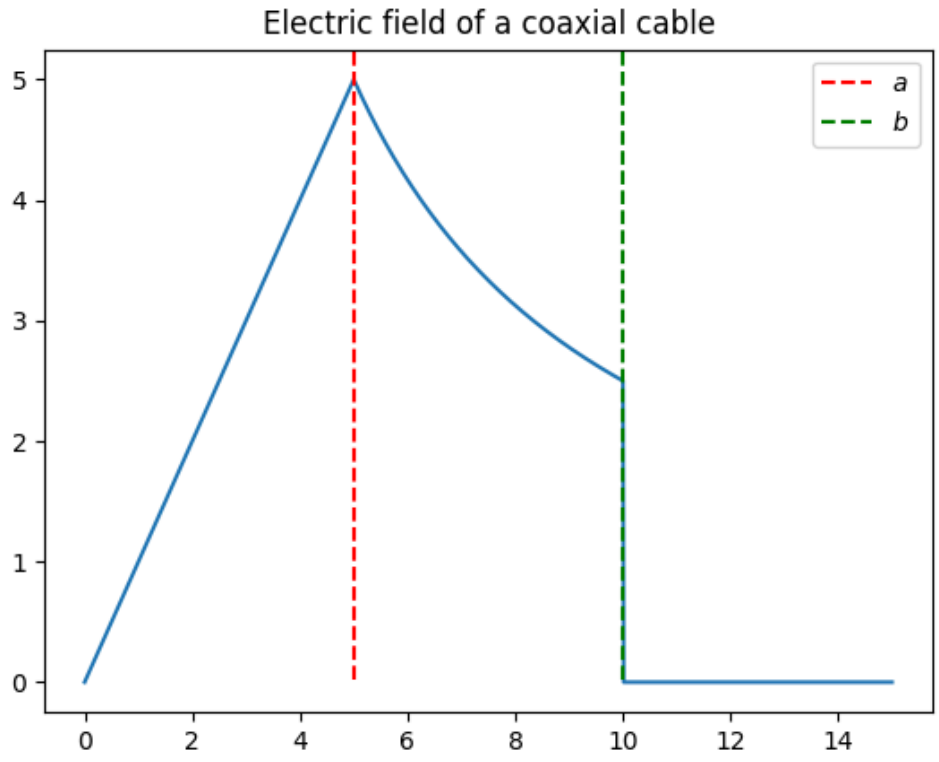
$$E 2\pi s z = \frac{\rho \pi a^2 z}{\epsilon_0}$$

$$\downarrow$$

$$\mathbf{E} = \frac{\rho a^2}{2\epsilon_0 s} \hat{\boldsymbol{s}}, \quad a < s < b$$

In region (iii), the total charge is zero since the positive and negative parts cancel,

$$\mathbf{E} = 0, \quad s > b$$



□

Problem 2.18

An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ (FIGURE). Find the electric field, as a function of y, where $y = 0$ at the center. Plot E versus y, calling E positive when it points in the $+y$ direction and negative when it points in the $-y$ direction.

Solution:

We create a Gaussian surface as a box. The enclosed charge is simply the box area A times the position y

$$\oint_{\mathcal{S}} \mathbf{E} \cdot \mathrm{d}\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$2EA = \frac{1}{\epsilon_0} \rho 2dA$$

$$\mathbf{E} = \frac{\rho d}{\epsilon_0}\hat{\boldsymbol{y}}, \qquad y > 0$$

$$\mathbf{E} = \frac{\rho d}{\epsilon_0}(-\hat{\boldsymbol{y}}), \quad y < 0$$

Inside:

$$2EA = \frac{1}{\epsilon_0}\rho 2yA$$

$$\mathbf{E} = \frac{\rho y}{\epsilon_0}\hat{\boldsymbol{y}}$$

□

Problem 2.19

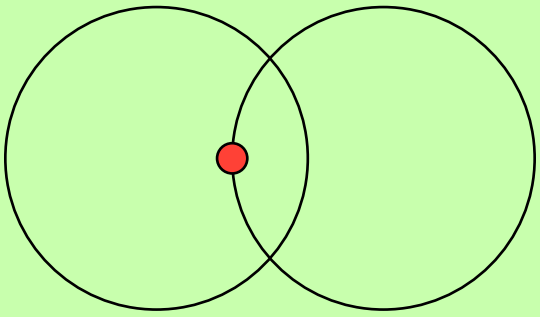
Two spheres, each of radius R and carrying uniform volume charge densities $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap (FIGURE). Call the vector from the positive center to the negative center d. Show that the field in the region of overlap is constant, and find its value. [Hint: Use the answer to Prob. 2.13.]

Solution:

The result of Prob. 2.13 is

$$\mathbf{E} = \frac{\rho a}{3\epsilon_0}\hat{\boldsymbol{r}}$$

Aligning the spheres in the $\hat{\boldsymbol{x}}$-direction and using superposition on the red point below:



Here, the field due to the positive sphere is

$$\mathbf{E}_+ = \frac{\rho(d-R)}{3\epsilon_0}\hat{\boldsymbol{x}}$$

and the field due to the negative sphere is

$$\mathbf{E}_{-} = \frac{\rho R}{3\epsilon_0}\hat{\boldsymbol{x}}$$

Since the field is constant in the overlapping region, it will be equal to the sum of these two contributions at all points in the overlap.

Adding them together gives

$$\boxed{\mathbf{E} = \frac{\rho d}{3\epsilon_0}\hat{\boldsymbol{x}}}$$

Generalizing in any direction simply gives

$$\boxed{\mathbf{E} = \frac{\rho d}{3\epsilon_0}\hat{\boldsymbol{d}}}$$

where $\hat{\boldsymbol{d}}$ is the vector pointing from the center of the positively charged sphere, towards the center of the negatively charged sphere. ! :) □

Problem 2.20

Problem 2.21

Problem 2.22

Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q. Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(\mathbf{r})$.

Solution:

Outside the sphere, where $r > R$:

$$V_{\text{o}} = -\frac{1}{4\pi\epsilon_0}\int_{\infty}^{r}\frac{q}{r^2}\,\mathrm{d}r = \frac{q}{4\pi\epsilon_0 r}$$

Inside the sphere, we have the result above at $r = R$ minus the integral from the sphere surface with $r < R$. The total charge is the ratio between the full sphere with radius R and the sphere with radius r: $q = q\frac{r^3}{R^3}$

$$V_{\mathrm{i}} = \frac{q}{4\pi\epsilon_0 R} - \frac{1}{4\pi\epsilon_0}\int_R^r \frac{q}{r^2}\,\mathrm{d}r =$$

$$\frac{q}{4\pi\epsilon_0 R} - \frac{1}{4\pi\epsilon_0}\int_R^r \frac{qr}{R^3}\,\mathrm{d}r = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R} - \left[\frac{r^2}{2R^3}\right]_R^r\right) =$$

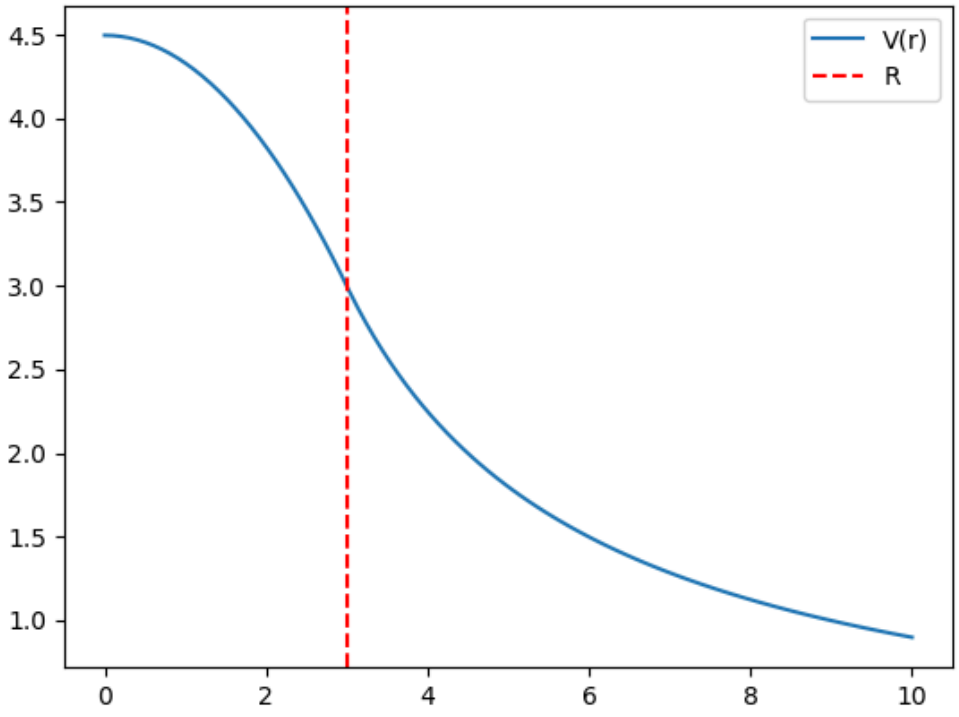
$$= \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R} - \frac{1}{R^3}\left(\frac{r^2 - R^2}{2}\right)\right)$$

The gradient of the potential outside the sphere gives the electric field outside:

$$\mathbf{E}_{\mathrm{o}} = -\boldsymbol{\nabla} V = \frac{q}{4\pi\epsilon_0 r^2}\hat{\boldsymbol{r}}$$

and inside gives the electric field

$$\mathbf{E}_{\mathrm{i}} = -\boldsymbol{\nabla} V = \frac{qr}{4\pi\epsilon_0 R^3}\hat{\boldsymbol{r}}$$



□

Problem 2.23

Find the potential a distance s from an infinitely long straight wire that carries a uniform line charge λ. Compute the gradient of your potential, and check that it yields the correct field.

Solution:

From Gauss' Law, the field is

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\boldsymbol{s}}$$

The potential is then

$$V(\mathbf{s}) = -\int_a^s \frac{\lambda}{2\pi\epsilon_0 s}\,\mathrm{d}s = \frac{\lambda}{2\pi\epsilon_0}[\ln(s)]_s^a =$$
$$= \frac{\lambda}{2\pi\epsilon_0}\ln\left(\frac{a}{s}\right)$$

The gradient of this potential, the electric field, is simply

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{\lambda}{2\pi\epsilon_0 s}\hat{\boldsymbol{s}}$$

□

Problem 2.24

For the charge configuration of Prob. 2.15, find the potential at the center, using infinity as your reference point.

Solution:

The electric field outside is

$$\mathbf{E} = \frac{k(b-a)}{r^2\epsilon_0}, \quad r > b$$

and in the shell

$$\mathbf{E} = \frac{k(r-a)}{r^2\epsilon_0}\hat{\boldsymbol{r}}$$

and zero inside.

The potential from ∞ to b is

$$V_{\infty\to b} = -\int_{\infty}^{b} \frac{k(b-a)}{r^2\epsilon_0}\,\mathrm{d}r = \left[\frac{k(b-a)}{r\epsilon_0}\right]_{\infty}^{b} = \frac{k(b-a)}{b\epsilon_0} = \frac{k}{\epsilon_0} - \frac{ka}{b\epsilon_0}$$

The potential from b to a is

$$V_{b\to a} = -\int_{b}^{a} \frac{k(r-a)}{r^2\epsilon_0}\,\mathrm{d}r = \int_{b}^{a} \frac{ka}{r^2\epsilon_0} - \frac{k}{r\epsilon_0}\,\mathrm{d}r =$$
$$= \left[-\frac{ka}{r\epsilon_0}\right]_{b}^{a} - \left[\frac{k}{\epsilon_0}\ln(r)\right]_{b}^{a} = -\frac{k}{\epsilon_0} + \frac{ka}{b\epsilon_0} + \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)$$

and the potential difference from a to 0 is zero. Adding up the potential difference from $\infty \to b, b \to a$ gives

$$\frac{k}{\epsilon_0} - \frac{ka}{b\epsilon_0} - \frac{k}{\epsilon_0} + \frac{ka}{b\epsilon_0} + \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)$$
$$\downarrow$$
$$V(0) = \frac{k}{\epsilon_0}\ln\left(\frac{b}{a}\right)$$

□

Problem 2.25

Problem 2.26

Problem 2.27

Problem 2.28

Problem 2.29

Problem 2.30

Problem 2.31

(a) Check that the results of Exs. 2.5 and 2.6, and Prob. 2.12, are consistent with

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0}\hat{\boldsymbol{n}} \tag{1.}$$

(b) Use Gauss' law to find the field inside and outside a long hollow cylindrical tube, which carries a uniform surface charge σ. Check that your result is consistent with Equation 1.

(c) Check that the result of Ex. 2.8 is consistent with boundary conditions

$$V_{\text{above}} = V_{\text{below}}$$
$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0}\sigma$$

Ex. 2.5 result

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0}\hat{\boldsymbol{n}}$$

and dividing it up between field above and field below

$$\mathbf{E}_{\text{above}} = \frac{\sigma}{\epsilon_0}\hat{\boldsymbol{n}}$$
$$\mathbf{E}_{\text{below}} = -\frac{\sigma}{\epsilon_0}\hat{\boldsymbol{n}}$$

gives

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0}\hat{\boldsymbol{n}}$$

Problem 2.32

(a) Three charges are situated at the corners of a square (side a), as shown in the figure. How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?

(b) How much work does it take to assemble the whole configuration of four charges?

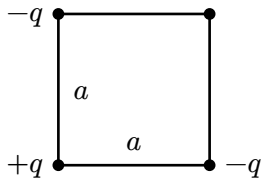


Figure 11:

Solution:

(a) The work required to bring in a charge q_4 is

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{\boldsymbol{r}_{14}} + \frac{q_2}{\boldsymbol{r}_{24}} + \frac{q_3}{\boldsymbol{r}_{34}} \right)$$

where each part is due to the three charges, q_1, q_2, q_3, already in the configuration. The separation distances $\boldsymbol{r}$ are $\boldsymbol{r}_{14} = a, \boldsymbol{r}_{24} = \sqrt{2}a, \boldsymbol{r}_{34} = a$

This gives

$$W_4 = \frac{1}{4\pi\epsilon_0} q^2 \left(-\frac{1}{a} + \frac{1}{\sqrt{2}a} - \frac{1}{a} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-2 + \frac{1}{\sqrt{2}} \right)$$

(b)

$$W_1 = 0$$

$$W_2 = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{\boldsymbol{r}_{12}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} (-1)$$

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{\boldsymbol{r}_{13}} + \frac{q_2}{\boldsymbol{r}_{23}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-1 + \frac{1}{\sqrt{2}} \right)$$

$$W_4 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-2 + \frac{1}{\sqrt{2}} \right)$$

$$W_1 + W_2 + W_3 + W_4 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-1 - 1 + \frac{1}{\sqrt{2}} - 2 + \frac{1}{\sqrt{2}} \right) = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-4 + \sqrt{2} \right)$$

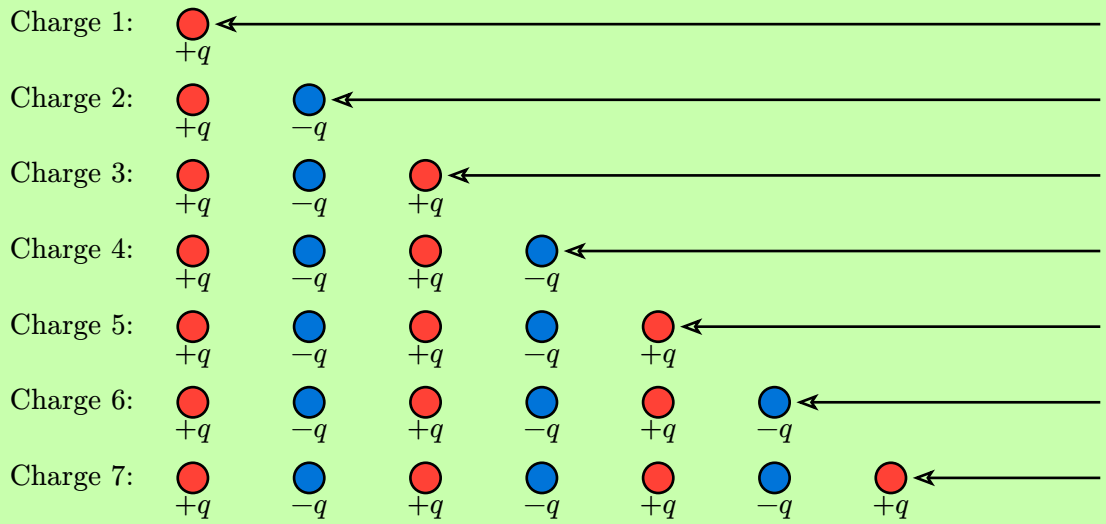
□

Problem 2.33

Problem 2.34

Consider an infinite chain of point charges, $\pm q$ (with alternating signs), strung out along the x axis, each a distance a from its nearest neighbors. Find the work per particle required to assemble this system.

Solution:



Adding charges 1-7 requires the work W_1-W_7:

$$
\begin{aligned}
W_1 &= 0 \\
W_2 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}(-1) \\
W_3 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}\right) \\
W_4 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}\right) \\
W_5 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}\right) \\
W_6 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}\right) \\
W_6 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}\right) \\
W_7 &= \frac{1}{4\pi\epsilon_0}\frac{q^2}{a}\left(-1+\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}-\frac{1}{7}\right)
\end{aligned}
$$

As we keep adding charges, we add the constant $(-1)^n\frac{1}{n}$ which means the Madelung constant is

$$\alpha = \sum_1^\infty \frac{(-1)^n}{n}$$

which is similar to the known Maclaurin series for $\ln(x+1)$:

$$\ln(x+1) = \sum_1^\infty \frac{(-1)^{n+1}}{n}x^n = -\sum_1^\infty \frac{(-1)^n}{n}x^n$$

Setting $x = 1$ and moving the negative sign to the LHS, we get

$$-\ln(2) = \sum_1^\infty \frac{(-1)^n}{n}$$

which is identical to the Madelung constant, which gives

$$\boxed{\alpha = -\ln(2)}$$

□

Problem 2.35

$$W = \frac{1}{2}\int \rho V \, \mathrm{d}\tau \qquad 2.$$

$$W = \frac{\epsilon_0}{2}\left(\int_{\mathcal{V}} E^2\,\mathrm{d}\tau + \oint_{\mathcal{S}} V\mathbf{E}\cdot\mathbf{da}\right) \tag{3.}$$

$$W = \frac{\epsilon_0}{2}\int E^2\,\mathrm{d}\tau \tag{4.}$$

Find the energy stored in a uniformly charged solid sphere of radius R and charge q. Do it three different ways:

(a) Use Equation 2 You found the potential in Prob. 2.22.

(b) Use Equation 4. Don't forget to integrate over all space.

(c) Use Equation 3. Take a spherical volume of radius a. What happens as $a \to \infty$?

Solution:

(a) Using Equation 2 with

$$V = \frac{q}{8\pi\epsilon_0 R}\left(3 - \frac{r^2}{R^2}\right)$$

gives the energy

$$W = \frac{1}{2}\int \frac{\rho q}{8\pi\epsilon_0 R}\left(3r^2 - \frac{r^4}{R^2}\right)\mathrm{d}\tau =$$

$$= \frac{1}{2}\frac{\rho q}{2\epsilon_0 R}\int_0^R \left(3r^2 - \frac{r^4}{R^2}\right)\mathrm{d}r = \frac{\rho q}{4\epsilon_0 R}\left[r^3 - \frac{r^5}{5R^2}\right]_0^R =$$

$$= \frac{\rho q}{4\epsilon_0 R}\left[R^3 - \frac{R^3}{5}\right] = \frac{\rho q}{4}\epsilon_0\left[R^2 - \frac{R^2}{5}\right] = \frac{\rho q}{5\epsilon_0}R^2$$

and with

$$\rho = \frac{q}{\frac{4}{3}\pi R^3}$$

we can simplify the expression to

$$\boxed{W = \frac{3q^2}{20\pi\epsilon_0 R} = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}}$$

(b) The electric field

$$\mathbf{E}_{\mathrm{o}} = \frac{q}{4\pi\epsilon_0 r^2}\hat{\boldsymbol{r}}$$

$$\mathbf{E}_{\mathrm{i}} = \frac{qr}{4\pi\epsilon_0 R^3}\hat{\boldsymbol{r}}$$

Starting with the outside, E^2 becomes $\frac{q^2}{16\pi^2\epsilon_0^2 r^4}$ giving the energy

$$W = \frac{\epsilon_0}{2}\int \frac{q^2}{16\pi^2\epsilon_0^2 r^4}\,\mathrm{d}\tau = \frac{q^2\epsilon_0 4\pi}{32\pi^2\epsilon_0^2}\int_R^\infty \frac{1}{r^2} = \frac{q^2\epsilon_0 4\pi}{32\pi^2\epsilon_0^2}\left[-\frac{1}{r}\right]_R^\infty =$$

$$= \frac{q^2}{8\pi\epsilon_0 R}$$

Doing the same for the inside, E^2 becomes $\frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}$ giving the energy

$$W = \frac{\epsilon_0}{2}\int \frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}\,\mathrm{d}\tau = \frac{4\pi\epsilon_0 q^2}{32\pi^2\epsilon_0^2 R^6}\int_0^R r^4 =$$

$$= \frac{q^2}{8\pi\epsilon_0}\left(\frac{1}{5R}\right)$$

Adding them up gives

$$\boxed{W = \frac{1}{4\pi\epsilon_0}q^2\left(\frac{1}{2R}+\frac{1}{10R}\right) = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}}$$

(c) Using Equation 3 we have the fields

$$E_i^2 = \frac{q^2 r^2}{16\pi^2\epsilon_0^2 R^6}$$

$$E_o^2 = \frac{q^2}{16\pi^2\epsilon_0^2 r^4}$$

$$V_o = \frac{q}{4\pi\epsilon_0 r}$$

This gives the energy

$$W = \frac{\epsilon_0}{2}\left(\frac{4\pi q^2}{16\pi^2\epsilon_0^2 R^6}\int_0^R r^4\,\mathrm{d}r + \frac{4\pi q^2}{16\pi^2\epsilon_0^2}\int_R^a \frac{1}{r^2}\,\mathrm{d}r + \frac{q^2}{16\pi^2\epsilon_0^2}\oint \frac{1}{r}\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi\right)$$

$$\downarrow$$

$$\frac{q^2}{2}\left(\frac{1}{4\pi\epsilon_0 R^6}\int_0^R r^4\,\mathrm{d}r + \frac{1}{4\pi\epsilon_0}\int_R^a \frac{1}{r^2}\,\mathrm{d}r + \frac{1}{4\pi\epsilon_0}\frac{1}{a}\right)$$

where the volume integrals were immediately simplified according to $\int_0^R\int_0^\pi\int_0^{2\pi}\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}\theta = 4\pi\int_0^R \mathrm{d}r$

$$\frac{q^2}{2}\left(\frac{1}{4\pi\epsilon_0 R}\frac{1}{5} - \frac{1}{4\pi\epsilon_0 a} + \frac{1}{4\pi\epsilon_0}\frac{1}{R} + \frac{1}{4\pi\epsilon_0}\frac{1}{a}\right)$$

$$\downarrow$$

$$\boxed{W = \frac{1}{4\pi\epsilon_0}\frac{3q^2}{5R}}$$

□

Problem 2.36

Problem 2.37

Problem 2.38

Problem 2.39

Problem 2.40

Problem 2.41

Problem 2.42

Problem 2.43

Problem 2.44

Problem 2.45

Problem 2.46

Find the electric field at a height z above the center of a square sheet (side length a) carrying a uniform surface charge σ. Check your result for the limiting cases $a \to \infty$ and $z \gg a$. [Answer: $\left(\frac{\sigma}{2\epsilon_0}\right)\left\{\left(\frac{4}{\pi}\right)\tan^{-1}\sqrt{1+\frac{a^2}{2z^2}}-1\right\}$.]

Problem 2.47

If the electric field in some region is given (in spherical coordinates) by the expression

$$\mathbf{E}(\mathbf{r}) = \frac{k}{r}\left[3\hat{\boldsymbol{r}} + 2\sin\theta\cos\theta\sin\varphi\hat{\boldsymbol{\theta}} + \sin\theta\cos\varphi\hat{\boldsymbol{\varphi}}\right],$$

for some constant k, what is the charge density? [Answer: $3k\epsilon_0(1 + \cos 2\theta \sin\varphi)/r^2$.]

Problem 2.48

Find the net force that the southern hemisphere of a uniformly charged solid sphere exerts on the northern hemisphere. Express your answer in terms of the radius R and the total charge Q. [Answer: $(1/4\pi\epsilon_0)(3Q^2/16R^2)$.]

Problem 2.49

An inverted hemispherical bowl of radius R carries a uniform surface charge density σ. Find the potential difference between the "north pole" and the center. [Answer: $(R\sigma/2\epsilon_0)\left(\sqrt{2}-1\right)$.]

Problem 2.50

A sphere of radius R carries a charge density $\rho(r) = kr$ (where k is a constant). Find the energy of the configuration. Check your answer by calculating it in at least two different ways. [Answer: $\pi k^2 R^7/7\epsilon_0$.]

Problem 2.51

The electric potential of some configuration is given by the expression

$$V(\mathbf{r}) = A\frac{e^{-\lambda r}}{r},$$

where A and λ are constants. Find the electric field $\mathbf{E}(\mathbf{r})$, the charge density $\rho(r)$, and the total charge Q. [Answer: $\rho = \epsilon_0 A\big(4\pi\delta^3(\mathbf{r}) - \lambda^2 e^{-\lambda r}/r\big)$.]

Problem 2.52

Find the potential on the rim of a uniformly charged disk (radius R, charge density σ). [Hint: First show that $V = k(\sigma R/\pi\epsilon_0)$, for some dimensionless number k, which you can express as an integral. Then evaluate k analytically, if you can, or by computer.]

Problem 2.53

Two infinitely long wires running parallel to the x-axis carry uniform charge densities $+\lambda$ and $-\lambda$ (see figure).

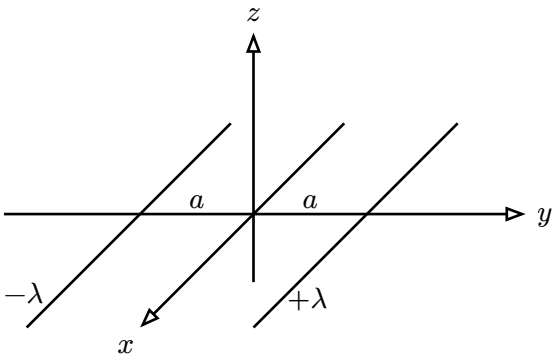


Figure 13:

(a) Find the potential at any point (x, y, z), using the origin as your reference.

(b) Show that the equipotential surfaces are circular cylinders, and locate the axis and radius of the cylinder corresponding to a given potential V_0.

Problem 2.54

In a vacuum diode, electrons "boil" off a hot **cathode**, at potential zero, and accelerate across a gap to the **anode**, which is held at positive potential V_0. The cloud of moving electrons within the gap (called **space charge**) quickly builds up to the point where it reduces the field at the surface of the cathode to zero. From then on, a steady current I flows between the plates.

Suppose the plates are large relative to the separation ($A \gg d^2$ in the figure), so that edge effects can be neglected. Then V, ρ, and v (the speed of the electrons) are all functions of x alone.

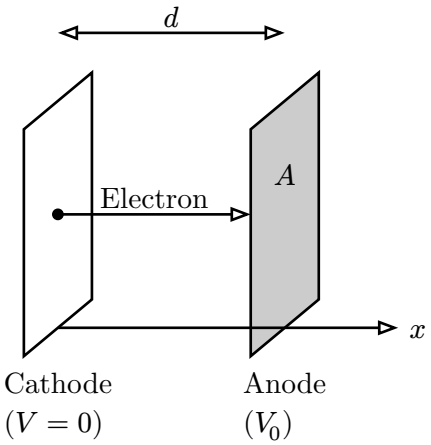


Figure 14:

(a) Write Poisson's equation for the region between the plates.

(b) Assuming the electrons start from rest at the cathode, what is their speed at point x, where the potential is $V(x)$?

(c) In the steady state, I is independent of x. What, then, is the relation between ρ and v?

(d) Use these three results to obtain a differential equation for V, by eliminating ρ and v.

(e) Solve this equation for V as a function of x, V_0, and d. Plot $V(x)$, and compare it to the potential *without* space charge. Also, find ρ and v as functions of x.

(f) Show that

$$I = KV_0^{3/2} \tag{5.}$$

and find the constant K. (Equation 5 is called the **Child–Langmuir law**. It holds for other geometries as well, whenever space charge limits the current. Notice that the space-charge-limited diode is *nonlinear* – it does not obey Ohm's law.)

Problem 2.55

Imagine that new and extraordinarily precise measurements have revealed an error in Coulomb's law. The *actual* force of interaction between two point charges is found to be

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{\imath^2}\left(1 + \frac{\imath}{\lambda}\right)e^{-\imath/\lambda}\hat{\boldsymbol{\imath}},$$

where λ is a new constant of nature (it has dimensions of length, obviously, and is a huge number – say half the radius of the known Universe – so the correction is small, which is why no one ever noticed the discrepancy before). You are charged with the task of reformulating electrostatics to accommodate the new discovery. Assume the principle of superposition still holds.

(a) What is the electric field of a charge distribution ρ (replacing Eq. 2.8)?

(b) Does this electric field admit a scalar potential? Explain briefly how you reached your conclusion. (No formal proof necessary – just a persuasive argument.)

(c) Find the potential of a point charge q – the analog to Eq. 2.26. (If your answer to (b) was "no," better go back and change it!) Use ∞ as your reference point.

(d) For a point charge q at the origin, show that

$$\oint_{\mathcal{S}} \mathbf{E}\cdot d\mathbf{a} + \frac{1}{\lambda^2}\int_{\mathcal{V}} V\, d\tau = \frac{1}{\epsilon_0}q,$$

where $\mathcal{S}$ is the surface and $\mathcal{V}$ the volume of any sphere centered at q.

(e) Show that this result generalizes:

$$\oint_{\mathcal{S}} \mathbf{E}\cdot d\mathbf{a} + \frac{1}{\lambda^2}\int_{\mathcal{V}} V\, d\tau = \frac{1}{\epsilon_0}Q_{\text{enc}},$$

for *any* charge distribution. (This is the next best thing to Gauss's law, in the new "electrostatics.")

(f) Draw the triangle diagram (like Fig. 2.35) for this world, putting in all the appropriate formulas. (Think of Poisson's equation as the formula for ρ in terms of V, and Gauss's law (differential form) as an equation for ρ in terms of $\mathbf{E}$.)

(g) Show that *some* of the charge on a conductor distributes itself (uniformly!) over the volume, with the remainder on the surface. [Hint: $\mathbf{E}$ is still zero, inside a conductor.]

Problem 2.56

Suppose an electric field $\mathbf{E}(x, y, z)$ has the form

$$E_x = ax, \quad E_y = 0, \quad E_z = 0,$$

where a is a constant. What is the charge density? How do you account for the fact that the field points in a particular direction, when the charge density is uniform? [This is a more subtle problem than it looks, and worthy of careful thought.]

Problem 2.57

All of electrostatics follows from the $1/r^2$ character of Coulomb's law, together with the principle of superposition. An analogous theory can therefore be constructed for Newton's law of universal gravitation. What is the gravitational energy of a sphere, of mass M and radius R, assuming the density is uniform? Use your result to estimate the gravitational energy of the Sun (look up the relevant numbers). Note that the energy is *negative* – masses *attract*, whereas (like) electric charges *repel*. As matter "falls in," to create the Sun, its kinetic energy is converted into other forms (typically thermal), and is eventually released as radiation. The Sun radiates at a rate of 3.86×10^{26} W; if all this came from gravitational energy, how long would the Sun last? [The Sun is in fact much older than that, so evidently this is *not* the source of its energy.]

Problem 2.58

We know that the charge on a conductor goes to the surface, but just how it distributes itself there is not easy to determine. One famous example in which the surface charge density can be calculated explicitly is the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

In this case,

$$\sigma = \frac{Q}{4\pi abc}\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right)^{-1/2} \tag{6.}$$

where Q is the total charge. By choosing appropriate values for a, b, and c, obtain (from Equation 6) (a) the net (both sides) surface charge density $\sigma(r)$ on a circular disk of radius R; (b) the net surface charge density $\sigma(x)$ on an infinite conducting "ribbon" in the xy plane, which straddles the y-axis from $x = -a$ to $x = a$ (let Λ be the total charge per unit length of ribbon); (c) the net charge per unit length $\lambda(x)$ on a conducting "needle," running from $x = -a$ to $x = a$. In each case, sketch the graph of your result.

Problem 2.59

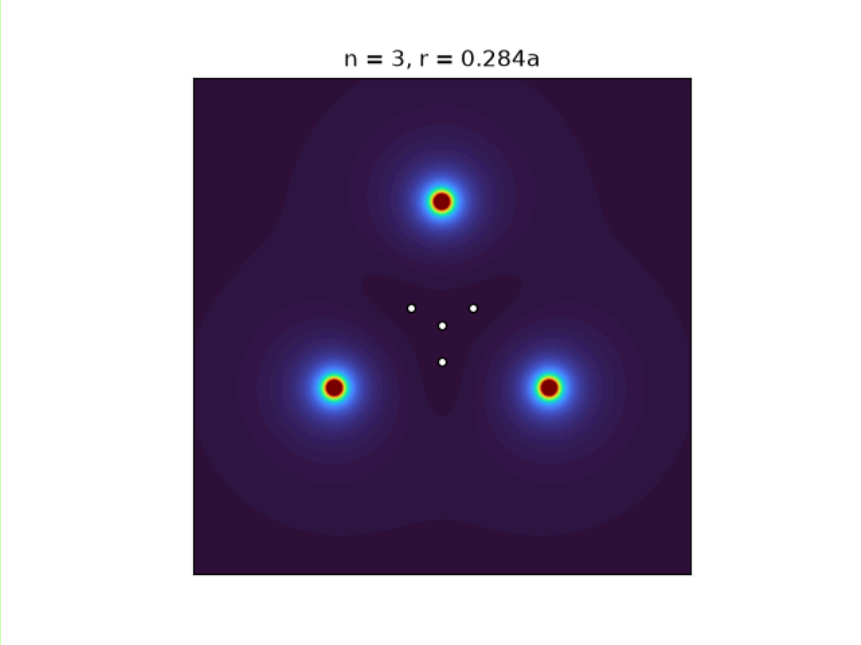
(a) Consider an equilateral triangle, inscribed in a circle of radius a, with a point charge q at each vertex. The electric field is zero (obviously) at the center, but (surprisingly) there are three

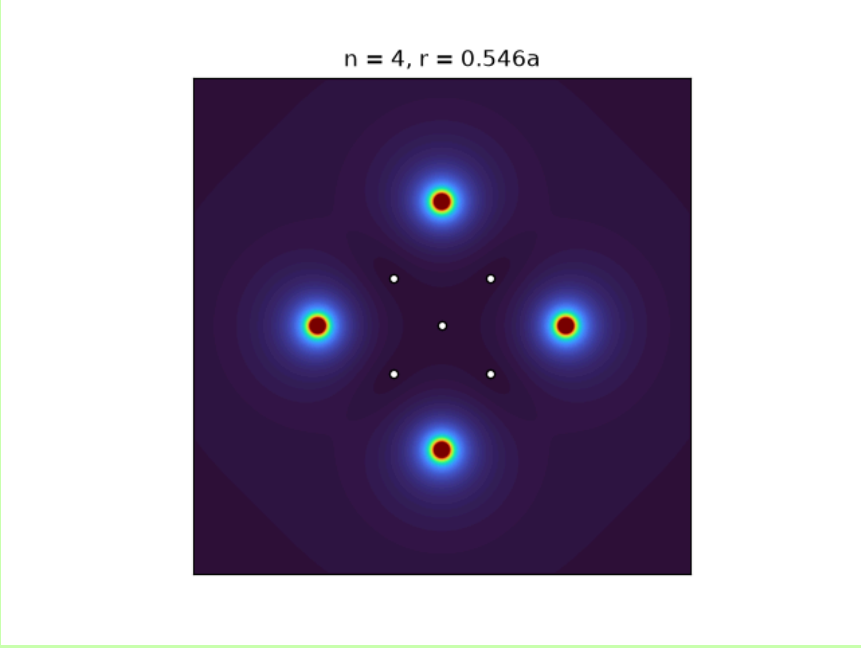
other points inside the triangle where the field is zero. Where are they? [Answer: $r = 0.285a$ – you'll probably need a computer to get it.]

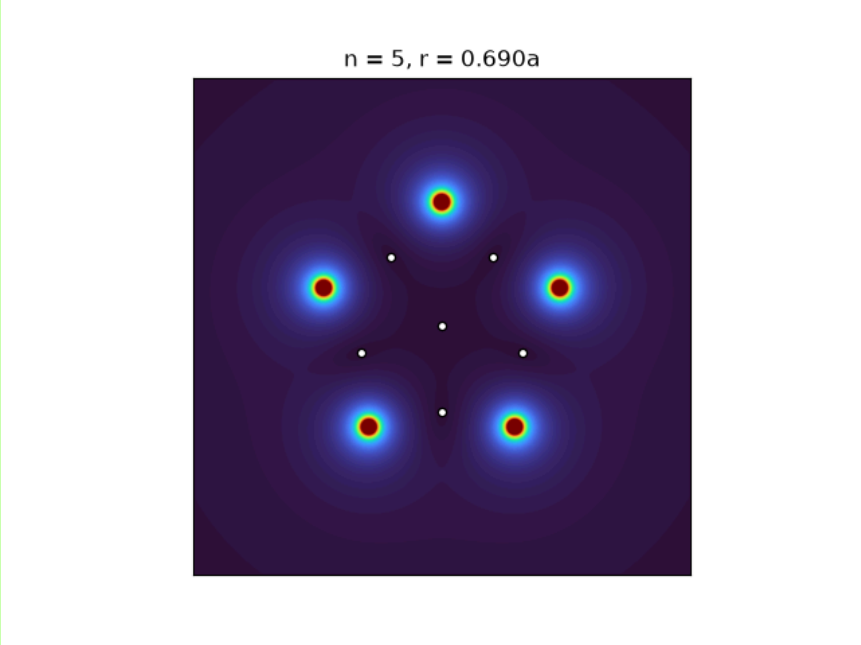
(b) For a regular n-sided polygon there are n points (in addition to the center) where the field is zero. Find their distance from the center for $n = 4$ and $n = 5$. What do you suppose happens as $n \to \infty$?

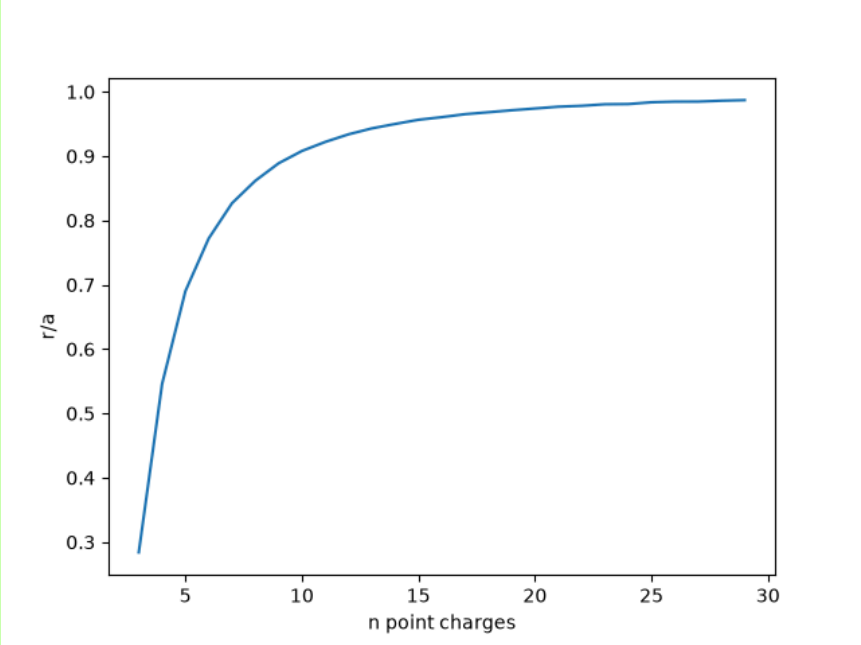
Solution:

The white parts of the plots are where the field is at its minimum, i.e. almost 0. For n charges, n additional points with $\mathbf{E} = 0$.









□

Problem 2.60

Prove or disprove (with a counterexample) the following:

Theorem. Suppose a conductor carrying a net charge Q, when placed in an external electric field $\mathbf{E}_e$, experiences a force $\mathbf{F}$; if the external field is now reversed ($\mathbf{E}_e \to -\mathbf{E}_e$), the force also reverses ($\mathbf{F} \to -\mathbf{F}$).

What if we stipulate that the external field is *uniform*?

Problem 2.61

A point charge q is at the center of an uncharged spherical conducting shell, of inner radius a and outer radius b. *Question:* How much work would it take to move the charge out to infinity (through a tiny hole drilled in the shell)? [Answer: $(q^2/8\pi\epsilon_0)(1/a - 1/b)$.]

Problem 2.62

What is the minimum-energy configuration for a system of N equal point charges placed on or inside a circle of radius R? Because the charge on a conductor goes to the surface, you might think the N charges would arrange themselves (uniformly) around the circumference. Show (to the contrary) that for $N = 12$ it is better to place 11 on the circumference and 1 at the center. How about for $N = 11$ (is the energy lower if you put all 11 around the circumference, or if you put 10 on the circumference and 1 at the center)? [Hint: Do it numerically – you'll need at least 4 significant digits. Express all energies as multiples of $q^2/4\pi\epsilon_0 R$.]

Problem 2.63

(a) Find the potential on the rim of a uniformly charged circular disk of radius r and charge density σ. [Hint: Chop the disk into circular arcs of radius $\varkappa$ and width $\mathrm{d}\varkappa$, running from $-\theta$ to $+\theta$, and integrate over θ.]

(b) Use your result in (a) to calculate the energy of a uniformly charged disk of radius R and total charge Q.

(c) A circular *conducting* disk of radius R carries a total charge Q. In Prob. 2.58a you showed that the charge density is

$$\sigma(r) = \frac{Q}{2\pi R}\frac{1}{\sqrt{R^2 - r^2}}.$$

Find the potential of the disk.

(d) Use your result in (c) to calculate the energy of a conducting disk of radius R and total charge Q. Compare your answer to (b); which is smaller, and why?

Problem 2.64

Many programming languages include packages for drawing field diagrams (like Fig. 2.12a). You input $\mathbf{E}(x, y, z)$, and the program generates a figure showing arrows representing the field at an array of representative points. In Mathematica it's called `VectorPlot`. Suppose, for example, you want to sketch the field of a point charge q at the origin; in the xy plane the field is

Problem 2.65

Problem 2.66

Problem 2.67

Chapter 3

Special Techniques

Problem 3.1

Find the average potential over a spherical surface of radius R due to a point charge q located inside (same as above, in other words, only with $z < R$). (In this case, of course, Laplace's equation does not hold within the sphere.) Show that, in general,

$$V_{\text{ave}} = V_{\text{center}} + \frac{Q_{\text{enc}}}{4\pi\epsilon_0 R}$$

where V_{center} is the potential at the center due to all the external charges, and Q_{enc} is the total enclosed charge.

Solution:

$$\boldsymbol{\imath} = R\hat{\mathbf{r}} - z\hat{\mathbf{z}}$$

$$\imath = \sqrt{R^2 + z^2 - 2Rz\cos\theta}$$

$$V_{\text{ave}} = \frac{1}{4\pi R^2}\frac{q}{4\pi\epsilon_0}\int \frac{1}{\sqrt{R^2 + z^2 - 2Rz\cos\theta}} R^2 \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi$$

$$\downarrow$$

$$V_{\text{ave}} = \frac{qR^2 2\pi}{4\pi R^2 4\pi\epsilon_0}\left[\frac{\sqrt{R^2 + z^2 - 2Rz\cos\theta}}{Rz}\right]_0^{\pi}$$

$$\downarrow$$

$$V_{\text{ave}} = \frac{q}{8\pi\epsilon_0 Rz}[(R+z) - (R-z)]$$

$$\downarrow$$

$$\boxed{V_{\text{ave}} = \frac{q}{4\pi\epsilon_0 R}}$$

Superposition gives that if there are more charges inside the sphere, V_{ave} is just the total charge (Q_{enc}) and therefore the general expression for V_{ave} is the contribution from chares outside (V_{center}) and the contribution from charges inside (V_{inside})

□

Problem 3.2

In one sentence, justify **Earnshaw's Theorem**: A charged particle cannot be held in a stable equilibrium by electrostatic forces alone. As an example, consider the cubical arrangement of fixed charges in Fig. 3.4. It looks, off hand, as though a positive charge at the center would be suspended in midair, since it is repelled away from each corner. Where is the leak in this "electrostatic bottle"? [To harness nuclear fusion as a practical energy source it is necessary to heat a plasma (soup of charged particles) to fantastic temperatures—so hot that contact would vaporize any ordinary pot. Earnshaw's theorem says that electrostatic containment is also out of the question. Fortunately, it is possible to confine a hot plasma magnetically.]

Solution:

There is no local maximum or minimum point in the solution to Laplace's equation. The apparent equilibrium point in the example must therefore be a saddle point. □

Problem 3.3

Find the general solution to Laplace's equation in spherical coordinates, for the case where V depends only on r. Do the same for cylindrical coordinates, assuming V depends only on s.

Solution:

$$\boldsymbol{\nabla}^2 V = 0$$

The laplacian in spherical coordinates with only r-dependence is

$$\boldsymbol{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)$$

$$\downarrow$$

$$\boldsymbol{\nabla}^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) = \frac{1}{r^2}\left(2r\frac{\partial V}{\partial r} + r^2\frac{\partial^2 V}{\partial r^2}\right) = \frac{\partial^2 V}{\partial r^2} + \frac{2}{r}\frac{\partial V}{\partial r} = 0$$

This is Bessel diff.eq with $\lambda = 0, l = 0$

$$V = a + \frac{b}{r}$$

For the cylindrical part, the laplacian is

$$\nabla^2 V = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial V}{\partial s}\right) = [\text{multiply by s and integrate}] = A$$

$$s\frac{\partial V}{\partial s} = A \rightarrow \frac{\partial V}{\partial s} = \frac{A}{s} \rightarrow V = A\ln(s) + B$$

□

Problem 3.4

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Problem 3.65

Chapter 4

Electric Fields in Matter

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Chapter 5

Magnetostatics

Problem 5.1

Problem 5.2

Problem 5.3

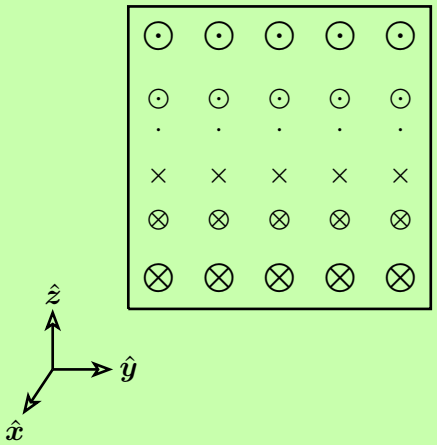
Problem 5.4

Suppose that the magnetic field in some region has the form

$$B = kz\hat{\boldsymbol{x}}$$

(where k is a constant). Find the force on a square loop (side a), lying in the yz plane and centered at the origin, if it carries a current I , flowing counterclockwise, when you look down the x axis.

Solution:



The force on a wire carrying a current $\mathbf{I}$ is

$$\mathbf{F} = \int I(\mathrm{d}\mathbf{l} \times \mathbf{B})$$

With current I, the forces on each of the four lines are, starting with the top line:

$$z = \frac{a}{2}, \quad y : \frac{a}{2} \to -\frac{a}{2}$$

$$B = k\frac{a}{2}\hat{\boldsymbol{x}}, \quad \mathrm{d}\mathbf{l} = \hat{\boldsymbol{y}}\,\mathrm{d}y$$

$$\mathbf{F} = Ik\frac{a}{2}\int_{\frac{a}{2}}^{-\frac{a}{2}} (\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}})\,\mathrm{d}y = -Ik\frac{a}{2}\hat{\boldsymbol{z}}\int_{\frac{a}{2}}^{-\frac{a}{2}} \mathrm{d}y = Ik\frac{a}{2}\hat{\boldsymbol{z}}\int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$

$$\downarrow$$

$$\mathbf{F} = \frac{Ika^2}{2}\hat{\boldsymbol{z}}$$

Bottom line:

$$B = -k\frac{a}{2}\hat{\boldsymbol{x}}, \quad \mathrm{d}\mathbf{l} = \hat{\boldsymbol{y}}\,\mathrm{d}y, y : -\frac{a}{2} \to \frac{a}{2}$$

$$\mathbf{F} = -Ik\frac{a}{2}\int_{-\frac{a}{2}}^{\frac{a}{2}} (\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}})\,\mathrm{d}y = Ik\frac{a}{2}\hat{\boldsymbol{z}}\int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$

$$\downarrow$$

$$\mathbf{F} = \frac{Ika^2}{2}\hat{\boldsymbol{z}}$$

The forces on the left and right wires are zero due to symmetry (there is equal parts of these wires in a $+\hat{\boldsymbol{x}}$ magnetic field as $-\hat{\boldsymbol{x}}$ magnetic field).

Total is top + Bottom

$$\mathbf{F}_{\text{tot}} = \mathbf{F}_{\text{top}} + \mathbf{F}_{\text{bot}} = \frac{Ika^2}{2}\hat{\boldsymbol{z}} + \frac{Ika^2}{2}\hat{\boldsymbol{z}} = \boxed{Ika^2\hat{\boldsymbol{z}}}$$

□

Problem 5.5

A current I flows down a wire of radius a.

(a) If it is uniformly distributed over the surface, what is the surface current density K ?

(b) If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is $J(s)$?

Solution:

(a) A wire of radius a has circumference $2\pi a$ so the current passing through each infitesimal part of the surface of the wire (circumference) is simply

$$K = \frac{I}{2\pi a}$$

(b) We want the volume current density to be

$$J = \frac{k}{s}$$

where k is the proportionality constant. If we were to integrate $J(s)$ over a cross section of the wire, we should recover I

$$I = \int J(s) = \int_0^a \int_0^{2\pi} \frac{k}{s} s \, \mathrm{d}s \, \mathrm{d}\varphi = \int_0^a \int_0^{2\pi} k \, \mathrm{d}s \, \mathrm{d}\varphi \rightarrow I = ka2\pi$$

$$\downarrow$$

$$k = \frac{I}{2\pi a}$$

so

$$J(s) = \frac{I}{2\pi a s}$$



□

Problem 5.6

Problem 5.7

Problem 5.8

(a) Find the magnetic field at the center of a square loop, which carries a steady current I . Let R be the distance from center to side.

(b) Find the field at the center of a regular n-sided polygon, carrying a steady current I. Again, let R be the distance from the center to any side.

(c) Check that your formula reduces to the field at the center of a circular loop, in the limit $n \to \infty$.

Solution:

(a) We can find the field produced by one of the sides, and due to symmetry the result is four times of that. Calculating the right wire gives

$$x = R, \quad y : -R \to R, \quad \mathrm{d}\boldsymbol{l}' = \hat{\boldsymbol{y}}\,\mathrm{d}y$$
$$\boldsymbol{\imath} = -R\hat{\boldsymbol{x}} - y\hat{\boldsymbol{y}}, \quad \imath = \sqrt{R^2 + y^2}$$
$$\hat{\boldsymbol{\imath}} = \frac{\boldsymbol{\imath}}{\imath}, \quad \mathrm{d}\boldsymbol{l}' \times \hat{\boldsymbol{\imath}} = \frac{R}{\sqrt{R^2 + y^2}^3}\,\mathrm{d}y\hat{\boldsymbol{z}}$$

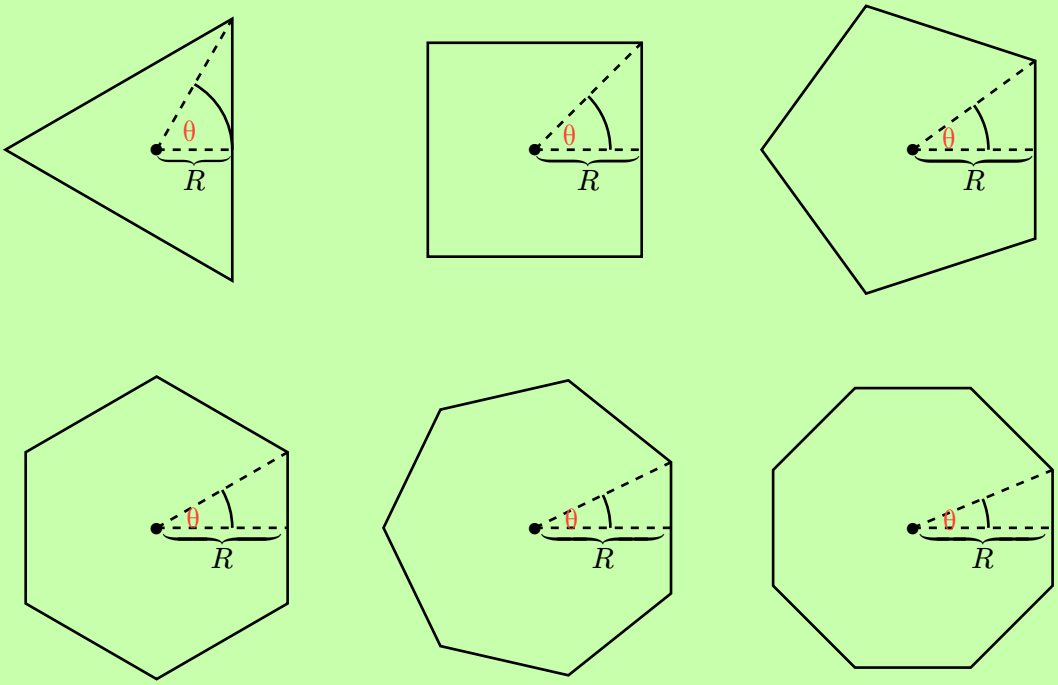
Using these expressions, Biot-Savart law becomes

$$\mathbf{B} = \frac{\mu_0 I}{4\pi}\hat{\boldsymbol{z}}\int_{-R}^{R}\frac{R}{\sqrt{R^2 + y^2}^3}\,\mathrm{d}y = \frac{\mu_0 I}{4\pi}\hat{\boldsymbol{z}}\left[\frac{Ry}{R^2\sqrt{R^2 + y^2}}\right]_{-R}^{R} =$$
$$= \frac{2\mu_0 I}{4\pi R\sqrt{2}}\hat{\boldsymbol{z}} = \frac{\sqrt{2}\mu_0 I}{4\pi R}\hat{\boldsymbol{z}}$$

Multiplying this result with 4 (since there are 4 straight wire segments) gives

$$\mathbf{B}(\mathbf{0}) = \frac{\sqrt{2}\mu_0 I}{\pi R}\hat{\boldsymbol{z}}$$

(b) For a n-sided polygon, the same principle holds, where if we can find the field from one of the straight segments, the total field is given by multiplying this result with n.



The total length of the straight right edges for all polygons is

$$L = 2R\tan(\theta), \quad \theta = \frac{\pi}{n}$$

so Biot-Savart becomes

$$\frac{R\mu_0 I}{4\pi}\hat{z}\int_{-R\tan\left(\frac{\pi}{n}\right)}^{R\tan\left(\frac{\pi}{n}\right)} \frac{1}{\sqrt{R^2+y^2}^3}\,\mathrm{d}y = \frac{R\mu_0 I}{4\pi}\left[\frac{y}{R^2\sqrt{R^2+y^2}}\right]_{-R\tan\left(\frac{\pi}{n}\right)}^{R\tan\left(\frac{\pi}{n}\right)} =$$

$$= \frac{\mu_0 I\tan\left(\frac{\pi}{n}\right)}{2\pi R\sqrt{1+\tan^2\left(\frac{\pi}{n}\right)}}\hat{z}$$

Now, $\tan(\theta) = \sin(\theta)/\cos(\theta)$ and $\sqrt{1+\tan^2(\theta)} = \sec(\theta) = 1/\cos(\theta)$, so the field from the right line segment is

$$\mathbf{B} = \frac{\mu_0 I\sin\left(\frac{\pi}{n}\right)}{2\pi R}\hat{z}$$

Multiplying by the amount of sides for the polygon, n, gives the total magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2R}\frac{\sin\left(\frac{\pi}{n}\right)n}{\pi}\hat{z} = \frac{\mu_0 I}{2R}\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}\hat{z}$$

(c) As $n\to\infty$, the term $\frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \to 1$ (since $\frac{\pi}{n}\to 0$ as $n\to\infty$, and we have the standard limit $\frac{\sin(x)}{x}\to 1$ as $x\to 0$). This gives the magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2R}\hat{z}$$

which is exactly the field given by a circular loop of radius R.

□

Problem 5.9

Problem 5.10

Problem 5.11

Problem 5.12

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Problem 5.14

A steady current I flows down a long cylindrical wire of radius a (Fig. 5.40). Find the magnetic field, both inside and outside the wire, if

(a) The current is uniformly distributed over the outside surface of the wire.

(b) The current is distributed in such a way that J is proportional to s, the distance from the axis.

Solution:

Amperian loop

$$\oint \mathbf{B} \cdot \mathrm{dl} = \mu_0 I_{\mathrm{enc}}$$

$$Bs2\pi = \mu_0 Ia$$

$$\downarrow$$

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\varphi}} \text{ outside}$$

$$\mathbf{B} = 0 \text{ inside (all current on the surface)}$$

(b)

$$B2\pi s = \mu_0 2\pi k s^3 \frac{1}{3}$$

$$B = \mu_0 k \frac{s^2}{3}$$

Now just find k

$$J = ks \to I = \int kss \, \mathrm{d}s \mathrm{d}\varphi = \frac{2\pi k s^3}{3} \to k = \frac{3I}{2\pi a^3}$$

and the final answer is

$$\mathbf{B} = \frac{\mu_0 I s^2}{2\pi a^3}\hat{\boldsymbol{\varphi}} \text{ inside}$$

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\varphi}} \text{ outside}$$

□

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Problem 5.28

(a) Check that Eq. 5.65 is consistent with Eq. 5.63, by applying the divergence.

(b) Check that Eq. 5.65 is consistent with Eq. 5.47, by applying the curl.

(c) Check that Eq. 5.65 is consistent with Eq. 5.64, by applying the Laplacian.

5.65:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{\imath} \, \mathrm{d}\tau'$$

5.63:

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

(a) Applying the divergence gives

$$\boldsymbol{\nabla} \cdot \mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}(\mathbf{r}')}{\imath} \right) \mathrm{d}\tau'$$

We will use the product rule

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath} \right) = \frac{\imath (\boldsymbol{\nabla} \cdot \mathbf{J}) - \mathbf{J} \cdot (\boldsymbol{\nabla} \imath)}{\imath^2}$$

where we have $\boldsymbol{\nabla} \cdot \mathbf{J} = 0$ (magnetostatics, steady currents) we are left with

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Chapter 6

Magnetic Fields in Matter

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Chapter 7

Electrodynamics

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Problem 7.69

Chapter 8

Conservation Laws

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Chapter 9

Electromagnetic Waves

Problem 9.1

By explicit differentiation, check that the functions $f_1, f_2,$ and f_3 satisfy the wave equation. Show that f_4 and f_5 do not.

$$f_1(z,t) = Ae^{-b(z-vt)^2}$$
$$f_2(z,t) = A\sin[b(z-vt)]$$
$$f_3(z,t) = \frac{A}{b(z-vt)^2+1}$$
$$f_4(z,t) = Ae^{-b(bz^2+vt)}$$
$$f_5(z,t) = A\sin(bz)\cos(bvt)^3$$

Solution:

The wave equation is

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}.$$

f_1:

$$\frac{\partial f_1}{\partial z} = -2bA(z-vt)e^{-b(z-vt)^2}, \quad \frac{\partial^2 f_1}{\partial z^2} = 4b^2A(z-vt)^2e^{-b(z-vt)^2}$$
$$\frac{\partial f_1}{\partial t} = 2vbA(z-vt)e^{-b(z-vt)^2}, \quad \frac{\partial^2 f_1}{\partial t^2} = 4v^2b^2A(z-vt)^2e^{-b(z-vt)^2}$$

$$\frac{\partial^2 f_1}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_1}{\partial t^2}$$
$$\downarrow$$
$$4b^2A(z-vt)^2e^{-b(z-vt)^2} = \frac{1}{v^2}4v^2b^2A(z-vt)^2e^{-b(z-vt)^2}\checkmark$$

f_2:

$$\frac{\partial f_2}{\partial z} = bA\cos[b(z-vt)], \qquad \frac{\partial^2 f_2}{\partial z^2} = -b^2 A\sin[b(z-vt)]$$
$$\frac{\partial f_2}{\partial t} = -vbA\cos[b(z-vt)], \qquad \frac{\partial^2 f_2}{\partial t^2} = -v^2 b^2 A\sin[b(z-vt)]$$

$$\frac{\partial^2 f_2}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_2}{\partial t^2}$$
$$\downarrow$$
$$-b^2 A\sin[b(z-vt)] = -\frac{1}{v^2}v^2 b^2 A\sin[b(z-vt)]\checkmark$$

f_3 using quotient rule:

$$\frac{\partial f_3}{\partial z} = \frac{-2bA(z-vt)}{(b(z-vt)^2+1)^2}$$
$$\frac{\partial^2 f_3}{\partial z^2} = \frac{-2bA}{(b(z-vt)^2+1)^2} - \frac{-2bA(z-vt)2(b(z-vt)^2+1)(2b(z-vt))}{(b(z-vt)^2+1)^4} =$$
$$= \frac{-2bA}{(b(z-vt)^2+1)^2} + \frac{8b^2A(z-vt)^2}{(b(z-vt)^2+1)^3}$$
$$\frac{\partial f_3}{\partial t} = \frac{2bvA(z-vt)}{(b(z-vt)^2+1)^2}$$
$$\frac{\partial^2 f_3}{\partial t^2} = \frac{-2bv^2A}{(b(z-vt)^2+1)^2} - \frac{2bvA(z-vt)2(b(z-vt)^2+1)(-2bv(z-vt))}{(b(z-vt)^2+1)^4} =$$
$$= \frac{-2bv^2A}{(b(z-vt)^2+1)^2} + \frac{8b^2v^2A(z-vt)^2}{(b(z-vt)^2+1)^3}$$

$$\frac{\partial^2 f_3}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f_3}{\partial t^2}\checkmark$$

f_4:

$$\frac{\partial f_4}{\partial z} = -2b^2 zAe^{-b(bz^2+vt)}, \qquad \frac{\partial^2 f_4}{\partial z^2} = (-2b^2+4b^4z^2)\left(Ae^{-b(bz^2+vt)}\right)$$
$$\frac{\partial f_4}{\partial t} = -bvAe^{-b(bz^2+vt)}, \qquad \frac{\partial^2 f_4}{\partial t^2} = b^2v^2Ae^{-b(bz^2+vt)}$$

$$\frac{\partial^2 f_4}{\partial z^2} \neq \frac{1}{v^2}\frac{\partial^2 f_4}{\partial t^2}\text{X}$$

f_5:

$$\frac{\partial f_5}{\partial z} = bA\cos(bz)\cos(bvt)^3, \quad \frac{\partial^2 f_5}{\partial z^2} = -b^2 A\sin(bz)\cos(bvt)^3$$
$$\frac{\partial f_5}{\partial t} = -3(b^3v^3t^2)\sin(bz)\sin(bvt)^3$$
$$\frac{\partial^2 f_5}{\partial t^2} = -9(b^6v^6t^4)\sin(bz)\cos(bvt)^3 - 6(b^3v^3t)\sin(bz)\sin(bvt)^3$$

$$\frac{\partial^2 f_5}{\partial z^2} \neq \frac{1}{v^2}\frac{\partial^2 f_5}{\partial t^2} \text{✗}$$

□

Problem 9.2

Show that the **standing wave** $f(z,t) = A\sin(kz)\cos(kvt)$ satisfies the wave equation, and express it as the sum of a wave traveling to the left and a wave traveling to the right.

Solution:

$$\frac{\partial^2 f}{\partial z^2} = -k^2 A\sin(kz)\cos(kvt)$$
$$\frac{\partial^2 f}{\partial t^2} = -v^2k^2 A\sin(kz)\cos(kvt)$$

The wave equation

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2}$$

is satisfied. Using the trigonometric identity

$$\sin(a)\cos(b) = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$$

we obtain

$$\frac{A}{2}[\sin(k(z+vt)) + \sin(k(z-vt))]$$

which is the sum of a wave traveling to the right and left. □

Problem 9.3

Problem 9.4

Problem 9.5

Problem 9.6

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Problem 9.42

Problem 9.43

Problem 9.44

Chapter 10

Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2\boldsymbol{A} - \boldsymbol{\nabla}L = -\mu_0\boldsymbol{J}$$

where

$$\Box^2 \equiv \boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \boldsymbol{\nabla}^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \boldsymbol{\nabla}^2 V + \frac{\partial}{\partial t}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

and for $\boldsymbol{A}$:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

□

Problem 10.2

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\boldsymbol{r},t) = 0, \quad \boldsymbol{A}(\boldsymbol{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The **E** and **B** fields are

$$\mathbf{E} = -\boldsymbol{\nabla}V - \frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$$

The derivative of **A** with respect to time is

$$\frac{\partial\mathbf{A}}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}},$$

while the curl of **A** in spherical coordinates is

$$\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_\phi) - \frac{\partial A_\theta}{\partial\phi}\right]\hat{\boldsymbol{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rv_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}.$$

Since $A_\phi = A_\theta = 0$, and $\frac{\partial A_r}{\partial\phi} = \frac{\partial A_r}{\partial\theta} = 0$, the curl of **A** is zero.

(a)

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

(b) Using the gauge function, V and $\mathbf{A}$ are transformed into

$$V'(\mathbf{r}, t) = V(\mathbf{r}, t) - \frac{\partial \lambda}{\partial t}$$
$$\mathbf{A}'(\mathbf{r}, t) = \mathbf{A}(\mathbf{r}, t) + \boldsymbol{\nabla}\lambda$$

Writing out the derivative and gradient of λ

$$\frac{\partial \lambda}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\boldsymbol{\nabla}\lambda = \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

gives the transformed potentials

$$V'(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$
$$\mathbf{A}'(\mathbf{r}, t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}} + \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2} = 0$$

$\mathbf{E}$ is now given by $-\boldsymbol{\nabla}V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\boldsymbol{\nabla} \times \mathbf{A} = \boldsymbol{\nabla} \times 0 = 0$.

$$\mathbf{E} = -\boldsymbol{\nabla}V = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\boldsymbol{r}}$$
$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$, where $A_0, \omega,$ and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k?

Solution:

Maxwell's equations in vacuum are

$$\text{(i)}\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho \qquad \text{(iii)}\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\text{(ii)}\nabla \cdot \mathbf{B} = 0 \qquad \text{(iv)}\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

where as before we set

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \nabla \times \mathbf{A}$$

This gives

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} = -\omega A_0 \cos(kx - \omega t)\hat{\boldsymbol{y}}$$
$$\mathbf{B} = \frac{\partial A_y}{\partial x}\hat{\boldsymbol{z}} = kA_0 \cos(kx - \omega t)\hat{\boldsymbol{z}}$$

Since $\rho = 0$, this directly gives $\nabla \cdot \mathbf{E} = 0$. (ii) gives $\nabla \cdot \mathbf{B} = 0$.

From (iv) we have the relation, since the current density $\mathbf{J} = 0$,

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{B} = -\frac{\partial B_z}{\partial x}\hat{\boldsymbol{y}} = k^2 A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$
$$\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \epsilon_0 \omega^2 A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$

From this we see that $k^2 = \mu_0 \epsilon_0 \omega^2 \rightarrow k = \sqrt{\mu_0 \epsilon_0}\omega \rightarrow k = \frac{\omega}{c}$

From (iii) we have the relation

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\downarrow$$
$$\nabla \times \mathbf{E} = \frac{\partial E_y}{\partial x}\hat{\boldsymbol{z}} = \omega k A_0 \sin(kx - \omega t)\hat{\boldsymbol{z}}$$
$$-\frac{\partial \mathbf{B}}{\partial t} = \omega k A_0 \sin(kx - \omega t)\hat{\boldsymbol{z}}$$

Equations (i)-(iv) are all verified with the condition $k = \frac{\omega}{c}$.

□

Problem 10.5

Which of the potentials

$$\text{(i) } V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\boldsymbol{z}}, & \text{for } |x| < ct \\ 0 & \text{for } |x| > ct \end{cases}$$

$$\text{(ii) } V = 0, \quad \mathbf{A} = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}}$$

$$\text{(iii) } V = 0, \quad \mathbf{A} = A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}$$

are in the Coulomb gauge? Which are in the Lorenz gauge?

Solution:

The Coulomb gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

and the Lorenz gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Starting with checking the Coulomb gauge, we get

$$\text{(i) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(\frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\boldsymbol{z}} \right) = 0$$

$$\text{(ii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(-\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\boldsymbol{r}} \right) = -\frac{qt}{4\pi\epsilon_0} \boldsymbol{\nabla} \cdot \left(\frac{\hat{\boldsymbol{r}}}{r^2} \right) = -\frac{qt}{4\pi\epsilon_0}\delta^3(\mathbf{r})$$

$$\text{(iii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot (A_0 \sin(kx - \omega t)\hat{\boldsymbol{y}}) = 0$$

(i) and (iii) are in the Coulomb gauge, but not (ii). Next, checking Lorenz gauge we see that the divergence of $\mathbf{A}$ should be proportional to the time derivative of V, but since $V = 0$ for (i) and (iii), they are also in the Lorenz gauge. This does not apply to (ii) since the divergence of $\mathbf{A} \neq 0$ while $\frac{\partial V}{\partial t} = 0$.

(i) Both Coulomb and Lorenz.

(ii) Neither Coulomb or Lorenz.

(iii) Both Coulomb and Lorenz. □

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

Problem 10.10

Confirm that the retarded potentials satisfy the Lorenz gauge condition. [Hint: First show that

$$\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})+\frac{1}{\imath}(\boldsymbol{\nabla}'\cdot\mathbf{J})-\boldsymbol{\nabla}'\cdot\left(\frac{\mathbf{J}}{\imath}\right)$$

where $\boldsymbol{\nabla}$ denotes derivatives with respect to $\mathbf{r}$, and $\boldsymbol{\nabla}'$ denotes derivatives with respect to $\mathbf{r}'$. Next, noting that $\mathbf{J}(\mathbf{r}',t-\imath/c)$ depends on $\mathbf{r}'$ both explicitly and through $\imath$, whereas it depends only on $\mathbf{r}$ through $\imath$, confirm that

$$\boldsymbol{\nabla}\cdot\mathbf{J}=-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}\imath),\boldsymbol{\nabla}'\cdot\mathbf{J}=-\dot{\rho}-\frac{1}{c}\dot{\mathbf{J}}\cdot(\boldsymbol{\nabla}'\imath)$$

Use this to calculate the divergence of $\mathbf{A}$ (Eq. 10.26.)]

Solution:

We want to show

$$\boldsymbol{\nabla}\cdot\mathbf{A}=-\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

where

$$\mathbf{A}=\frac{\mu_0}{4\pi}\int\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\,\mathrm{d}\tau'$$

Now, taking the divergence of $\mathbf{A}$ gives

$$\boldsymbol{\nabla}\cdot\mathbf{A}=\frac{\mu_0}{4\pi}\int\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath}\right)\mathrm{d}\tau'$$

From now on, $\mathbf{J}(\mathbf{r}',t_r)=\mathbf{J}$. Using a divergence identity gives

$$\boldsymbol{\nabla}\cdot\left(\frac{\mathbf{J}}{\imath}\right)=\frac{1}{\imath}(\boldsymbol{\nabla}\cdot\mathbf{J})+\mathbf{J}\cdot\left(\boldsymbol{\nabla}\frac{1}{\imath}\right)$$

and

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) + \mathbf{J} \cdot \left(\boldsymbol{\nabla}' \frac{1}{\imath}\right)$$

Since $\frac{1}{\imath} = \frac{1}{|\mathbf{r}-\mathbf{r}'|}$, we have

$$\boldsymbol{\nabla}' \frac{1}{\imath} = -\boldsymbol{\nabla} \frac{1}{\imath}$$

which means we can rewrite $\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$ as

$$\boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath}\right)$$

$$\downarrow$$

$$\mathbf{J} \cdot \left(\boldsymbol{\nabla} \frac{1}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$$

Putting this expression into $\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right)$ gives

$$\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{J}}{\imath}\right) = \frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)$$

which is what the hint wanted us to show. Putting this into the expression for $\boldsymbol{\nabla} \cdot \mathbf{A}$ gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi} \int \left[\frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) + \frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) - \boldsymbol{\nabla}' \cdot \left(\frac{\mathbf{J}}{\imath}\right)\right] \mathrm{d}\tau' \tag{1}$$

According to the Lorenz gauge condition,

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t} = -\frac{\mu_0}{4\pi} \int \frac{\dot{\rho}}{\imath} \, \mathrm{d}\tau'$$

With this information, we now evaluate the three terms in Equation 1

Starting with the first term, and using the chain rule on $\mathbf{J}$. Since $\mathbf{J}$ depends on $\mathbf{r}$ only through t_r,

$$\boldsymbol{\nabla} \cdot \mathbf{J} = \dot{\mathbf{J}} \cdot \boldsymbol{\nabla} t_r = -\frac{1}{c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

$$\downarrow \tag{2}$$

$$\frac{1}{\imath}(\boldsymbol{\nabla} \cdot \mathbf{J}) = -\frac{1}{\imath c} \dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

The next term is the divergence of $\mathbf{J}$ with respect to $\mathbf{r}'$. $\mathbf{J}$ depends on $\mathbf{r}'$ both explicitly and through t_r, so

$$\boldsymbol{\nabla}' \cdot \mathbf{J} = (\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r} + \dot{\mathbf{J}} \cdot \boldsymbol{\nabla}' t_r = -\dot{\rho} - \frac{1}{c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath)$$
$$\downarrow \tag{3}$$
$$\frac{1}{\imath}(\boldsymbol{\nabla}' \cdot \mathbf{J}) = -\frac{\dot{\rho}}{\imath} - \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla}' \imath) = -\frac{\dot{\rho}}{\imath} + \frac{1}{\imath c}\dot{\mathbf{J}} \cdot (\boldsymbol{\nabla} \imath)$$

where $(\boldsymbol{\nabla}' \cdot \mathbf{J})_{t_r}$ is the divergence taken at fixed t_r, which equals $-\dot{\rho}$ by the continuity equation.

The last term integrates to 0 which I will explain later.

Adding Equation 2 and Equation 3, the only term left is

$$-\frac{\dot{\rho}}{\imath}$$

which gives

$$\boldsymbol{\nabla} \cdot \mathbf{A} = \frac{\mu_0}{4\pi}\int\left(-\frac{\dot{\rho}}{\imath}\right)\mathrm{d}\tau' = -\frac{\mu_0\epsilon_0}{4\pi\epsilon_0}\int\frac{\dot{\rho}}{\imath}\,\mathrm{d}\tau' =$$
$$= -\mu_0\epsilon_0\frac{\partial}{\partial t}\left[\frac{1}{4\pi\epsilon_0}\int\frac{\rho}{\imath}\,\mathrm{d}\tau'\right] = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$$
$$\downarrow$$
$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}}$$

□

Problem 10.11

Problem 10.12

Problem 10.13

Solution:

We have

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}', t_r)}{\imath^2}\hat{\boldsymbol{\imath}} + \frac{\dot{\rho}(\mathbf{r}', t_r)}{c\imath}\hat{\boldsymbol{\imath}} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t_r)}{c^2\imath}\right]\mathrm{d}\tau' \tag{4}$$

and

$$\rho(\mathbf{r},t) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)t_r = \rho(\mathbf{r},0) + \dot{\rho}(\mathbf{r},0)\left(t - \frac{\imath}{c}\right)$$

$$\downarrow$$

$$\rho(\mathbf{r},t_r) = \rho(\mathbf{r},t) - \dot{\rho}(\mathbf{r},0)\frac{\imath}{c}$$

$$\downarrow$$

$$\frac{\rho(\mathbf{r},t_r)}{\imath^2} = \frac{\rho(\mathbf{r},t)}{\imath^2} - \frac{\dot{\rho}(\mathbf{r},0)}{c\imath}$$

Putting this into Equation 4, and noting that $\dot{\mathbf{J}} = 0$ since $\mathbf{J}$ is constant in time, we get

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\imath^2}\hat{\boldsymbol{\imath}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\imath}\hat{\boldsymbol{\imath}} + \frac{\dot{\rho}(\mathbf{r}',t_r)}{c\imath}\hat{\boldsymbol{\imath}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{J} \text{ constant in time} \leftrightarrow \dot{\rho} \text{ constant in time } [\boldsymbol{\nabla}\cdot\mathbf{J} = -\dot{\rho}] \rightarrow \dot{\rho}(\mathbf{r},t) = \dot{\rho}(\mathbf{r},0)$$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho(\mathbf{r}',t)}{\imath^2}\hat{\boldsymbol{\imath}} - \frac{\dot{\rho}(\mathbf{r}',0)}{c\imath}\hat{\boldsymbol{\imath}} + \frac{\dot{\rho}(\mathbf{r}',0)}{c\imath}\hat{\boldsymbol{\imath}}\right]\mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int\frac{\rho(\mathbf{r}',t)}{\imath^2}\hat{\boldsymbol{\imath}}\,\mathrm{d}\tau'$$

□

Problem 10.14

Show that the Biot-Savart law holds, with $\mathbf{J}$ evaluated at the nonretarded time, considering only the first two terms in the Taylor expansion for $\mathbf{J}$ is

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t) + ...$$

where the $\mathbf{r}$ dependence in $\mathbf{J}(\mathbf{r},t_r)$ has been omitted for compactness.

Solution:

$$\mathbf{B}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int\left[\frac{\mathbf{J}(\mathbf{r}',t_r)}{\imath^2} + \frac{\dot{\mathbf{J}}(\mathbf{r}',t_r)}{c\imath}\right]\times\hat{\boldsymbol{\imath}}\,\mathrm{d}\tau' \tag{5}$$

We use the given Taylor expansion on the first term in Equation 5:

$$\mathbf{J}(t_r) = \mathbf{J}(t) + (t_r - t)\dot{\mathbf{J}}(t)$$

$$t_r = t - \frac{\mathscr{r}}{c} \rightarrow t_r - t = -\frac{\mathscr{r}}{c}$$

$$\downarrow$$

$$\mathbf{J}(t_r) = \mathbf{J}(t) - \frac{\mathscr{r}}{c}\dot{\mathbf{J}}(t)$$

$$\downarrow$$

$$\frac{\mathbf{J}(t_r)}{\mathscr{r}^2} = \frac{\mathbf{J}(t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(t)}{c\mathscr{r}} \tag{6}$$

and on the second term of Equation 5:

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) + (t_r - t)\ddot{\mathbf{J}}(t)$$

But since all higher order terms are neglected, $\ddot{\mathbf{J}}(t) = 0$ so

$$\dot{\mathbf{J}}(t_r) = \dot{\mathbf{J}}(t) \tag{7}$$

Putting Equation 6 and Equation 7 into Equation 5 gives

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \left[\frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} - \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} + \frac{\dot{\mathbf{J}}(\mathbf{r}', t)}{c\mathscr{r}} \right] \times \hat{\boldsymbol{\mathscr{r}}}\, \mathrm{d}\tau'$$

$$\downarrow$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t)}{\mathscr{r}^2} \times \hat{\boldsymbol{\mathscr{r}}}\, \mathrm{d}\tau'$$

□

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Chapter 11

Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r,\theta,t) = \frac{p_0 \cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$

$$\boldsymbol{A}(r,\theta,t) = -\frac{\mu_0 p_0 \omega}{4\pi r}\sin\left[\omega\left(t-\frac{r}{c}\right)\right]\hat{z}$$

satisfy the Lorenz gauge condition.

Solution:

The Lorenz gauge condition is

$$\boldsymbol{\nabla}\cdot\mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Calculating the time derivative of V

$$\frac{\partial V}{\partial t} = \frac{p_0 \cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$

Calculating the divergence of $\mathbf{A}$

$$\boldsymbol{\nabla}\cdot\mathbf{A} =$$

□

Problem 11.2

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Problem 11.4

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Problem 11.36

Chapter 12

Electrodynamics and Relativity

Problem 12.1

Let $\mathcal{S}$ be an inertial reference system. Use Galileo's velocity addition rule.

(a) Suppose that $\overline{\mathcal{S}}$ moves with constant velocity relative to $\mathcal{S}$. Show that $\overline{\mathcal{S}}$ is also an inertial reference system. [Hint: Use the definition in footnote 1.]

(b) Conversely, show that if $\overline{\mathcal{S}}$ is an inertial system, then it moves with respect to $\mathcal{S}$ at constant velocity.

Problem 12.2

Problem 12.3

(a) What's the percent error introduced when you use Galileo's rule, instead of Einstein's, with $v_{AB} = 5$ mi/h and $v_{BC} = 60$ mi/h?

(b) Suppose you could run at half the speed of light down the corridor of a train going three-quarters the speed of light. What would your speed be relative to the ground?

(c) Prove, using Eq. 12.3, that if $v_{AB} < c$ and $v_{BC} < c$ then $v_{AC} < c$. Interpret this result.

Solution:

(a) Galileo's rule is simply

$$v_{AC} = v_{AB} + v_{BC} = 60 + 5 = 65 \text{ mi/h}$$

while Einstein's rule gives

$$v_{AC} = \frac{v_{AB} + v_{BC}}{1 + \frac{v_{AB} v_{BC}}{c^2}}$$

where the denominator is

$$1+\frac{v_{AB}v_{BC}}{c^2}=1+\frac{60\text{mi/h}\times 5\text{mi/h}}{(670,616,629\text{ mi/h})^2}=1+6.67\times 10^{-16}$$

so the percentage error is

$$6.67\times 10^{-16}\times 100$$
$$\downarrow$$
$$\boxed{6.67\times 10^{-14}}$$

(b) With Einstein's velocity addition rule and $v_{AB}=\frac{1}{2}c, v_{BC}=\frac{3}{4}c$, the result is

$$v_{AC}=\frac{v_{AB}+v_{BC}}{1+\frac{v_{AB}v_{BC}}{c^2}}=\frac{\frac{1}{2}c+\frac{3}{4}c}{1+\frac{3}{8}}=\frac{5}{4}\frac{8}{11}c=\boxed{\frac{10}{11}c}$$

(c) Introducing variables $\alpha=\frac{v_{AC}}{c}, \beta_1=\frac{v_{AB}}{c}, \beta_2=\frac{v_{BC}}{c}$ makes this proof easier. The idea for these variables comes from manually solving without variable substitution (which is done below) and recognizing how it can be written easier. The equation becomes

$$\alpha=\frac{\beta_1+\beta_2}{1+\beta_1\beta_2}$$

Squaring this gives

$$\alpha^2=\frac{\beta_1^2+2\beta_1\beta_2+\beta_2^2}{1+2\beta_1\beta_2+\beta_1^2\beta_2^2}$$

Extending the numerator with $1+\beta_1^2\beta_2^2-1-\beta_1^2\beta_2^2$ gives

$$\alpha^2=\frac{1+2\beta_1\beta_2+\beta_1^2\beta_2^2+\beta_1^2+\beta_2^2-1-\beta_1^2\beta_2^2}{1+2\beta_1\beta_2+\beta_1^2\beta_2^2}=$$
$$=1+\frac{\beta_1^2+\beta_2^2-1-\beta_1^2\beta_2^2}{1+2\beta_1\beta_2+\beta_1^2\beta_2^2}$$

What's left is to showthat the numerator above is negative (the denominator is positive)

$$\beta_1^2+\beta_2^2-1-\beta_1^2\beta_2^2<0$$
$$\downarrow$$
$$\beta_1^2(1-\beta_2^2)-(1-\beta_2^2)=(\beta_1^2-1)(1-\beta_2^2)$$

Since $\beta_1,\beta_2<1$, the above equation is

$$(\beta_1^2-1)(1-\beta_2^2)=(\text{negative})(\text{positive})=\text{negative}$$

and the proof is done.

Extra: Doing this from the start without variable substitution gives: Squaring the equation gives

$$v_{AC}^2 = \frac{(v_{AB} + v_{BC})^2}{\left(1 + \frac{v_{AB}v_{BC}}{c^2}\right)^2} = \frac{v_{AB}^2 + 2v_{AB}v_{BC} + v_{BC}^2}{1 + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{AB}^2 v_{BC}^2}{c^4}}$$

Dividing both sides by c^2 gives

$$\frac{v_{AC}^2}{c^2} = \frac{\frac{v_{AB}^2}{c^2} + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{BC}^2}{c^2}}{1 + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{AB}^2 v_{BC}^2}{c^4}}$$

Extending the numerator with $1 + \frac{v_{AB}^2 v_{BC}^2}{c^4} - 1 - \frac{v_{AB}^2 v_{BC}^2}{c^4}$ gives

$$\frac{v_{AC}^2}{c^2} = \frac{1 + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{AB}^2 v_{BC}^2}{c^4} - 1 - \frac{v_{AB}^2 v_{BC}^2}{c^4} + \frac{v_{AB}^2}{c^2} + \frac{v_{BC}^2}{c^2}}{1 + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{AB}^2 v_{BC}^2}{c^4}} =$$

$$= 1 + \frac{\frac{v_{AB}^2}{c^2} + \frac{v_{BC}^2}{c^2} - 1 - \frac{v_{AB}^2 v_{BC}^2}{c^4}}{1 + 2\frac{v_{AB}v_{BC}}{c^2} + \frac{v_{AB}^2 v_{BC}^2}{c^4}}$$

What's left to prove the statement is showing that the numerator part is negative (since the denominator is always positive):

$$\frac{v_{AB}^2}{c^2} + \frac{v_{BC}^2}{c^2} - 1 - \frac{v_{AB}^2 v_{BC}^2}{c^4} < 0$$

$$\frac{v_{AB}^2}{c^2}\left(1 - \frac{v_{BC}^2}{c^2}\right) + \frac{v_{BC}^2}{c^2} - 1$$

$$\downarrow$$

$$\frac{v_{AB}^2}{c^2}\left(1 - \frac{v_{BC}^2}{c^2}\right) - \left(1 - \frac{v_{BC}^2}{c^2}\right) = \left(\frac{v_{AB}^2}{c^2} - 1\right)\left(1 - \frac{v_{BC}^2}{c^2}\right)$$

Since $v_{AB} < c \Rightarrow \frac{v_{AB}^2}{c^2} - 1 < 0$, and $v_{BC} < c \Rightarrow 1 - \frac{v_{BC}^2}{c^2} > 0$, the product is strictly negative, hence $\frac{v_{AC}^2}{c^2} < 1 \Rightarrow v_{AC} < c$. We can now recognize the variable substitution for easier calculations. □

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target. □

Problem 12.5

Synchronized clocks are stationed at regular intervals, a million km apart, along a straight line. When the clock next to you reads 12 noon:

(a) What time do you see on the 90th clock down the line?

(b) What time do you observe on that clock?

Solution:

(a) The 90th clock down the line is a distance $d = 90 \times 10^6$ km $= 90 \times 10^9$ m. The time it takes for light to travel this distance is

$$t = \frac{d}{c} = \frac{90 \times 10^9 \text{ m}}{3 \times 10^8 \text{ m/s}} = 300 \text{ s}.$$

This means that what you see on the 90th clock is what it showed 300 seconds ago, so the clock reads 11:55.

(b) The time you **observe** on that clock is taking into account the time it takes for the information to travel. Therefore, what you observe is what the clock really is, 12:00. □

Problem 12.6

Every 2 years, more or less, The New York Times publishes an article in which some astronomer claims to have found an object traveling faster than the speed of light. Many of these reports result from a failure to distinguish what is seen from what is observed — that is, from a failure to account for light travel time. Here's an example: A star is traveling with speed v at an angle θ to the line of sight. What is its apparent speed across the sky? (Suppose the light signal from b reaches the earth at a time Δt after the signal from a, and the star has meanwhile advanced a distance Δs across the celestial sphere; by "apparent speed," I mean $\frac{\Delta s}{\Delta t}$.) What angle θ gives

the maximum apparent speed? Show that the apparent speed can be much greater than c, even if v itself is less than c.

Problem 12.7

In a laboratory experiment, a muon is observed to travel 800 m before disintegrating. A graduate student looks up the lifetime of a muon (2×10^{-6}s) and concludes that its speed was

$$v = \frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = 4 \times 10^8 \text{ m/s}.$$

Faster than light! Identify the student's error, and find the actual speed of this muon.

Solution:

The student did not take into account the lifetime of the muon in the muons reference time. The muons lifetime in its own reference time is

$$\gamma t = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} t$$

If it travels for 800m then its speed is

$$\frac{800 \text{ m}}{\gamma \times 2 \times 10^{-6}\text{s}} = v$$

Simplifying the distance over time gives

$$\frac{800 \text{ m}}{2 \times 10^{-6}\text{s}} = \frac{4}{3}c$$

Solving for v

$$\frac{4}{3}c = \gamma v = \frac{v}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2 = \frac{v^2}{1-\left(\frac{v}{c}\right)^2}$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2\left(1-\left(\frac{v}{c}\right)^2\right) = v^2$$

$$\downarrow$$

$$\left(\frac{4}{3}c\right)^2 = v^2\left(1+\left(\frac{4}{3}c\right)^2\frac{1}{c^2}\right) = v^2\left(1+\frac{16}{9}\right)$$

$$\downarrow$$

$$v = \frac{4}{3}c\frac{1}{\sqrt{1+\frac{16}{9}}} = \frac{4}{3}c\frac{1}{\sqrt{\frac{25}{9}}} = \frac{12}{15}c = \boxed{\frac{4}{5}c}$$

□

Problem 12.8

A rocket ship leaves earth at a speed of $\frac{3}{5}c$. When a clock on the rocket says 1 hour has elapsed, the rocket ship sends a light signal back to earth.

(a) According to earth clocks, when was the signal sent?

(b) According to earth clocks, how long after the rocket left did the signal arrive back on earth?

(c) According to the rocket observer, how long after the rocket left did the signal arrive back on earth?

Solution:

(a) The time it takes for one hour to pass on the rocket, $t_r = 1\text{h}$, is according to earth the time t_e

$$t_e = \gamma t_r = \frac{1}{\sqrt{1-\frac{3^2}{5^2}}}t_r = \sqrt{\frac{25}{16}}t_r = \frac{5}{4}t_r = \frac{5}{4}\text{h}$$

(b) The time the signal from the rocket was sent, according to earth, was $\frac{5}{4}\text{h}$. This means the rocket has travelled a distance $d = t_e/\left(\frac{3}{5}c\right) = 4500\text{s} \times \frac{3}{5}c = 8.1 \times 10^{10}$ m. This distance takes 2700s to reach earth, with a total time of $4500 + 2700$ s $= 7200$ s $= 2\text{h}$

(c) We can use the result in (b), where the time on earth was 2 hours. According to the rocket, earth's clock run slow, so the time according to the rocket is

$$\gamma \times 2\text{h} = \frac{5}{4} \times 2\text{h} = 2.5\text{h}$$

Another solution, if the result for (b) is missing. Imagine that the rocket is now stationary and earth is moving away at $\frac{3}{5}c$. The signal is sent after 1 hour, and in this time the earth has travelled $60\text{min} \times \frac{3}{5}c = 6.48 \times 10^9$ m. This distance now needs to be covered by the signal, which takes $6.48 \times 10^9\text{m} \times \frac{1}{c} = 36\text{min}$. Repeating the argument, in those 36 minutes, the earth has travelled another distance, so we get $36\text{min} \times \frac{3}{5}c$. We can model this as an infinite sum, with the factor $3/5$:

$$t = 60\text{min} + 36\text{min} \times \sum_0^\infty \left(\frac{3}{5}\right)^n$$

which is a geometric series with solution

$$60\text{min} + 36\text{min}\ \frac{1}{1-\frac{3}{5}} = 60\text{min} + 90\text{min} = 150\text{min} = 2\text{h}30\text{min}$$

□

Problem 12.9

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L. With length contraction due to the velocity accounted for the length is denoted L'. With v_{vw} given and $L_{\text{lc}} = 2L_{\text{vw}}$, the solution is obtained by setting $L'_{\text{lc}} = L'_{\text{vw}}$ and solving for v_{lc}.

$$L_{\text{lc}} = 2L_{\text{vw}}$$

$$L'_{\text{vw}} = L_{\text{vw}}\sqrt{1 - \frac{v_{\text{vw}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

$$L'_{\text{vw}} = L'_{\text{lc}} = L_{\text{lc}}\sqrt{1 - \frac{v_{\text{lc}}^2}{c^2}} = L_{\text{vw}}\frac{\sqrt{3}}{2}$$

Using $L_{\text{lc}} = 2L_{\text{vw}}$ and solving for v_{lc}, the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{\text{lc}} = \frac{\sqrt{13}}{4}c$$

□

Problem 12.10

A sailboat is manufactured so that the mast leans at an angle $\overline{\theta}$ with respect to the deck. An observer standing on a dock sees the boat go by at speed v. What angle does this observer say the mast makes?

Solution:

The direction of the speed is Lorentz contracted by $\frac{1}{\gamma}$. The horizontal and vertical components of the mast is

$$\overline{x} = \overline{l}\cos\left(\overline{\theta}\right), \quad \overline{y} = \overline{l}\sin\left(\overline{\theta}\right)$$

Transforming these components with regard to the velocity gives

$$x = l\frac{\cos(\theta)}{\gamma}, \quad y = l\sin(\theta)$$

This gives

$$\frac{\overline{y}}{\overline{x}} = \frac{y}{x}$$

$$\downarrow$$

$$\boxed{\tan\left(\overline{\theta}\right) = \gamma\tan(\theta)}$$

□

Problem 12.11

Problem 12.12

Solve

$$\text{(i)}\ \overline{x} = \gamma(x - vt)$$
$$\text{(ii)}\ \overline{y} = y$$
$$\text{(iii)}\ \overline{z} = z$$
$$\text{(iv)}\ \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

for x, y, z, t in terms of $\overline{x}, \overline{y}, \overline{z}, \overline{t}$, and check that you recover

$$\text{(i')}\ x = \gamma(\overline{x} + v\overline{t})$$
$$\text{(ii')}\ y = \overline{y}$$
$$\text{(iii')}\ z = \overline{z}$$
$$\text{(iv')}\ t = \gamma\left(\overline{t} + \frac{v}{c^2}\overline{x}\right)$$

Solution:

From (i) and (iv) we have

$$\text{(i)}\quad \overline{x} = \gamma(x - vt) \rightarrow x = \frac{\overline{x}}{\gamma} + vt$$
$$\text{(iv)}\ \overline{t} = \gamma\left(t - \frac{v}{c^2}x\right) \rightarrow t = \frac{\overline{t}}{\gamma} + \frac{v}{c^2}x$$

Inserting t from (iv) into the expression for x in (i) gives

$$x = \frac{\overline{x}}{\gamma} + v\left(\frac{\overline{t}}{\gamma} + \frac{v}{c^2}x\right) \rightarrow x\left(1 - \frac{v^2}{c^2}\right) = \frac{\overline{x}}{\gamma} + \frac{v\overline{t}}{\gamma}$$
$$\downarrow$$
$$x\frac{1}{\gamma^2} = \frac{1}{\gamma}(\overline{x} + v\overline{t}) \rightarrow \boxed{x = \gamma(\overline{x} + v\overline{t}) = \text{(i')}}$$

Putting this expression for x into the expression for t in (iv) gives

$$t = \frac{\overline{t}}{\gamma} + \frac{v}{c^2}(\gamma(\overline{x} + v\overline{t})) = \gamma\left(\frac{\overline{t}}{\gamma^2} + \frac{v}{c^2}\overline{x} + \frac{v^2}{c^2}\overline{t}\right) = \gamma\left(\left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right)\overline{t} + \frac{v}{c^2}\overline{x}\right)$$
$$\downarrow$$
$$\boxed{t = \gamma\left(\overline{t} + \frac{v}{c^2}\overline{x}\right) = \text{(iv')}}$$

(ii) $\longleftrightarrow$ (ii') and (iii) $\longleftrightarrow$ (iii') follow directly from the equations.

□

Problem 12.13

Sophie Zabar, clairvoyante, cried out in pain at precisely the instant her twin brother, 500 km away, hit his thumb with a hammer. A skeptical scientist observed both events (brother's accident, Sophie's cry) from an airplane traveling at $\frac{12}{13}c$ to the right. Which event occurred first, according to the scientist? How much earlier was it, in seconds?

Solution:

Using the Lorentz transformations

$$\overline{x} = \gamma(x - vt)$$
$$\bar{t} = \gamma\left(t - \frac{v}{c^2}x\right)$$

and setting Sophie's position to $x = 500\text{km}$, and her brothers position to $x = 0\text{km}$. The time of the brothers event according to the scientist is

$$\bar{t}_1 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}0\right) = 0\text{s}$$

The time of Sophie's event is

$$\bar{t}_2 = \frac{1}{\sqrt{1 - \frac{12^2}{13^2}}}\left(0 - \frac{12}{13c}500\text{ km}\right) = \frac{1}{\sqrt{\frac{25}{169}}}\left(-\frac{12}{13c}500\text{km}\right) =$$
$$= -\frac{13}{5}\frac{12}{13c}500\text{km} = -\frac{1200000}{c}\text{ m} = -0.004\text{ s} = -4\text{ms}$$

The twin brothers event happened 4ms earlier according to the scientist. □

Problem 12.14

Problem 12.15

You probably did Prob. 12.4 from the point of view of an observer on the *ground*. Now do it from the point of view of the police car, the outlaws, and the bullet. That is, fill in the gaps in the following table:

	Ground	**Police**	**Outlaws**	**Bullet**	**Do they escape?**
Ground	0	$\frac{1}{2}c$	$\frac{3}{4}c$		
Police		0		$\frac{1}{3}c$	

	Ground	Police	Outlaws	Bullet	Do they escape?
Outlaws			0		
Bullet				0	

Solution:

The speed of the bullet relative to the ground is obtained by using Einsteins rule.

$$v = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}} = \frac{\frac{5}{6}}{1 + \frac{1}{6}} = \frac{\frac{5}{6}}{\frac{7}{6}} = \frac{5}{7}c$$

With this we can also directly fill in the full first column, the speed of the ground relative to everything else.

	Ground	Police	Outlaws	Bullet	Do they escape?
Ground	0	$\frac{1}{2}c$	$\frac{3}{4}c$	$\frac{5}{7}c$	**No**
Police	$-\frac{1}{2}c$	0		$\frac{1}{3}c$	**No**
Outlaws	$-\frac{3}{4}c$		0		**No**
Bullet	$-\frac{5}{7}c$			0	**No**

Speed of outlaws relative to police is Einstein rule on speed of outlaws relative to ground and speed of ground relative to police.

$$v = \frac{\frac{3}{4}c - \frac{1}{2}c}{1 - \frac{3}{8}} = \frac{\frac{2}{8}}{\frac{5}{8}}c = \frac{2}{5}c$$

With this we can also fill in the full column of the speed of the police relative to everything.

	Ground	Police	Outlaws	Bullet	Do they escape?
Ground	0	$\frac{1}{2}c$	$\frac{3}{4}c$	$\frac{5}{7}c$	**No**
Police	$-\frac{1}{2}c$	0	$\frac{2}{5}c$	$\frac{1}{3}c$	**No**

	Ground	Police	Outlaws	Bullet	Do they escape?
Outlaws	$-\frac{3}{4}c$	$-\frac{2}{5}c$	0		**No**
Bullet	$-\frac{5}{7}c$	$-\frac{1}{3}$		0	**No**

Finally, the speed of the bullet relative to the outlaws is obtained by using Einstein rule on the bullet relative to the police and the police relative to the outlaws

$$v = \frac{\frac{1}{3} - \frac{2}{5}}{1 - \frac{2}{15}}c = \frac{-\frac{1}{15}}{\frac{13}{15}}c = -\frac{1}{13}c$$

With this we can also directly fill in the speed of the outlaws relative to the bullet.

	Ground	Police	Outlaws	Bullet	Do they escape?
Ground	0	$\frac{1}{2}c$	$\frac{3}{4}c$	$\frac{5}{7}c$	**No**
Police	$-\frac{1}{2}c$	0	$\frac{2}{5}c$	$\frac{1}{3}c$	**No**
Outlaws	$-\frac{3}{4}c$	$-\frac{2}{5}c$	0	$-\frac{1}{13}c$	**No**
Bullet	$-\frac{5}{7}c$	$-\frac{1}{3}$	$\frac{1}{13}c$	0	**No**

In all cases, the speed of the bullet is less than the speed of the outlaws, so the bullet does not reach the outlaws. □

Problem 12.16

Problem 12.17

Check

$$-\overline{a}^0\overline{b}^0 + \overline{a}^1\overline{b}^1 + \overline{a}^2\overline{b}^2 + \overline{a}^3\overline{b}^3 = -a^0b^0 + a^1b^1 + a^2b^2 + a^3b^3$$

using

$$\begin{cases} \overline{a}^0 = \gamma(a^0 - \beta a^1) \\ \overline{a}^1 = \gamma(a^1 - \beta a^0) \\ \overline{a}^2 = a^2 \\ \overline{a}^3 = a^3 \end{cases}$$

Solution:

Computing the first term gives

$$-\overline{a}^0\overline{b}^0 = -\gamma^2(a^0 - \beta a^1)(b^0 - \beta b^1) = -\gamma^2(a^0b^0 - a^0b^1\beta - a^1b^0\beta + a^1b^1\beta^2)$$

and the second term gives

$$\overline{a}^1\overline{b}^1 = \gamma^2(a^1 - \beta a^0)(b^1 - \beta b^0) = \gamma^2(a^1b^1 - a^1b^0\beta - a^0b^1\beta + a^0b^0\beta^2)$$

Adding these two terms gives

$$-\overline{a}^0\overline{b}^0 + \overline{a}^1\overline{b}^1 = \gamma^2(-a^0b^0 + a^0b^1\beta + a^1b^0\beta - a^1b^1\beta^2 + a^1b^1 - a^1b^0\beta - a^0b^1\beta + a^0b^0\beta^2) =$$

$$= \gamma^2(a^0b^0(\beta^2 - 1) + a^1b^1(1 - \beta^2)) = \gamma^2\left(-a^0b^0\frac{1}{\gamma^2} + a^1b^1\frac{1}{\gamma^2}\right) = \boxed{-a^0b^0 + a^1b^1}$$

The terms

$$\overline{a}^2\overline{b}^2 = a^2b^2$$
$$\overline{a}^3\overline{b}^3 = a^3b^3$$

Follow directly since $\overline{a}^2 = a^2, \overline{a}^3 = a^3$. ✓

□

Problem 12.18

(a) Write out the matrix that describes a Galilean transformation.

(b) Write out the matrix describing a Lorentz transformation along the y-axis.

(c) Find the matrix describing a Lorentz transformation with velocity v along the x-axis followed by a Lorentz transformation with velocity $\overline{v}$ along the y-axis. Does it matter in what order the transformations are carried out?

Solution:

(a) Mapping the time variable $t \to ct$ and setting $\beta = \frac{v}{c}$ gives the Galilean transformation

$$\begin{pmatrix} c\overline{t} \\ \overline{x} \\ \overline{y} \\ \overline{z} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

(b)

$$\begin{pmatrix} c\bar{t} \\ \bar{x} \\ \bar{y} \\ \bar{z} \end{pmatrix} = \begin{pmatrix} \gamma & 0 & -\gamma\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma\beta & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

(c) First writing out the transformation with velocity v along the x-axis, setting $\beta = \frac{v}{c}$

$$\Lambda_x = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

And now with velocity $\overline{v}$ in the y-axis, with $\overline{\beta} = \frac{\overline{v}}{c}$, $\overline{\gamma} = \frac{1}{\sqrt{1-\overline{\beta}^2}}$

$$\Lambda_y = \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Transforming first along the x-axis and then the y-axis gives the matrix

$$\Lambda_y\Lambda_x = \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \overline{\gamma}\gamma & -\overline{\gamma}\gamma\beta & -\overline{\gamma}\overline{\beta} & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ -\overline{\gamma}\overline{\beta}\gamma & \overline{\gamma}\overline{\beta}\gamma\beta & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Doing it the other way around gives another matrix:

$$\Lambda_x\Lambda_y = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \overline{\gamma} & 0 & -\overline{\gamma}\overline{\beta} & 0 \\ 0 & 1 & 0 & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \gamma\overline{\gamma} & -\gamma\beta & -\gamma\overline{\gamma}\overline{\beta} & 0 \\ -\gamma\beta\overline{\gamma} & \gamma & \gamma\beta\overline{\gamma}\overline{\beta} & 0 \\ -\overline{\gamma}\overline{\beta} & 0 & \overline{\gamma} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and we see that $\Lambda_y\Lambda_x \neq \Lambda_x\Lambda_y$

□

Problem 12.19

The parallel between rotations and Lorentz transformations is even more striking if we introduce the **rapidity**:

$$\theta \equiv \tanh^{-1}\left(\frac{v}{c}\right)$$

(a) Express the Lorentz transformation matrix Λ (Eq. 12.24) in terms of θ, and compare it to the rotation matrix (Eq. 1.29).

In some respects, rapidity is a more natural way to describe motion than velocity. For one thing, it ranges from $-\infty$ to ∞, instead of $-c$ to c. More significantly, rapidities add, whereas velocities do not.

(b) Express the Einstein velocity addition law in terms of rapidity.

Solution:

(a) Rewriting γ and β in terms of the rapidity gives

$$\tanh^{-1}\beta = \theta \rightarrow \beta = \tanh(\theta)$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \rightarrow \gamma = \frac{1}{\sqrt{1-\tanh^2(\theta)}} = \frac{1}{\operatorname{sech}(\theta)} \rightarrow \gamma = \cosh(\theta)$$

This makes the transformation matrix

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} \cosh(\theta) & -\cosh(\theta)\tanh(\theta) & 0 & 0 \\ -\cosh(\theta)\tanh(\theta) & \cosh(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since $\tanh(\theta) = \frac{\sinh(\theta)}{\cosh(\theta)}$, the matrix can be further simplified to

$$\Lambda = \begin{pmatrix} \cosh(\theta) & -\sinh(\theta) & 0 & 0 \\ -\sinh(\theta) & \cosh(\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Comparing it to the rotation matrix (Eq. 1.29) which is

$$\begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(b) The Einstein velocity addition law is

$$v_{AC} = \frac{v_{AB} + v_{BC}}{1 + \frac{v_{AB}v_{BC}}{c^2}}$$

$$\theta_{AB} = \tanh^{-1}\left(\frac{v_{AB}}{c}\right) \rightarrow \tanh(\theta_{AB}) = \frac{v_{AB}}{c}$$

$$\theta_{BC} = \tanh^{-1}\left(\frac{v_{BC}}{c}\right) \rightarrow \tanh(\theta_{BC}) = \frac{v_{BC}}{c}$$

$$\theta_{AC} = \tanh^{-1}\left(\frac{v_{AC}}{c}\right) \rightarrow \tanh(\theta_{AC}) = \frac{v_{AC}}{c}$$

This gives the Einstein velocity addition law in terms of the rapidity

$$c\tanh(\theta_{AC}) = \frac{c(\tanh(\theta_{AB}) + \tanh(\theta_{BC}))}{1 + \tanh(\theta_{AB})\tanh(\theta_{BC})}$$

$$\downarrow$$

$$\tanh(\theta_{AC}) = \frac{\tanh(\theta_{AB}) + \tanh(\theta_{BC})}{1 + \tanh(\theta_{AB})\tanh(\theta_{BC})}$$

Using the identity

$$\tanh(x+y) = \frac{\tanh(x) + \tanh(y)}{1 + \tanh(x)\tanh(y)}$$

gives

$$\tanh(\theta_{AC}) = \tanh(\theta_{AB} + \theta_{BC}) \rightarrow \boxed{\theta_{AC} = \theta_{AB} + \theta_{BC}}$$

□

Problem 12.20

(a) Event A happens at point $(x_A = 5, y_A = 3, z_A = 0)$ and at time t_A given by $ct_A = 15$; event B occurs at (10, 8, 0) and $ct_B = 5$, both in system $\mathcal{S}$.

(i) What is the invariant interval between A and B?

(ii) Is there an inertial system in which they occur simultaneously? If so, find its velocity (magnitude and direction) relative to $\mathcal{S}$.

(iii) Is there an inertial system in which they occur at the same point? If so, find its velocity relative to $\mathcal{S}$.

(b) Repeat part (a) for $A = (2, 0, 0), ct = 1$; and $B = (5, 0, 0), ct = 3$.

Solution:

(a) The displacement 4-vector is

$$\Delta x^\mu = x_A^\mu - x_B^\mu$$

The scalar product of Δx^μ with itself gives the invariant interval between two events:

$$I \equiv (\Delta x)^\mu (\Delta x)_\mu = -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2$$

The two events A and B are

$$x_A^\mu = (15, 5, 3, 0)$$
$$x_B^\mu = (5, 10, 8, 0)$$
$$\downarrow$$
$$\Delta x^\mu = (15-5, 5-10, 3-8, 0) = (10, -5, -5, 0)$$

This gives the invariant interval

$$I = (\Delta x)^\mu (\Delta x)_\mu = (10, -5, -5, 0) \cdot (-10, -5, -5, 0) =$$
$$= -(10)^2 + (-5)^2 + (-5)^2 = -50$$

Since $I < 0$, the invariant interval is timelike, so there exist an inertial system in which they occur at the same point, but not at the same time.

Using the Lorentz transformation, we want to find the velocity of the frame in which they occur at the same point. This is equal to setting the difference in displacement to zero after Lorentz transformation.

$$\Delta\overline{x} = \gamma(\Delta x - v_x \Delta t) = 0 \rightarrow v_x = \frac{\Delta x}{\Delta t} = \frac{-5}{\frac{10}{c}} = -\frac{c}{2}$$

$$\Delta\overline{y} = \gamma\big(\Delta y - v_y \Delta t\big) = 0 \rightarrow v_y = \frac{\Delta y}{\Delta t} = \frac{-5}{10c} = -\frac{c}{2}$$

So the speed of the frame is

$$v = -\frac{c}{2}\hat{\boldsymbol{x}} - \frac{c}{2}\hat{\boldsymbol{y}}$$

(b) The displacement 4-vector is now

$$x_A^\mu = (1, 2, 0, 0)$$
$$x_B^\mu = (3, 5, 0, 0)$$
$$\downarrow$$
$$\Delta x^\mu = (1-3, 2-5, 0, 0)$$

This gives the invariant interval

$$I = (\Delta x)^\mu (\Delta x)_\mu = (-2, -3, 0, 0) \cdot (2, -3, 0, 0) =$$
$$= -(2)^2 + (-3)^2 = 5$$

Since $I > 0$, the invariant interval is spacelike so there exists a frame in which they occur at the same time, but not at the same point.

To find the velocity of the frame in which they occur at the same time, we set the difference in time to zero after Lorentz transformation.

$$\Delta\bar{t} = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right) = 0 \rightarrow v = \frac{\Delta t}{\Delta x}c^2$$

$$v = \frac{ct_A - ct_B}{x_A - x_B}c^2 = \frac{1-3}{2-5}c = \frac{-2}{-3}c = \frac{2}{3}c$$

□

Problem 12.21

The coordinates of event A are $(x_A, 0, 0), t_A$, and the coordinates of event B are $(x_B, 0, 0), t_B$. Assuming the displacement between them is spacelike, find the velocity of the system in which they are simultaneous.

Solution:

To find the velocity of the frame in which two events happen at the same time, we set the difference in time to zero after Lorentz transformation.

$$\Delta\bar{t} = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right) = 0 \rightarrow v = \frac{\Delta t}{\Delta x}c^2$$

$$v = \frac{t_A - t_B}{x_A - x_B}c^2$$

□

Problem 12.22

(a) Draw a space-time diagram representing a game of catch (or a conversation) between two people at rest, 10 ft apart. How is it possible for them to communicate, given that their separation is spacelike?

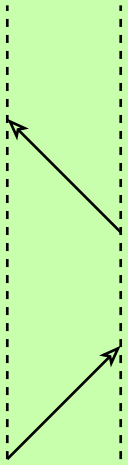
(b) There's an old limerick that runs as follows:

> There once was a girl named Ms. Bright,
> Who could travel much faster than light.
> She departed one day,
> The Einsteinian way,
> And returned on the previous night.

What do you think? Even if she could travel faster than light, could she return before she set out? Could she arrive at some intermediate destination before she set out? Draw a space-time diagram representing this trip.

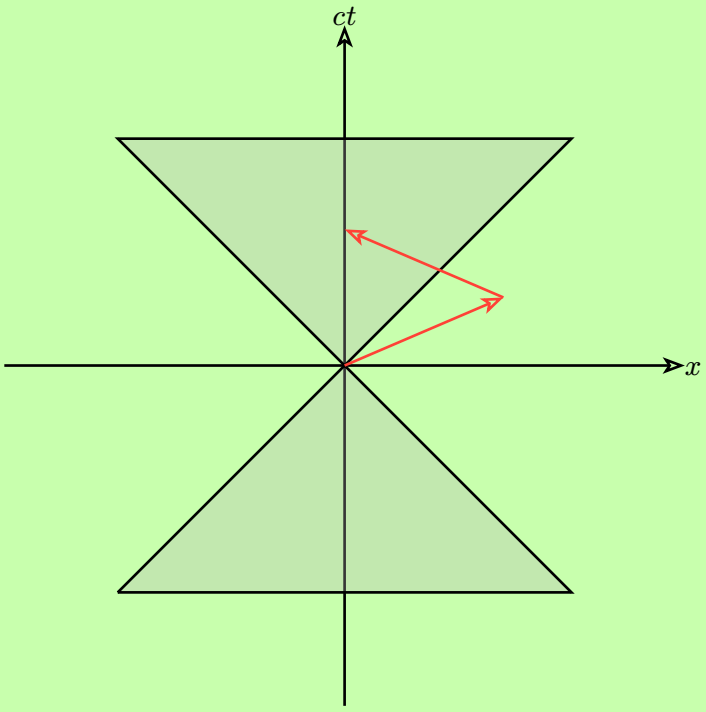
Solution:

(a)



You don't communicate with the person at the insant, but a future version of the person.

(b) No, travelling faster than light still makes you advance upwards in time.



(Unless it's possible to travel downwards?)

□

Problem 12.23

Inertial system $\overline{\mathcal{S}}$ moves in the x direction at speed $\frac{3}{5}c$ relative to system S. (The $\overline{x}$ axis slides along the x axis, and the origins coincide at $t = \overline{t} = 0$, as usual.)

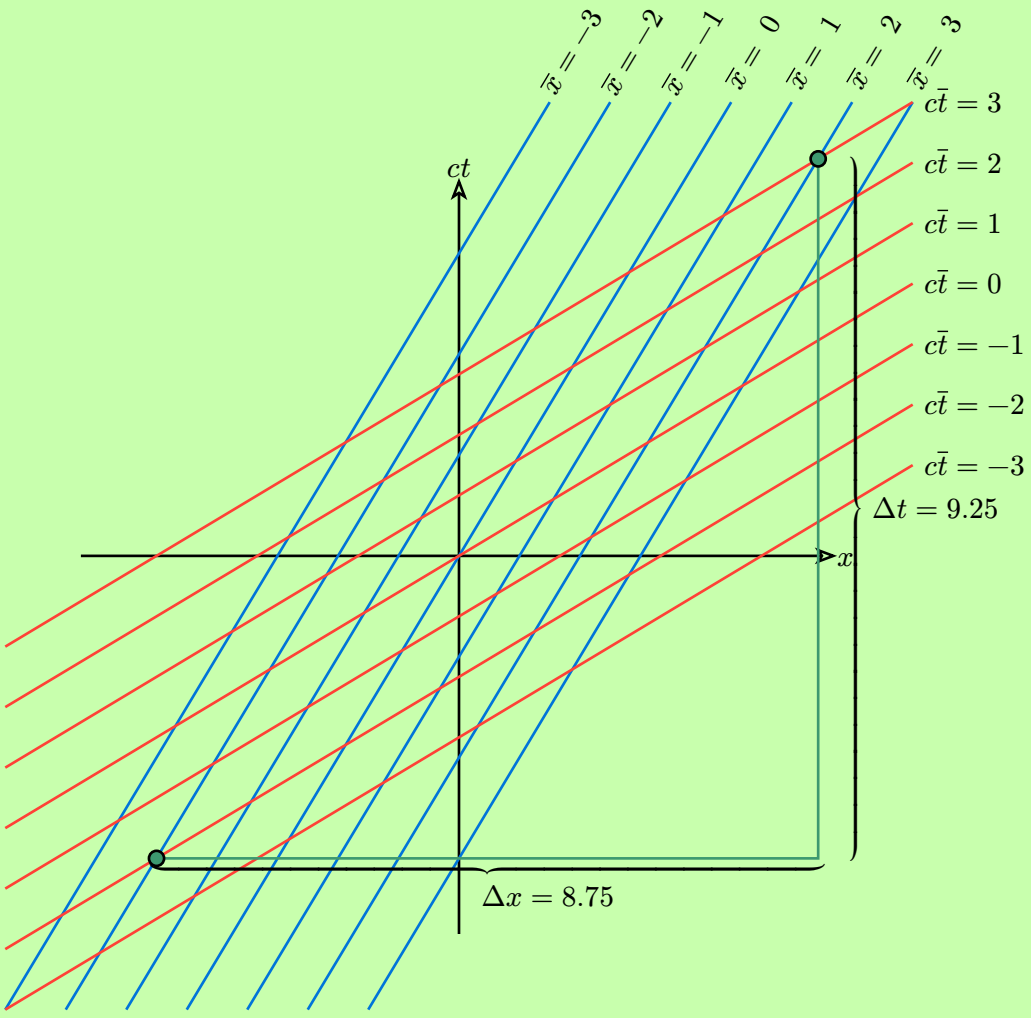
(a) On graph paper set up a Cartesian coordinate system with axes ct and x. Carefully draw in lines representing $\overline{x} = -3, -2, -1, 0, 1, 2,$ and 3. Also draw in the lines corresponding to $c\overline{t} = -3, -2, -1, 0, 1, 2,$ and 3. Label your lines clearly.

(b) In $\overline{\mathcal{S}}$, a free particle is observed to travel from the point $\overline{x} = -2$ at time $c\overline{t} = -2$ to the point $\overline{x} = 2$ at $c\overline{t} = 3$. Indicate this displacement on your graph. From the slope of this line, determine the particle's speed in $\mathcal{S}$.

(c) Use the velocity addition rule to determine the velocity in $\mathcal{S}$ algebraically.

Solution:

The graph will look like the following



(b) The displacement in $\mathcal{S}$ due to the particles movement in $\overline{\mathcal{S}}$, $\overline{x} : -2 \to 2, c\overline{t} : -2 \to 3$ is marked by the left $((\overline{x}, c\overline{t}) = (-2, -2))$ and right $((\overline{x}, c\overline{t}) = (2, 3))$ green dots. The slope

of this line is $\frac{\Delta ct}{\Delta x} = \frac{9.25}{8.75} = \frac{37}{35}$, and the speed is given by the inverse of this slope, which is $\frac{35}{37}c$. Looking at the graph this makes immediate sense since the usual notion of speed is $\frac{\Delta x}{\Delta t}$ which in our case gives

$$\frac{8.75}{9.25}c = \boxed{\frac{35}{37}c}$$

(c) The speed of $\overline{\mathcal{S}}$ is $\frac{3}{5}c$, and from (b) we are given $\Delta c\bar{t} = 5, \Delta\overline{x} = 4 \rightarrow v_p = \frac{4}{5}c$ where v_p denotes the particles speed in the frame $\overline{\mathcal{S}}$. Einsteins velocity addition rule then gives

$$v = \frac{\frac{3}{5}c + \frac{4}{5}c}{1 + \frac{3}{5}c\frac{4}{5}c\frac{1}{c^2}} = \frac{\frac{7}{5}c}{\frac{25}{25} + \frac{12}{25}} = \frac{\frac{25\times 7}{5}}{25 + 12}c = \boxed{\frac{35}{37}c}$$

which is exactly the result in (b). □

Problem 12.24

(a) Equation 12.40 defines proper velocity in terms of ordinary velocity. Invert that equation to get the formula for u in terms of η.

(b) What is the relation between proper velocity and rapidity (Eq. 12.34)? Assume the velocity is along the x direction, and find η as a function of θ.

Solution:

(a) The proper velocity is

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\mathbf{u}$$

Squaring both sides gives

$$\eta^2 = \frac{1}{1 - \frac{u^2}{c^2}}u^2$$

Solving for u

$$\eta^2\left(1-\frac{u^2}{c^2}\right)=u^2$$

$$\eta^2=u^2+\eta^2\frac{u^2}{c^2}$$

$$\eta^2=u^2\left(1+\frac{\eta^2}{c^2}\right)$$

$$\frac{\eta^2}{1+\frac{\eta^2}{c^2}}=u^2$$

Now taking the square root on both sides gives the ordinary velocity u in terms of the proper velocity η

$$\boxed{\mathbf{u}=\frac{1}{\sqrt{1+\eta^2/c^2}}\boldsymbol{\eta}}$$

(b) The rapidity θ is defined as

$$\theta=\tanh^{-1}\left(\frac{u}{c}\right)\rightarrow\frac{u}{c}=\tanh(\theta),\quad u=c\tanh(\theta)$$

Rewriting the proper velocity in terms of rapidity then gives

$$\boldsymbol{\eta}=\frac{1}{\sqrt{1-\tanh^2(\theta)}}c\tanh(\theta)$$

Using the hyperbolic relations

$$\text{(i) } 1-\tanh^2(\theta)=\operatorname{sech}^2(\theta)=\frac{1}{\cosh^2(\theta)}$$

$$\text{(ii) } \tanh(\theta)=\frac{\sinh(\theta)}{\cosh(\theta)}$$

we can rewrite the proper velocity as

$$\boldsymbol{\eta}=c\frac{\sinh(\theta)}{\cosh(\theta)\frac{1}{\cosh(\theta)}}=\boxed{c\sinh(\theta)}$$

□

Problem 12.25

Problem 12.26

Find the invariant product of the 4-velocity with itself, $\eta^\mu \eta_\mu$. Is η^μ timelike, spacelike, or lightlike?

Solution:

The proper velocity is

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}\mathbf{u}$$

This is the spatial component of a 4-vector which has the zeroth component

$$\eta^0 = \frac{c}{\sqrt{1-\frac{u^2}{c^2}}}$$

The invariant product is

$$\eta^\mu \eta_\mu = (\eta^0, \eta)\cdot(-\eta^0, \eta) = -\frac{c^2}{1-\frac{u^2}{c^2}} + \frac{u^2}{1-\frac{u^2}{c^2}} = -c^2\frac{1-\frac{u^2}{c^2}}{1-\frac{u^2}{c^2}} = \boxed{-c^2}$$

The invariant product is therefore timelike ($I < 0$). □

Problem 12.27

A cop pulls you over and asks what speed you were going. "Well, officer, I cannot tell a lie: the speedometer read 4×10^8 m/s." He gives you a ticket, because the speed limit on this highway is 2.5×10^8 m/s. In court, your lawyer (who, luckily, has studied physics) points out that a car's speedometer measures proper velocity, whereas the speed limit is ordinary velocity. Guilty, or innocent?

Solution:

The relation between proper velocity and ordinary velocity is

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\mathbf{u}$$

$$\downarrow$$

$$\mathbf{u} = \frac{1}{\sqrt{1 + \frac{\eta^2}{c^2}}}\boldsymbol{\eta}$$

Solving for $\mathbf{u}$ with $\boldsymbol{\eta} = \frac{4}{3}c$ gives

$$\mathbf{u} = \frac{1}{\sqrt{1 + \frac{4^2}{3^2}}}\frac{4}{3}c = \frac{1}{\sqrt{\frac{25}{9}}}\frac{4}{3}c = \frac{3}{5}\frac{4}{3}c = \frac{4}{5}c = \boxed{2.4 \times 10^8 \text{ m/s}}$$

Since $\mathbf{u} < 2.5 \times 10^8$ m/s, the driver is innocent. □

Problem 12.28

Problem 12.29

(a) Repeat Prob. 12.2(a) using the (incorrect) definition $\mathbf{p} = m\mathbf{u}$, but with the (correct) Einstein velocity addition rule. Notice that if momentum (so defined) is conserved in $\mathcal{S}$, it is not conserved in $\overline{\mathcal{S}}$. Assume all motion is along the x axis.

(b) Now do the same using the correct definition, $\mathbf{p} = m\boldsymbol{\eta}$. Notice that if momentum (so defined) is conserved in $\mathcal{S}$, it is automatically also conserved in $\overline{\mathcal{S}}$. [Hint: Use Eq. 12.43 to transform the proper velocity.] What must you assume about relativistic energy?

Problem 12.30

If a particle's kinetic energy is n times its rest energy, what is its speed?

Solution:

The kinetic energy is given by

$$E_{\text{kin}} = mc^2\left(\frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} - 1\right)$$

and the rest energy is

$$E_{\text{rest}} = mc^2$$

If we want the kinetic energy to be n times its rest energy, we set

$$\frac{1}{\sqrt{1-\frac{u^2}{c^2}}} - 1 = n$$

Solving for u gives

$$\frac{1}{1-\frac{u^2}{c^2}} = (n+1)^2$$

$$\frac{1}{(n+1)^2} = 1 - \frac{u^2}{c^2}$$

$$\frac{u^2}{c^2} = 1 - \frac{1}{(n+1)^2}$$

$$u^2 = \left(1 - \frac{1}{(n+1)^2}\right)c^2$$

$$u = \sqrt{\frac{(n+1)^2 - 1}{(n+1)^2}}c$$

$$u = \frac{\sqrt{n^2+2n}}{(n+1)^2}c$$

$$\boxed{u = \frac{\sqrt{n(n+2)}}{n+1}c}$$

□

Problem 12.31

Problem 12.32

Find the velocity of the muon in Ex. 12.8.

Solution:

$$E_{\text{before}} = m_\pi c^2, \qquad \mathbf{p}_{\text{before}} = 0$$
$$E_{\text{after}} = E_\mu + E_\nu, \quad \mathbf{p}_{\text{after}} = \mathbf{p}_\mu + \mathbf{p}_\nu$$

Conservation of energy gives

$$m_\pi c^2 = E_\mu + E_\nu$$

while conservation of momentum gives

$$\mathbf{p}_\mu = -\mathbf{p}_\nu$$

From example 12.8 we got

$$E_\mu = \frac{(m_\pi^2 + m_\mu^2)c^2}{2m_\pi}$$

and we also have

$$E_\mu = \frac{m_\mu c^2}{\sqrt{1 - \frac{u_\mu^2}{c^2}}}$$

With these two equations we can solve for u_μ

$$\frac{(m_\pi^2 + m_\mu^2)c^2}{2m_\pi} = \frac{m_\mu c^2}{\sqrt{1 - \frac{u_\mu^2}{c^2}}}$$

$$\downarrow$$

$$1 - \frac{u_\mu^2}{c^2} = \frac{4m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2}$$

$$\downarrow$$

$$u_\mu^2 = c^2 - \frac{4m_\pi^2 m_\mu^2 c^2}{\left(m_\pi^2 + m_\mu^2\right)^2} = \frac{\left(m_\pi^2 + m_\mu^2\right)^2 - \left(4m_\pi^2 m_\mu^2\right)}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 =$$

$$= \frac{m_\pi^4 + 2m_\pi^2 m_\mu^2 + m_\mu^4 - 4m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 = \frac{m_\pi^4 + m_\mu^4 - 2m_\pi^2 m_\mu^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2 =$$

$$= \frac{\left(m_\pi^2 - m_\mu^2\right)^2}{\left(m_\pi^2 + m_\mu^2\right)^2} c^2$$

$$\downarrow$$

$$\boxed{u_\mu = \frac{m_\pi^2 - m_\mu^2}{m_\pi^2 + m_\mu^2} c}$$

□

Problem 12.33

A particle of mass m whose total energy is twice its rest energy collides with an identical particle at rest. If they stick together, what is the mass of the resulting composite particle? What is its velocity?

Solution:

A particle of mass m has rest energy

$$E_{\text{rest}} = mc^2$$

and total energy

$$E^2 - p^2c^2 = m^2c^4 \rightarrow E^2 = m^2c^4 + p^2c^2$$

Twice gives

$$\left(2mc^2\right)^2 = m^2c^4 + p^2c^2$$

Solving for (initial) momentum p

$$p = \sqrt{3}mc$$

The initial energy is the energy of the moving particle plus the energy of the stationary particle:

$$2mc^2 + mc^2 = 3mc^2$$

Both energy and momentum are conserved so we can write

$$E^2 - p^2c^2 = \left(3mc^2\right)^2 - \left(\sqrt{3}mc\right)^2c^2 = 9m^2c^4 - 3m^2c^4 = 6m^2c^4$$

This means the mass of the composite particle is $6m^2$ which gives each particle a mass of

$$m_p = \sqrt{6m^2} = \sqrt{6}m$$

Using

$$E = \frac{m_p c^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$
$$\downarrow$$
$$3mc^2 = \frac{\sqrt{6}mc^2}{\sqrt{1 - \frac{u^2}{c^2}}}$$
$$\downarrow$$
$$9\left(1 - \frac{u^2}{c^2}\right) = 6$$
$$\downarrow$$
$$1 - \frac{u^2}{c^2} = \frac{6}{9}$$
$$\downarrow$$
$$u = \sqrt{1 - \frac{6}{9}}c = \sqrt{\frac{1}{3}}c$$
$$\boxed{u = \frac{c}{\sqrt{3}}}$$

□

Problem 12.34

A neutral pion of (rest) mass m and (relativistic) momentum $p = \frac{3}{4}mc$ decays into two photons. One of the photons is emitted in the same direction as the original pion, and the other in the opposite direction. Find the (relativistic) energy of each photon.

Solution:

Momentum and energy are conserved. We have the energy of the pion

$$E^2 = p^2c^2 + m^2c^4 = \frac{9}{16}m^2c^4 + m^2c^4 = \frac{5}{4}m^2c^4$$

Conservation of energy gives that the total energy of the two photons is

$$E_1 + E_2 = \frac{5}{4}mc^2$$

Conservation of momentum gives

$$p_1 + p_2 = \frac{3}{4}mc$$

and using the relation

$$E = pc$$

gives ($+c$ for photon 1 and $-c$ for photon 2)

$$p_1 + p_2 = \frac{E_1}{c} - \frac{E_2}{c} = \frac{3}{4}mc \rightarrow E_1 - E_2 = \frac{3}{4}mc^2$$

$$2E_1 = mc^2\left(\frac{5}{4} + \frac{3}{4}\right)$$

Solving for E_1 and E_2 gives the answer

$$E_1 = mc^2, \quad E_2 = \frac{1}{4}mc^2$$

□

Problem 12.35

Problem 12.36

Problem 12.37

In classical mechanics, Newton's law can be written in the more familiar form $\mathbf{F} = m\mathbf{a}$. The relativistic equation, $\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$, cannot be so simply expressed. Show, rather, that

$$\mathbf{F} = \frac{m}{\sqrt{1 - \frac{u^2}{c^2}}}\left[\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2 - u^2}\right]$$

where $\mathbf{a} \equiv \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t}$ is the ordinary acceleration.

Solution:

Newton's first law, which holds for relativistic speeds, is

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$$

Rewriting the momuentum $\mathbf{p}$ in terms of the relativistic speed gives

$$\mathbf{p} = m\gamma\mathbf{u} \rightarrow \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(m\gamma\mathbf{u}) = m(\gamma'\mathbf{u} + \gamma\mathbf{u}')$$

where, as usual,

$$\gamma = \frac{1}{\sqrt{1 - \frac{\mathbf{u}\cdot\mathbf{u}}{c^2}}}$$

The derivative of γ is

$$\gamma' = -\frac{1}{2}\frac{1}{\sqrt{1 - \frac{\mathbf{u}\cdot\mathbf{u}}{c^2}}^3} \times \left(-\frac{\mathbf{u}\cdot\mathbf{u}}{c^2}\right)' = \frac{1}{2c^2}\frac{1}{\sqrt{1 - \frac{\mathbf{u}\cdot\mathbf{u}}{c^2}}}(\mathbf{u}\cdot\mathbf{u})'$$

with

$$(\mathbf{u}\cdot\mathbf{u})' = \mathbf{u}'\cdot\mathbf{u} + \mathbf{u}\cdot\mathbf{u}' = 2\mathbf{u}\cdot\mathbf{u}'$$

so the derivative is simplified as

$$\gamma' = \frac{\mathbf{u}\cdot\mathbf{u}'}{c^2\sqrt{1 - \frac{u^2}{c^2}}^3}$$

Putting this into Newton's first law gives

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m(\gamma'\mathbf{u} + \gamma\mathbf{u}') = m\left(\frac{\mathbf{u}\cdot\mathbf{u}'}{c^2\sqrt{1 - \frac{u^2}{c^2}}^3}\mathbf{u} + \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\mathbf{u}'\right) =$$

$$= \frac{m}{\sqrt{1 - \frac{u^2}{c^2}}}\left(\frac{\mathbf{u}\cdot\mathbf{u}'}{c^2 - u^2}\mathbf{u} + \mathbf{u}'\right)$$

Finally, replacing $\mathbf{u}'$ with $\mathbf{a}$ we get

$$\mathbf{F} = \frac{m}{\sqrt{1 - \frac{u^2}{c^2}}}\left[\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2 - u^2}\right]$$

□

Problem 12.38

Show that it is possible to outrun a light ray, if you're given a sufficient head start, and your feet generate a constant force.

Problem 12.39

Define proper acceleration in the obvious way:

$$\alpha^\mu \equiv \frac{\mathrm{d}\eta^\mu}{\mathrm{d}\tau} = \frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2}$$

(a) Find α^0 and $\boldsymbol{\alpha}$ in terms of $\mathbf{u}$ and $\mathbf{a}$ (the ordinary acceleration).

(b) Express $\alpha_\mu \alpha^\mu$ in terms of $\mathbf{u}$ and $\mathbf{a}$.

(c) Show that $\eta^\mu \alpha_\mu = 0$.

(d) Write the Minkowski version of Newton's second law,

$$K^\mu \equiv \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau}$$

in terms of α^μ. Evaluate the invariant product $K^\mu \eta_\mu$.

Solution:

$$\alpha^0 = \frac{\mathrm{d}\eta^0}{\mathrm{d}\tau} = \frac{\mathrm{d}^2 x^0}{\mathrm{d}\tau^2}$$

$$\eta^0 = \frac{c}{\sqrt{1 - \frac{\mathbf{u}\cdot\mathbf{u}}{c^2}}}$$

$$\mathrm{d}\tau = \sqrt{1 - \frac{u^2}{c^2}}\,\mathrm{d}t$$

$$\frac{\mathrm{d}\eta^0}{\mathrm{d}\tau} = \frac{\mathrm{d}\eta^0}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\frac{\mathrm{d}\eta^0}{\mathrm{d}t}$$

$$\frac{\mathrm{d}\eta^0}{\mathrm{d}t} = \frac{\mathbf{u}\cdot\mathbf{u}'}{c}\frac{1}{\sqrt{1 - \frac{u^2}{c^2}}^3}$$

$$\boxed{\alpha^0 = \frac{\mathbf{u}\cdot\mathbf{a}}{c\left(1 - \frac{u^2}{c^2}\right)^2}}$$

$$\boldsymbol{\alpha} = \frac{\mathrm{d}\boldsymbol{\eta}}{\mathrm{d}\tau} = \frac{\mathrm{d}\boldsymbol{\eta}}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\tau}$$

$$\boldsymbol{\eta} = \frac{1}{\sqrt{1 - \frac{\mathbf{u}\cdot\mathbf{u}}{c^2}}}\mathbf{u} = \gamma\mathbf{u}$$

$$\frac{\mathrm{d}\boldsymbol{\eta}}{\mathrm{d}t} = (\gamma\mathbf{u})' = \gamma'\mathbf{u} + \gamma\mathbf{u}'$$

We have

$$\gamma' = \frac{\mathbf{u}\cdot\mathbf{u}'}{c^2\sqrt{1 - \frac{u^2}{c^2}}^3}$$

$$\gamma'\mathbf{u} + \gamma\mathbf{u}' = \left(\mathbf{u}\frac{\mathbf{u}\cdot\mathbf{a}}{c^2\sqrt{1 - \frac{u^2}{c^2}}^3} + \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\mathbf{a}\right) = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}\left(\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2 - u^2}\right) = \frac{\mathrm{d}\boldsymbol{\eta}}{\mathrm{d}t}$$

Now

$$\boldsymbol{\alpha} = \frac{\mathrm{d}\boldsymbol{\eta}}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}\left(\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2-u^2}\right)\frac{1}{\sqrt{1-\frac{u^2}{c^2}}} = \frac{1}{1-\frac{u^2}{c^2}}\left(\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2-u^2}\right)$$

So

$$\alpha^0 = \frac{\mathbf{u}\cdot\mathbf{a}}{c\left(1-\frac{u^2}{c^2}\right)^2}$$

$$\boldsymbol{\alpha} = \frac{1}{1-\frac{u^2}{c^2}}\left(\mathbf{a} + \frac{\mathbf{u}(\mathbf{u}\cdot\mathbf{a})}{c^2-u^2}\right)$$

(b) Incomplete

$\alpha_\mu\alpha^\mu = -(\alpha^0)^2 + \boldsymbol{\alpha}\cdot\boldsymbol{\alpha}$

(c) We will use

$$\alpha^\mu = \frac{\mathrm{d}\eta^\mu}{\mathrm{d}\tau}, \quad \eta^\mu\eta_\mu = -c^2$$

Taking the derivative of $\eta^\mu\eta_\mu$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(\eta^\mu\eta_\mu) = \frac{\mathrm{d}}{\mathrm{d}\tau}(-c^2) = 0$$

$$\downarrow$$

$$\frac{\mathrm{d}\eta^\mu}{\mathrm{d}\tau}\eta_\mu + \eta^\mu\frac{\mathrm{d}\eta_\mu}{\mathrm{d}\tau} = \alpha^\mu\eta_\mu + \eta^\mu\alpha_\mu$$

We also have

$$\alpha^\mu\eta_\mu = \eta^\mu\alpha_\mu$$

so this gives

$$2\eta^\mu\alpha_\mu = 0$$

$$\downarrow$$

$$\eta^\mu\alpha_\mu = 0$$

□

Problem 12.40

Problem 12.41

Problem 12.42

Problem 12.43

Problem 12.44

Problem 12.45

Problem 12.46

Problem 12.47

Problem 12.48

Problem 12.49

Problem 12.50

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Problem 12.74

Problem 12.75

Problem 12.76